Thin Flament Mechanics

Wilson Lough

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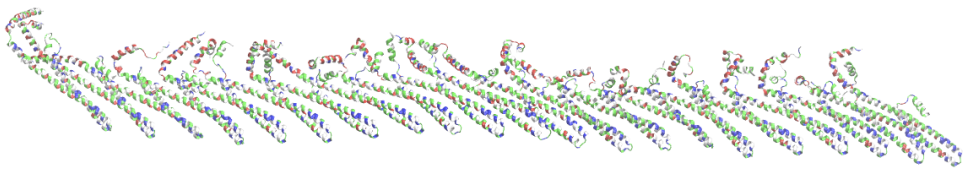
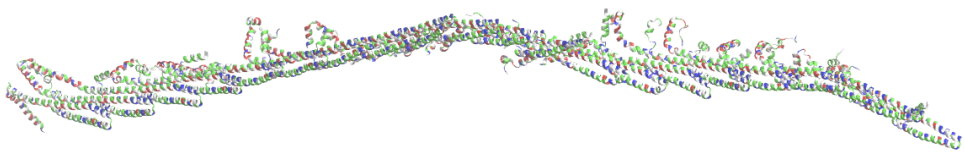


(a) Top view



(b) Side view

Figure 1: Images from Tarak

This stuff:

$$-B_3\alpha_1^3\alpha_2^2 + 2B_3\alpha_1^3\alpha_2\beta\,\mathrm{K_{22}}^2(s) + B_3\alpha_1^3\beta^2\,\mathrm{K_{22}}^4(s) + 3B_3\alpha_1^2\alpha_2^2\beta\,\mathrm{K_{12}}^2(s) + 4B_3\alpha_1^2\alpha_2\beta^2\,\mathrm{K_{12}}^2(s)\,\mathrm{K_{22}}^2(s) \tag{1}$$

$$+B_3\alpha_1^2\beta^3\,\mathrm{K_{12}}^2(s)\,\mathrm{K_{22}}^4(s) + 3B_3\alpha_1\alpha_2^2\beta^2\,\mathrm{K_{12}}^4(s) + 2B_3\alpha_1\alpha_2\beta^3\,\mathrm{K_{12}}^4(s)\,\mathrm{K_{22}}^2(s) + B_3\alpha_2^2\beta^3\,\mathrm{K_{12}}^6(s) \tag{2}$$

divided by this stuff:

$$\left(-\beta^2\,\mathrm{K_{12}}^2(s)\,\mathrm{K_{22}}^2(s) + \left(\alpha_1 + \beta\,\mathrm{K_{12}}^2(s)\right)\left(\alpha_2 + \beta\,\mathrm{K_{22}}^2(s)\right)\right)^3 \tag{3}$$

1 ESCRT

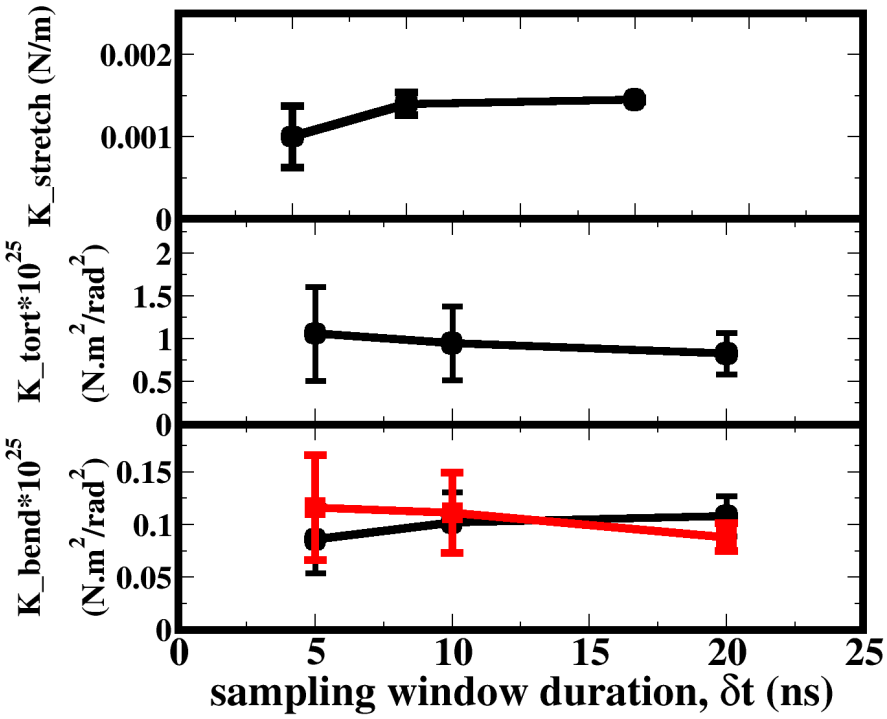


Figure 2: Bending and twisting moduli for finite filament from Tarak.

- Twisting Stiffness: $B_1 \approx .75 * 10^{-25}\ \mathrm{N} \cdot \mathrm{m}^2 = 75 \frac{\mathrm{kg \cdot nm^3}}{\mathrm{s^2}}$

- Bending stiffness: $B_2 \approx B_2 \approx .1 * 10^{-25}\ \mathrm{N} \cdot \mathrm{m}^2 = 10 \frac{\mathrm{kg \cdot nm^3}}{\mathrm{s^2}}$

$$B_2 \approx B_3 < B_1 \tag{4}$$

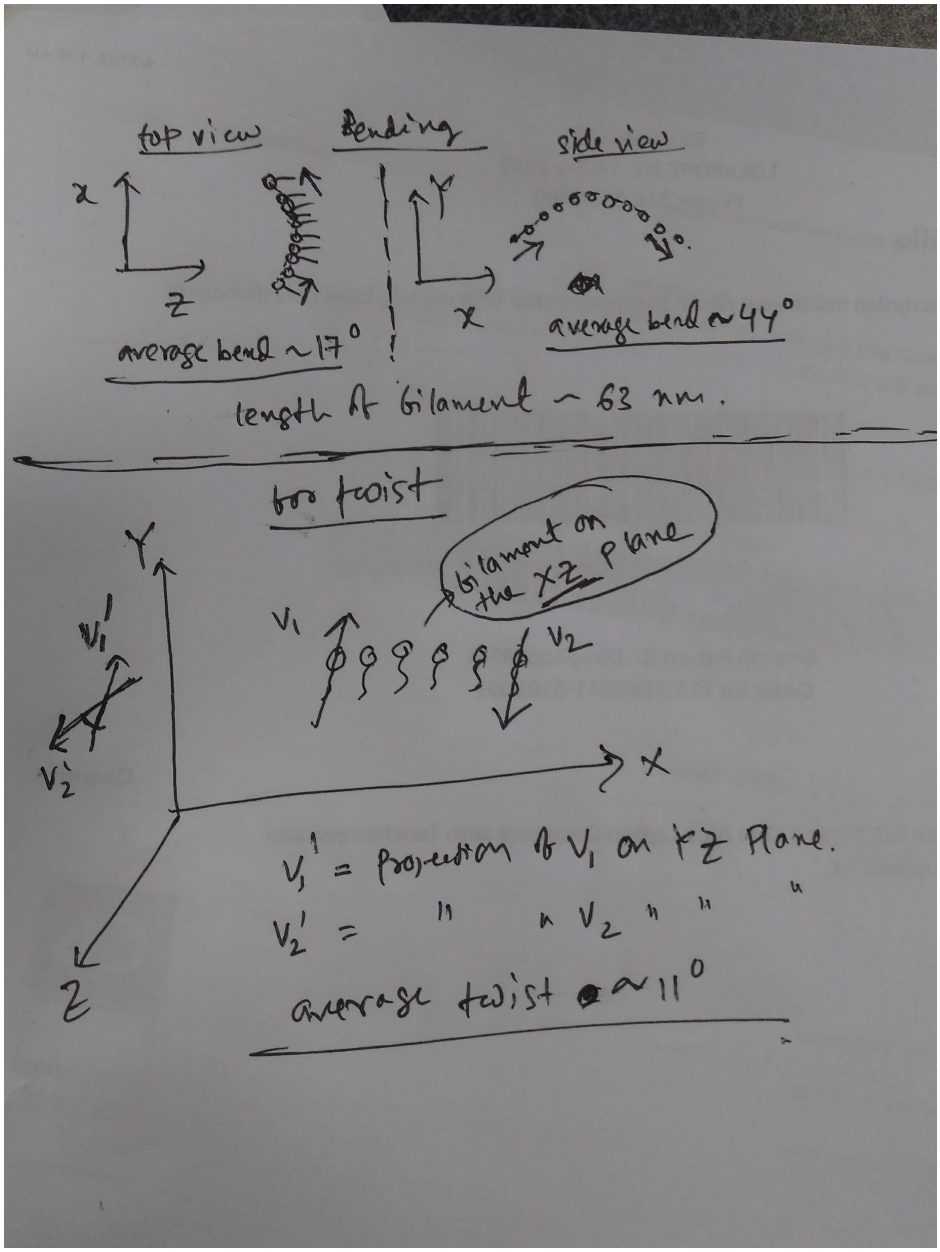


Figure 3: ESCRT sketch from Tarak.

- Spontaneous twist: $\sigma_1 = -\frac{11\pi}{180*63}$ nm^{-1} ≈ -0.00305 nm^{-1}.
- Out of plane bending: $\sigma_2 = \frac{44\pi}{180*63}$ nm^{-1} ≈ 0.0122 nm^{-1}

- In plane bending: $\sigma_3 = \frac{17\pi}{180*63}$ nm$^{-1} \approx 0.00471$ nm^{-1}

$$|\sigma_1| < |\sigma_3| < |\sigma_2| \tag{5}$$

2 Filament Kinematics

An inextensible and unshearable filament Γ can be thought of as a 1-dimensional continuum of rigid cross-sections labelled by their distance s along the filament (or some other material coordinate). Each cross-section is completely described the position of its' centroid/center of mass

$$\boldsymbol{r}(s) \in \mathbb{R}^3 \tag{6}$$

and an orthonormal frame

$$D(s) = \{\boldsymbol{d}_1(s), \boldsymbol{d}_2(s), \boldsymbol{d}_3(s)\} \in SO(3) \tag{7}$$

describing its' orientation. Call $\boldsymbol{r}(s)$ the *centerline* and $\{\boldsymbol{d}_i(s)\}$ the *Darboux frame.*

$$\Gamma : [0, L] \to \mathbb{R}^3 \times SO(3), \qquad \Gamma(s) = (\boldsymbol{r}(s), D(s)) \tag{8}$$

Cross-sections are assumed to be normal to the curve centerline, and the frame vectors are chosen to be smooth functions of s with $\boldsymbol{d}_1(s)$ tangent to $\boldsymbol{r}(s)$. It will be convenient to express derivatives of the Darboux frame in terms of the *Darboux vector* $\boldsymbol{\omega}$.

$$\boldsymbol{d}_i' = \boldsymbol{\omega} \times \boldsymbol{d}_i \tag{9}$$

Keep in mind that we are not introducing additional degrees of freedom or constraints to be enforced. We already know that such a relationship between the frame vectors exists since it is a requirement for the frame to be a curve through $SO(3)$. This is really just notation and at the end of the day, the Darboux vector is a function of the frame vectors and their derivatives.

$$\boldsymbol{\omega} = \frac{1}{2} \sum_i \boldsymbol{d}_i \times \boldsymbol{d}_i' \tag{10}$$

2.1 Some more notation

$$\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix} \tag{11}$$

$$\boldsymbol{d}_i(s) = D(s)\boldsymbol{e}_i = R(s)\boldsymbol{d}_i(0) \qquad D(s) = R(s)D(0) \tag{12}$$

$$\boldsymbol{d}_i' = \frac{\mathrm{d}}{\mathrm{d}s}\boldsymbol{d}_i \tag{13}$$

$$\mathring{v} = D^T\boldsymbol{v} = \begin{bmatrix}\boldsymbol{d}_1 \cdot \boldsymbol{v}\\ \boldsymbol{d}_2 \cdot \boldsymbol{v}\\ \boldsymbol{d}_3 \cdot \boldsymbol{v}\end{bmatrix} \tag{14}$$

2.2 Variations and virtual displacements

Orthonormality constraints are preserverd under variation of the Darboux frame as long as $\delta D(s)$ is restricted to the tangent space of $SO(3)$ at $D(s)$.

$$T_{D(s)}SO(3) = \{\hat{\boldsymbol{\alpha}} \cdot D(s) | \hat{\boldsymbol{\alpha}} \in \mathfrak{so}(3)\} \tag{15}$$

Such a variation is equivalent to performing an infinitesimal rotation of the frame at each point along the filament. This ensures that the variations don't take D out of $SO(3)$. Frame variations can be written in terms of some $\hat{\boldsymbol{\psi}}(s) \in \mathfrak{so}(3)$ which can be thought of as the variational analogue of the Darboux vector.

$$\delta D = \hat{\boldsymbol{\psi}} \cdot D \qquad \delta\boldsymbol{d}_i = \boldsymbol{\psi} \times \boldsymbol{d}_i \tag{16}$$

Proof. Let $D(\epsilon; s)$ be a family of rotation operators parametrized by ϵ which satisfies $D(0; s) = D(s)$. Then, define $U(\epsilon; s) := D(\epsilon; s)D(s)^T$ and note that it satisfies $U(0; s) = 1$. So, assuming everything is sufficiently smooth, $\hat{\boldsymbol{\psi}}(s) := \frac{\partial}{\partial\epsilon}U(\epsilon; s)\big|_{\epsilon=0}$ defines a map through $\mathfrak{so}(3)$.

$$\begin{aligned}
D(\epsilon; s) &= D(s) + \epsilon\left[\frac{\partial}{\partial\epsilon}D(\epsilon; s)\Big|_{\epsilon=0}\right] + O(\epsilon^2)\\
&= D(s) + \epsilon\left[\frac{\partial}{\partial\epsilon}\left(D(\epsilon; s)D(s)^T\right)\Big|_{\epsilon=0}\right]D(s) + O(\epsilon^2)\\
&= D(s) + \epsilon\hat{\boldsymbol{\psi}}(s)D(s) + O(\epsilon^2)
\end{aligned}$$

□

Fact 1. *The requirement that derivatives and variations to commute is equivalent to*[1]

$$\delta\hat{\boldsymbol{\omega}} = \hat{\boldsymbol{\psi}}' + \left[\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\omega}}\right] \qquad \text{or} \qquad \delta\boldsymbol{\omega} = \boldsymbol{\psi}' + \boldsymbol{\psi} \times \boldsymbol{\omega}. \tag{17}$$

Their componenets in the Darboux basis are related by

$$\delta\omega_i = \boldsymbol{d}_i \cdot \boldsymbol{\psi}' \qquad \text{or} \qquad \psi_i' = \boldsymbol{d}_i \cdot \delta\boldsymbol{\omega}. \tag{18}$$

Proof. $\delta\left(\boldsymbol{d}_i'\right) = \left(\delta\boldsymbol{d}_i\right)'$, then

$$\begin{aligned}\boldsymbol{d}_i \times \delta\boldsymbol{\omega} &= \boldsymbol{d}_i \times \boldsymbol{\psi}' - \boldsymbol{\omega} \times \left(\boldsymbol{d}_i \times \boldsymbol{\psi}\right) - \boldsymbol{\psi} \times \left(\boldsymbol{\omega} \times \boldsymbol{d}_i\right) \\ &= \boldsymbol{d}_i \times \boldsymbol{\psi}' + \boldsymbol{d}_i \times \left(\boldsymbol{\psi} \times \boldsymbol{\omega}\right) \qquad \text{(Jacobi identity)}\end{aligned}$$

and since the $\boldsymbol{d}_i$'s form a basis, we get $\delta\boldsymbol{\omega} = \boldsymbol{\psi}' + \boldsymbol{\psi} \times \boldsymbol{\omega}$. You can also show this by using the family of rotation operators defined in the previous proof. □

Fact 2. *Both derivatives and variations of the Darboux frame commute with the hat map.*

Proof.

$$\begin{aligned}\widehat{\left(\boldsymbol{d}_i'\right)} &= \widehat{\boldsymbol{\omega} \times \boldsymbol{d}}_i \\ &= \left[\hat{\boldsymbol{\omega}}, \hat{\boldsymbol{d}}_i\right] \\ &= \varepsilon_{in}^{m}\hat{\boldsymbol{\omega}} \cdot \boldsymbol{d}_m\boldsymbol{d}^n - \varepsilon_{in}^{m}\boldsymbol{d}_m\boldsymbol{d}^n \cdot \hat{\boldsymbol{\omega}} \\ &= \varepsilon_{in}^{m}\boldsymbol{d}_m'\boldsymbol{d}^n + \varepsilon_{in}^{m}\boldsymbol{d}_m\boldsymbol{d}'^n \\ &= \left(\hat{\boldsymbol{d}}_i\right)'\end{aligned}$$

□

2.3 Alternative descriptions of the filament

Once the position of the centerline is known, the configuration of the filament can be determined by specifying the orientation of the Darboux frame

[1] for non-invariant Darboux vector, use $\boldsymbol{\nabla}_\Theta$ and variations to get $\boldsymbol{\nabla}_\Theta\hat{\boldsymbol{\psi}} + \left[\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\omega}}_\Theta\right] = \delta\hat{\boldsymbol{\omega}}_\Theta$?

with respect to some ‘reference‘ frame (e.g. Frenet-Serret or Bishop). Since addapted frames can only differ by a rotation about the tangent[2], non-redundant description of the filament is given by $\Gamma = (\kappa, \tau, \phi)$ where κ, τ are curvature and torsion and ϕ is the angle which the Darboux frame is rotated with respect to the Frenet-Serret/Bishop frame.

[2]See Appendix B.

3 Energy and forces

Let the filament have a stress free configuration characterized by *spontaneous curvatures*

$$\sigma_1, \sigma_2, \sigma_3. \tag{19}$$

and let $\boldsymbol{f}$ and $\boldsymbol{m}$ be the force and moment exerted on the cross-section at s by the upstream piece of filament [3].

Fact 3. *In equilibrium, local force and torque balance imply*

$$\boldsymbol{f}' + \boldsymbol{F} = 0 \tag{20}$$

$$\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f} + \boldsymbol{M} = 0. \tag{21}$$

where $\boldsymbol{F}$ and $\boldsymbol{M}$ are any additional force and moment (external stresses, steric forces, etc...) per unit length applied to the filament.

Proof. Consider a slice with thickness $\mathrm{d}s$, calculate total force/moment, divide by $\mathrm{d}s$ and let $\mathrm{d}s \to 0$. □

Fact 4. *The force and moment banlance equations can be obtained by minimizing the energy*

$$E = \int \frac{1}{2} \sum_i B_i \left(\omega_i - \sigma_i\right)^2 + \boldsymbol{f} \cdot \left(\boldsymbol{r}' - \boldsymbol{d}_1\right) \mathrm{d}s\,, \tag{22}$$

interpreting the quantity

$$\boldsymbol{m} = \sum_i B_i \left(\omega_i - \sigma_i\right) \boldsymbol{d}_i. \tag{23}$$

as the moment on cross-sections, and interpreting the Lagrange multiplier enforcing inextensibility/unshearbility

$$\boldsymbol{f} \tag{24}$$

[3] Technically, the pair $(\boldsymbol{f}, \boldsymbol{m})$ is the decomposition of the actual stress over the surface into a force acting at the centroid (so it doesn't cause a torque about the centroid) and a couple (or pure moment). You can think of the couple as the torque caused by a pair (a 'couple'!) of equal and opposite forces (so the net force is zero) acting on different points of the surface. Keep in mind that $\boldsymbol{f}(s)$ is only torque free when you are calculating torques about the centroid of the face at s.

as the force on cross-sections. The first term

$$\int \frac{1}{2} \sum_i B_i \left(\omega_i - \sigma_i\right)^2 \mathrm{d}s \tag{25}$$

is the elastic energy of the filament.

Proof. In general, there may be external forces and moments acting on the filament. Let $\boldsymbol{F}, \boldsymbol{M}$ be the external force and moment applied per unit length and let $\boldsymbol{f}_{\text{ext}}(0), \boldsymbol{f}_{\text{ext}}(L)$, $\boldsymbol{m}_{\text{ext}}(0), \boldsymbol{m}_{\text{ext}}(L)$ be the external force and moment applied to the ends of the filament. The work done by external forces/moments is

$$\delta W_{\text{ext}} = \boldsymbol{m}_{\text{ext}}(0) \cdot \boldsymbol{\psi}(0) + \boldsymbol{m}_{\text{ext}}(L) \cdot \boldsymbol{\psi}(L) + \boldsymbol{f}_{\text{ext}}(0) \cdot \delta \boldsymbol{r}(0) + \boldsymbol{f}_{\text{ext}}(L) \cdot \delta \boldsymbol{r}(L) + \int \boldsymbol{F} \cdot \delta \boldsymbol{r} + \boldsymbol{M} \cdot \boldsymbol{\psi} \, \mathrm{d}s \tag{26}$$

and the change in energy is

$$\begin{aligned}
\delta E &= \int \sum_i B_i \left(\omega_i - \sigma_i\right) \delta \omega_i + \boldsymbol{f} \cdot \left(\delta \boldsymbol{r}' - \delta \boldsymbol{d}_1\right) && (27)\\
&= \int \boldsymbol{m} \cdot \boldsymbol{\psi}' - \left(\boldsymbol{d}_1 \times \boldsymbol{f}\right) \cdot \boldsymbol{\psi} + \boldsymbol{f} \cdot \delta \boldsymbol{r}' \, \mathrm{d}s && (28)\\
&= \int \boldsymbol{m} \cdot \boldsymbol{\psi}' - \left(\boldsymbol{d}_1 \times \boldsymbol{f}\right) \cdot \boldsymbol{\psi} + \boldsymbol{f} \cdot \delta \boldsymbol{r}' \, \mathrm{d}s && (29)\\
&= \left(\boldsymbol{m} \cdot \boldsymbol{\psi} + \boldsymbol{f} \cdot \delta \boldsymbol{r}\right)\Big|_0^L - \int \left(\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f}\right) \cdot \boldsymbol{\psi} + \boldsymbol{f}' \cdot \delta \boldsymbol{r} \, \mathrm{d}s \,. && (30)
\end{aligned}$$

In equilibrium, variation of the total energy is zero for arbitrary virtual displacements

$$\delta E_{\text{tot}} = \delta E - \delta W, \qquad \text{for all } \delta \boldsymbol{r}, \boldsymbol{\psi}. \tag{31}$$

Coefficients of $\boldsymbol{\psi}(s)$ and $\delta \boldsymbol{r}(s)$ give the force and moment balance equations. Vanishing of the boundary terms requires

$$\begin{aligned}
\boldsymbol{m}(0) &= -\boldsymbol{m}_{\text{ext}}(0) && (32)\\
\boldsymbol{m}(L) &= \boldsymbol{m}_{\text{ext}}(L) && (33)\\
\boldsymbol{f}(0) &= -\boldsymbol{f}_{\text{ext}}(0) && (34)\\
\boldsymbol{f}(L) &= \boldsymbol{f}_{\text{ext}}(L) && (35)
\end{aligned}$$

These boundary conditions just state that the net force/moment on each of the ends is zero. The difference in signs for the $s = 0, L$ is due to the fact that we defined $\boldsymbol{f}, \boldsymbol{m}$ to be the force and moment acting on the cross-section by the upstream piece of the filament. The force that the rest of the filament exerts on the $s = 0$ end is $\boldsymbol{f}(0)$, but the force that the rest of the filament exerts on the $s = L$ end is actually $-\boldsymbol{f}(L)$ (i.e the negative of the force exerted on the cross-section at $s = L - \mathrm{d}s$ by the upstream piece at $s = L$). $\square$

3.1 Steric Interactions

Suppose the filament has a charge per unit length $c(s)$ and charge distribution

$$\rho(\boldsymbol{x}) = \int_0^L c(s)\delta(\boldsymbol{x} - \boldsymbol{r}(s))\,\mathrm{d}s \tag{36}$$

which produces an electric potential $\phi(\boldsymbol{x})$ with effective range a.

$$\left[\nabla^2 - 1/a^2\right]\phi = -\rho \tag{37}$$

$$\phi(\boldsymbol{x}) = \int_0^L c(s)G(\boldsymbol{x} - \boldsymbol{r}(s)) \tag{38}$$

Maybe we should use something like

$$c(s) = \sum_k c_k \delta(s - s_k) \tag{39}$$

The self energy of the distribution is

$$E_s[\boldsymbol{r}] = \frac{1}{2}\int\int \frac{C(s,s')e^{|\boldsymbol{r}(s)-\boldsymbol{r}(s')|/a}}{4\pi|\boldsymbol{r}(s) - \boldsymbol{r}(s')|}\,\mathrm{d}s\,\mathrm{d}s' \tag{40}$$

$$= \frac{1}{2}\int\int C(s,s')G(\boldsymbol{r}(s) - \boldsymbol{r}(s'))\,\mathrm{d}s\,\mathrm{d}s' \tag{41}$$

where maybe $C(s,s')$ is something to deal with infinite energies (symmetric in s, s'). Variation of the centerline position gives

$$\delta E_s = \frac{1}{2}\int\int C(s,s')\boldsymbol{\nabla}G(\boldsymbol{r}(s) - \boldsymbol{r}(s'))\cdot\delta\boldsymbol{r}(s)\,\mathrm{d}s\,\mathrm{d}s' - \frac{1}{2}\int\int C(s,s')\boldsymbol{\nabla}G(\boldsymbol{r}(s) - \boldsymbol{r}(s'))\cdot\delta\boldsymbol{r}(s' \tag{42}$$

$$= \int\int C(s,s')\boldsymbol{\nabla}G(\boldsymbol{r}(s) - \boldsymbol{r}(s'))\cdot\delta\boldsymbol{r}(s)\,\mathrm{d}s\,\mathrm{d}s' \tag{43}$$

This contributes a term to the force balance equation for the filament.

$$\boldsymbol{F}_s = -\int C(s,s')\boldsymbol{\nabla} G(\boldsymbol{r}(s)-\boldsymbol{r}(s'))\,\mathrm{d}s' \tag{44}$$

$$= \int \frac{C(s,s')\left(|\boldsymbol{r}(s)-\boldsymbol{r}(s')|+a\right)e^{-|\boldsymbol{r}(s)-\boldsymbol{r}(s')|/a}}{4\pi a|\boldsymbol{r}(s)-\boldsymbol{r}(s')|^3}\left(\boldsymbol{r}(s)-\boldsymbol{r}(s')\right)\mathrm{d}s' \tag{45}$$

The minus sign comes from integration by parts of the other stuff in the force balance equation.

A new conservation law

If $C(s, s') = c^2 = \text{constant}$

$$\boldsymbol{F}_s(s) \cdot \boldsymbol{d}_1(s) = \boldsymbol{r}'(s) \cdot \boldsymbol{F}_s(s) \tag{46}$$

$$= -\int_0^L c^2 \boldsymbol{r}'(s) \cdot \boldsymbol{\nabla} G(\boldsymbol{r}(s) - \boldsymbol{r}(\xi))\, \mathrm{d}\xi \tag{47}$$

$$= -\frac{\mathrm{d}}{\mathrm{d}s}\left[\int_0^L c^2 G(\boldsymbol{r}(s) - \boldsymbol{r}(\xi))\, \mathrm{d}\xi\right] \tag{48}$$

$$\boldsymbol{\omega} \cdot \boldsymbol{m}' + \boldsymbol{d}_1' \cdot \boldsymbol{f} + \boldsymbol{M}_{ext} \cdot \boldsymbol{\omega} = 0 \tag{49}$$

$$\Rightarrow \sum_i B_i \omega_i \omega_i' + f_1' - \boldsymbol{d}_1 \cdot \boldsymbol{f}' + \boldsymbol{M}_{ext} \cdot \boldsymbol{\omega} = 0 \tag{50}$$

$$\Rightarrow \sum_i B_i \omega_i \omega_i' + f_1' + \boldsymbol{d}_1 \cdot \boldsymbol{F}_S + \boldsymbol{d}_1 \cdot \boldsymbol{F}_{ext} + \boldsymbol{M}_{ext} \cdot \boldsymbol{\omega} = 0 \tag{51}$$

$$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}\sum_i B_i \omega_i^2 + f_1 - \int_0^L c^2 G(\boldsymbol{r} - \boldsymbol{r}(\xi))\, \mathrm{d}\xi\right] + \boldsymbol{d}_1 \cdot \boldsymbol{F}_{ext} + \boldsymbol{M}_{ext} \cdot \boldsymbol{\omega} = 0 \tag{52}$$

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}\sum_i B_i \omega_i^2 + f_1 - \int_0^L c^2 G(\boldsymbol{r} - \boldsymbol{r}(\xi))\, \mathrm{d}\xi\right] = -\left[\boldsymbol{d}_1 \cdot \boldsymbol{F}_{ext} + \boldsymbol{M}_{ext} \cdot \boldsymbol{\omega}\right]} \tag{53}$$

4 Filament Dynamics

$$\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f} + \boldsymbol{M}_v = 0 \tag{54}$$
$$\boldsymbol{f}' + \boldsymbol{F}_s + \boldsymbol{F}_v = 0 \tag{55}$$
$$\boldsymbol{d}_i' = \boldsymbol{\omega} \times \boldsymbol{d}_i \tag{56}$$
$$\boldsymbol{r}' = \boldsymbol{d}_1 \tag{57}$$
$$\dot{\boldsymbol{r}} = \boldsymbol{v} \tag{58}$$
$$\dot{\boldsymbol{d}}_i = \boldsymbol{\Omega} \times \boldsymbol{d}_i \tag{59}$$
$$\dot{\boldsymbol{\omega}} = \boldsymbol{\Omega}' + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{60}$$

Where $\boldsymbol{m} = \sum_i (\omega_i - \sigma_i)\boldsymbol{d}_i$ is the internal moment. Let $\boldsymbol{\delta}_\perp = \boldsymbol{d}_2\boldsymbol{d}_2 + \boldsymbol{d}_3\boldsymbol{d}_3$ be the projection onto the subspace perpendicular to $\boldsymbol{d}_1$. The viscous moment and force are

$$\boldsymbol{M}_v = -\beta\Omega_1\boldsymbol{d}_1 \tag{61}$$
$$\boldsymbol{F}_v = -\alpha_\parallel v_1\boldsymbol{d}_1 - \alpha_\perp\boldsymbol{\delta}_\perp \cdot \boldsymbol{v} \tag{62}$$

Given the filament configuration $\boldsymbol{r}, \boldsymbol{d}_i, \boldsymbol{\omega}$ at time t you can get the configuration at $t + \Delta t$ by time evolving the Darboux vector $\boldsymbol{\omega}$ and then solving the kinematic equations

$$\boldsymbol{d}_i' = \boldsymbol{\omega} \times \boldsymbol{d}_i \tag{63}$$
$$\boldsymbol{r}' = \boldsymbol{d}_1. \tag{64}$$

To time evolve $\boldsymbol{\omega}$, you first need to express the internal force $\boldsymbol{f}$, the velocity $\boldsymbol{v}$, and the angular velocity $\boldsymbol{\Omega}$ in terms of the filament configuration.

- <u>Internal force</u>:

$$\boxed{\boldsymbol{f} = f_1\boldsymbol{d}_1 + \boldsymbol{d}_1 \times \boldsymbol{m}'} \tag{65}$$

  The second term $\boldsymbol{d}_2 \times \boldsymbol{m}'$ can be immediately obtained from the filament configuration. The $\boldsymbol{d}_1$ component (tension) satisfies the second order DE

$$\boxed{f_1'' - \frac{\alpha_\parallel}{\alpha_\perp}|\boldsymbol{r}''|^2 f_1 = \chi(s)} \tag{66}$$

$$\boxed{f_1' + (\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' + \boldsymbol{r}' \cdot \boldsymbol{F}_s|_{s=0} = 0} \tag{67}$$

$$\boxed{f_1|_{s=L} = 0} \tag{68}$$

where $\chi(s)$ is a known function which depends ononly the filament configuration,

$$\chi(s) = -\left[(\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' + \boldsymbol{r}' \cdot \boldsymbol{F}_s\right]' - \frac{\alpha_\parallel}{\alpha_\perp}\left[(\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' - \boldsymbol{r}'' \cdot \boldsymbol{F}_s\right], \tag{69}$$

and boundary conditions are from $\boldsymbol{v}|_{s=0} = 0,\ \boldsymbol{f}|_{s=L} = 0$.

- <u>Velocity</u>:

$$\boxed{\boldsymbol{v} = \left(\frac{1}{\alpha_\parallel}\boldsymbol{d}_1\boldsymbol{d}_1 + \frac{1}{\alpha_\perp}\boldsymbol{\delta}_\perp\right) \cdot (\boldsymbol{f}' + \boldsymbol{F}_s)} \tag{70}$$

- <u>Angular velocity</u>:

$$\boxed{\boldsymbol{\Omega} = \frac{1}{\beta}\left(\boldsymbol{d}_1 \cdot \boldsymbol{m}'\right)\boldsymbol{d}_1 + \boldsymbol{d}_1 \times \boldsymbol{v}'} \tag{71}$$

- <u>Time evolution</u>:

$$\dot{\boldsymbol{\omega}} = \boldsymbol{\Omega}' + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{72}$$

Dissipation

The dissipation rate is the negative of instantaneous power of viscous/external forces.

$$\boxed{P_D = -\left[\boldsymbol{f}_{\text{ext}}(0) \cdot \boldsymbol{v}(0) + \boldsymbol{f}_{\text{ext}}(L) \cdot \boldsymbol{v}(L) + \boldsymbol{m}_{\text{ext}}(0) \cdot \boldsymbol{\Omega}(0) + \boldsymbol{m}_{\text{ext}}(L) \cdot \boldsymbol{\Omega}(L)\right] - \int \boldsymbol{F}_s \cdot \boldsymbol{v} + \boldsymbol{M}} \tag{73}$$

$$
\begin{aligned}
\dot{E} &= \frac{\mathrm{d}}{\mathrm{d}t}\int \frac{1}{2}\sum_i B_i(\omega_i - \sigma_i)^2 \,\mathrm{d}s + \frac{\mathrm{d}}{\mathrm{d}t}\int\int \frac{1}{2}c^2 G(\boldsymbol{r}(s) - \boldsymbol{r}\xi)\,\mathrm{d}s\,\mathrm{d}\xi \\
&= \int \sum_i m_i\dot{\omega}_i - \boldsymbol{F}_s \cdot \boldsymbol{v}\,\mathrm{d}s \\
&= \int \boldsymbol{m}\cdot\boldsymbol{\Omega}' - \boldsymbol{F}_s\cdot\boldsymbol{v}\,\mathrm{d}s \\
&= \boldsymbol{m}\cdot\boldsymbol{\Omega}|_0^L + \int -\boldsymbol{m}'\cdot\boldsymbol{\Omega} - \boldsymbol{F}_s\cdot\boldsymbol{v}\,\mathrm{d}s \\
&= \boldsymbol{m}\cdot\boldsymbol{\Omega}|_0^L + \int (\boldsymbol{d}_1\times\boldsymbol{f} + \boldsymbol{M}_v)\cdot\boldsymbol{\Omega} - \boldsymbol{F}_s\cdot\boldsymbol{v}\,\mathrm{d}s \\
&= \boldsymbol{m}\cdot\boldsymbol{\Omega}|_0^L + \int \boldsymbol{f}\cdot(\boldsymbol{\Omega}\times\boldsymbol{d}_1) + \boldsymbol{M}_v\cdot\boldsymbol{\Omega} - \boldsymbol{F}_s\cdot\boldsymbol{v}\,\mathrm{d}s \\
&= \boldsymbol{m}\cdot\boldsymbol{\Omega}|_0^L + \int \boldsymbol{f}\cdot\dot{\boldsymbol{d}}_1 + \boldsymbol{M}_v\cdot\boldsymbol{\Omega} - \boldsymbol{F}_s\cdot\boldsymbol{v}\,\mathrm{d}s \\
&= \boldsymbol{m}\cdot\boldsymbol{\Omega}|_0^L + \int \boldsymbol{f}\cdot\boldsymbol{v}' + \boldsymbol{M}_v\cdot\boldsymbol{\Omega} - \boldsymbol{F}_v\cdot\boldsymbol{v}\,\mathrm{d}s \\
&= \boldsymbol{f}\cdot\boldsymbol{v} + \boldsymbol{m}\cdot\boldsymbol{\Omega}|_0^L + \int -\boldsymbol{f}'\cdot\boldsymbol{v} + \boldsymbol{M}_v\cdot\boldsymbol{\Omega} - \boldsymbol{F}_s\cdot\boldsymbol{v}\,\mathrm{d}s \\
&= \boldsymbol{f}\cdot\boldsymbol{v} + \boldsymbol{m}\cdot\boldsymbol{\Omega}|_0^L + \int \boldsymbol{F}_v\cdot\boldsymbol{v} + \boldsymbol{M}_v\cdot\boldsymbol{\Omega}\,\mathrm{d}s \\
&= \boldsymbol{f}_{\mathrm{ext}}(0)\cdot\boldsymbol{v}(0) + \boldsymbol{f}_{\mathrm{ext}}(L)\cdot\boldsymbol{v}(L) + \boldsymbol{m}_{\mathrm{ext}}(0)\cdot\boldsymbol{\Omega}(0) + \boldsymbol{m}_{\mathrm{ext}}(L)\cdot\boldsymbol{\Omega}(L) + \int \boldsymbol{F}_s\cdot\boldsymbol{v} + \boldsymbol{M}_v\cdot\boldsymbol{\Omega}\,\mathrm{d}s
\end{aligned}
$$

This holds even for surface bound filaments since the associated constraint forces/moments don't do any work

$$
P_C = -\lambda\boldsymbol{n}\cdot\boldsymbol{v} + ... = 0 \tag{74}
$$

Derivation of some of the stuff above

- <u>Internal force $\boldsymbol{f}$</u>:

$$
\boldsymbol{f} = f_1\boldsymbol{d}_1 + \boldsymbol{d}_1\times\boldsymbol{m}' = f_1\boldsymbol{d}_1 + \boldsymbol{f}_\perp \tag{75}
$$

Take an s-derivative of $\boldsymbol{d}_1\cdot$(force balance) to get a DE for f_1.

$$f_1' + \boldsymbol{d}_1 \cdot (\boldsymbol{\omega} \times \boldsymbol{f}) + \boldsymbol{d}_1 \cdot \boldsymbol{F}_s - \alpha_\parallel \boldsymbol{d}_1 \cdot \boldsymbol{v} = 0 \tag{76}$$

$$\Rightarrow f_1'' + [\boldsymbol{d}_1 \cdot (\boldsymbol{\omega} \times \boldsymbol{f}) + \boldsymbol{d}_1 \cdot \boldsymbol{F}_s]' - \alpha_\parallel (\boldsymbol{\omega} \times \boldsymbol{d}_1) \cdot \boldsymbol{v} - \alpha_\parallel \boldsymbol{d}_1 \cdot \boldsymbol{v}' = 0 \tag{77}$$

Using $0 = (\boldsymbol{d}_1 \cdot \boldsymbol{d}_1)' = 2\boldsymbol{d}_1 \cdot \boldsymbol{v}'$ and

$$-\alpha_\parallel (\boldsymbol{\omega} \times \boldsymbol{d}_1) \cdot \boldsymbol{v} = -\alpha_\parallel (\boldsymbol{\omega} \times \boldsymbol{d}_1) \cdot \boldsymbol{v}_\perp \tag{78}$$

$$= -\frac{\alpha_\parallel}{\alpha_\perp} (\boldsymbol{\omega} \times \boldsymbol{d}_1) \cdot (\boldsymbol{f}' + \boldsymbol{F}_s) \tag{79}$$

this can be written as

$$f_1'' + [\boldsymbol{d}_1 \cdot (\boldsymbol{\omega} \times \boldsymbol{f}) + \boldsymbol{d}_1 \cdot \boldsymbol{F}_s]' - \frac{\alpha_\parallel}{\alpha_\perp} (\boldsymbol{\omega} \times \boldsymbol{d}_1) \cdot (\boldsymbol{f}' + \boldsymbol{F}_s) = 0 \tag{80}$$

$$\Rightarrow f_1'' + [-\boldsymbol{\omega} \cdot (\boldsymbol{d}_1 \times \boldsymbol{f}) + \boldsymbol{d}_1 \cdot \boldsymbol{F}_s]' - \frac{\alpha_\parallel}{\alpha_\perp} (\boldsymbol{\omega} \times \boldsymbol{d}_1) \cdot (f_1 \boldsymbol{d}_1' + \boldsymbol{f}_\perp' + \boldsymbol{F}_s) = 0 \tag{81}$$

$$\Rightarrow f_1'' + [(\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' + \boldsymbol{r}' \cdot \boldsymbol{F}_s]' + \frac{\alpha_\parallel}{\alpha_\perp} [(\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' - \boldsymbol{r}'' \cdot \boldsymbol{F}_s] - \frac{\alpha_\parallel}{\alpha_\perp} |\boldsymbol{r}''|^2 f_1 = 0 \tag{82}$$

$$\boxed{f_1'' - \frac{\alpha_\parallel}{\alpha_\perp} |\boldsymbol{r}''|^2 f_1 = \chi(s)} \tag{83}$$

$$\boxed{f_1' + (\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' + \boldsymbol{r}' \cdot \boldsymbol{F}_s|_{s=0} = 0} \tag{84}$$

$$\boxed{f_1|_{s=L} = 0} \tag{85}$$

where $\chi(s)$ is a known function which depends ononly the filament configuration,

$$\chi(s) = -[(\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' + \boldsymbol{r}' \cdot \boldsymbol{F}_s]' - \frac{\alpha_\parallel}{\alpha_\perp} [(\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{m}' - \boldsymbol{r}'' \cdot \boldsymbol{F}_s], \tag{86}$$

and boundary conditions are from $\boldsymbol{v}|_{s=0} = 0,\ \boldsymbol{f}|_{s=L} = 0$.

- <u>Angular velocity $\boldsymbol{\Omega}$</u>:

$$\begin{aligned} \boldsymbol{v}' = \dot{\boldsymbol{r}}' = \dot{\boldsymbol{d}}_1 = \boldsymbol{\Omega} \times \boldsymbol{d}_1 \\ \Rightarrow \boldsymbol{\Omega}_\perp = \boldsymbol{d}_1 \times \boldsymbol{v}' \end{aligned}$$

$$\boldsymbol{f} = f_1 \boldsymbol{d}_1 + \boldsymbol{d}_1 \times \boldsymbol{m}' = f_1 \boldsymbol{r}' + \boldsymbol{r}' \times \boldsymbol{m}' \tag{87}$$

$$\boldsymbol{f}' = f_1' \boldsymbol{d}_1 + f_1 \boldsymbol{d}_1' + (\boldsymbol{d}_1 \times \boldsymbol{m}')' \tag{88}$$

$$\boldsymbol{f}'' = f_1'' \boldsymbol{d}_1 + 2 f_1' \boldsymbol{d}_1' + f_1 \boldsymbol{d}_1'' + (\boldsymbol{d}_1 \times \boldsymbol{m}')'' \tag{89}$$

4.1 2D example

Suppose a filament with spontaneous curvature σ lies in the plane $z = 0$ with its' $s = 0$ end clamped.

- Darboux frame: $\boldsymbol{d}_1, \boldsymbol{d}_2, \boldsymbol{d}_3 = \boldsymbol{r}', \boldsymbol{e}_z \times \boldsymbol{r}', \boldsymbol{e}_z$
- Darboux vector: $\boldsymbol{\omega}_3 = \omega \boldsymbol{e}_z$
- Angular velocity: $\boldsymbol{\Omega} = \Omega \boldsymbol{e}_z$
- Velocity: $\boldsymbol{v} = v_1 \boldsymbol{d}_1 + v_2 \boldsymbol{d}_2$
- Internal moment: $\boldsymbol{m} = B(\omega - \sigma)\boldsymbol{e}_z$
- Internal force: $\boldsymbol{f} = \Lambda \boldsymbol{d}_1 - B\omega' \boldsymbol{d}_2$ ($f_2 = -B\omega'$ from moment balance eq.)
- Steric force: $\boldsymbol{F}_s = F_1 \boldsymbol{d}_1 + F_2 \boldsymbol{d}_2$
- Drag force: $\boldsymbol{F}_d = -\alpha_\parallel v_1 \boldsymbol{d}_1 - \alpha_\perp v_2 \boldsymbol{d}_2$
- Force Balance: $\boldsymbol{f}' + \boldsymbol{F}_s + \boldsymbol{F}_d = 0$
- $\boldsymbol{\omega}/\boldsymbol{\Omega}$ consistency: $\dot{\omega} = \Omega'$

Tension

$$\boldsymbol{f}' = (\Lambda' + B\omega\omega')\boldsymbol{d}_1 + (\Lambda\omega - B\omega'')\boldsymbol{d}_2 \tag{90}$$

Taking Darboux components of the force balance eq. gives

$$\Lambda' + B\omega\omega' + F_1 = \alpha_\parallel v_1 \tag{91}$$
$$\omega\Lambda - B\omega'' + F_2 = \alpha_\perp v_2 \tag{92}$$

From inextensibility we know $\boldsymbol{d}_1 \cdot \boldsymbol{v}' = 0$, so that $v_1' = \omega v_2$

$$v_1' = \frac{1}{\alpha_\parallel}\left(\Lambda' + B\omega\omega' + F_1\right)' = \frac{1}{\alpha_\perp}\omega\left(\omega\Lambda - B\omega'' + F_2\right) = \omega v_2 \tag{93}$$

$$\Lambda'' - \frac{\alpha_\parallel}{\alpha_\perp}\omega^2\Lambda = \frac{\alpha_\parallel}{\alpha_\perp}\omega\left(F_2 - B\omega''\right) - \left(B\omega\omega' + F_1\right)' \tag{94}$$

$$\boxed{\Lambda'' - \frac{\alpha_\parallel}{\alpha_\perp}\omega^2\Lambda = -F_1' + \frac{\alpha_\parallel}{\alpha_\perp}\omega F_2 - \left(1 + \frac{\alpha_\parallel}{\alpha_\perp}\right)B\omega\omega'' - B\omega'^2} \tag{95}$$

Or choose units with $L = 1, B = 1, \alpha_\perp = 1$ to get

$$\boxed{\Lambda'' - \alpha\omega^2\Lambda = -F_1' + \alpha\omega F_2 - (1+\alpha)\,\omega\omega'' - \omega'^2} \tag{96}$$

I dropped the subscript on $\alpha_\parallel$ because.

Time evolution

The $\boldsymbol{d}_2$ component of $\boldsymbol{v}' = \dot{\boldsymbol{d}}_1 = \boldsymbol{\Omega} \times \boldsymbol{d}_1$ says

$$\Omega = v_2' + \omega v_1 \tag{97}$$

which can be combined with the ω/Ω consistency condition to give an equation for the time derivative of ω.

$$\dot{\omega} = \left(v_2' + \omega v_1\right)' \tag{98}$$

Using expressions for the velocity obtained above,

$$v_1 = \frac{1}{\alpha_\parallel}\left(F_1 + \Lambda' + B\omega\omega'\right) \tag{99}$$

$$v_2 = \frac{1}{\alpha_\perp}\left(F_2 + \omega\Lambda - B\omega''\right) \tag{100}$$

we can write this as

$$\boxed{\dot{\omega} = -\frac{B}{\alpha_\perp}\omega'''' + \frac{1}{\alpha_\perp}\left(F_2'' + \omega''\Lambda + \omega\Lambda'' + 2\omega'\Lambda'\right) + \frac{1}{\alpha_\parallel}\left(\omega' F_1 + \omega'\Lambda' + 2B\omega\omega'^2 + \omega F_1' + \omega\Lambda'' + B\omega^2\omega''\right.} \tag{101}$$

or after nondimensionalizing,

$$\boxed{\dot{\omega} = -\omega'''' + F_2'' + \omega''\Lambda + \omega\Lambda'' + 2\omega'\Lambda' + \frac{1}{\alpha_\parallel}\left(\omega' F_1 + \omega'\Lambda' + 2\omega\omega'^2 + \omega F_1' + \omega\Lambda'' + \omega^2\omega''\right)} \tag{102}$$

If we let $u = F_2 + \omega\Lambda$ then

$$v_2 = -\omega'' + u \tag{103}$$

$$Q = (u' + \omega v_1)' \tag{104}$$

$$\dot{\omega} = -\omega'''' + Q \tag{105}$$

Boundary conditions

The clamped end must be stationary and the free end must be force/torque free.

$$v_1, v_2, \Omega\big|_{s=0} = 0, \qquad \boldsymbol{f}, \boldsymbol{m}\big|_{s=L} = 0 \tag{106}$$

The free end conditions are equivalent to

$$\Lambda, B(\omega - \sigma), \omega'\big|_{s=L} = 0 \tag{107}$$

For an initial shape to satisfy BCs you need

$$-B\omega''' + \frac{\omega'}{\omega}(B\omega'' - F_{s2}) + \omega(-B\omega\omega' - F_{s1}) + F_{s2}'\Big|_{s=0} = 0 \tag{108}$$

$$\omega(L), \omega'(L) = 0 \tag{109}$$

Numerics

$$s_0 = 0, s_1 = \Delta s, \ldots, s_n = n\Delta s, \ldots, s_N = N\Delta s = L \tag{110}$$
$$t_0 = 0, t_1 = \Delta t, \ldots, t_m = m\Delta t, \ldots, t_M = M\Delta t = T \tag{111}$$

$$\Lambda^{[mn]} = \Lambda(t_m, s_n), \qquad \omega^{[mn]} = \omega(t_m, s_n), \qquad \text{etc}\ldots \tag{112}$$

Tension

$$\frac{\Lambda^{[m,n-1]} - 2\Lambda^{[m,n]} + \Lambda^{[m,n+1]}}{\Delta s^2} - \alpha\left(\omega^{[mn]}\right)^2 \Lambda^{[m,n]} = -F_1'^{[mn]} + \alpha\omega^{[mn]}F_2^{[mn]} - (1+\alpha)\omega^{[mn]}\omega''^{[mn]} - \left(\omega'^{[mn]}\right)^2 + O(\Delta s^2) \tag{113}$$

$$\Lambda^{[m,n-1]} - \left(2 + \Delta s^2\alpha\left(\omega^{[mn]}\right)^2\right)\Lambda^{[m,n]} + \Lambda^{[m,n+1]} = b^{[mn]} \tag{114}$$

where $b^{[mn]} = \Delta s^2\left(-F_1'^{[mn]} + \alpha\omega^{[mn]}F_2^{[mn]} - (1+\alpha)\omega^{[mn]}\omega''^{[mn]} - \left(\omega'^{[mn]}\right)^2\right)$

From boundary conditions we get

$$\Lambda^{[mN]} = 0 \tag{115}$$

$$-\Lambda^{[m,-1]} + \Lambda^{[m,1]} = -2\Delta s\left(\omega^{[m0]}\omega'^{[m0]} + F_1^{[m0]}\right) \tag{116}$$

$$\boldsymbol{A}^{[m]}\boldsymbol{\Lambda}^{[m]} = \boldsymbol{b}^{[m]} \tag{117}$$

$$\boldsymbol{A}^{[m]} = \begin{bmatrix} -1 & 0 & 1 & 0 & \ldots \\ 1 & -\left(2 + \Delta s^2\alpha\left(\omega^{[m0]}\right)^2\right) & 1 & 0 & \ddots \\ 0 & 1 & -\left(2 + \Delta s^2\alpha\left(\omega^{[m1]}\right)^2\right) & 1 & \ddots \\ \vdots & \ddots & \ddots & \ddots & \ddots \\ 0 & \ddots & \ddots & 1 & -\left(2 + \Delta s^2\alpha\left(\omega^{[m,N-1]}\right)^2\right) \\ 0 & \ldots & 0 & 0 & 0 \end{bmatrix} \tag{118}$$

$\boldsymbol{A}^{[m]}$ is $(N+2)$-by-$(N+2)$

- +2 diagonal: $[1, 0, \ldots, 0]$
- +1 diagonal: $[0, 1, \ldots, 1]$
- Main diagonal: $\left[-1, -\left(2 + \Delta s^2 \alpha \left(\omega^{[m0]}\right)^2\right), \ldots, -\left(2 + \Delta s^2 \alpha \left(\omega^{[m,N-1]}\right)^2\right), 1\right]$
- -1 diagonal: $[1, \ldots, 1, 0]$

$$\boldsymbol{\Lambda}^{[m]} = \begin{bmatrix} \Lambda^{[m,-1]} \\ \Lambda^{[m0]} \\ \Lambda^{[m,1]} \\ \vdots \\ \Lambda^{[mN]} \end{bmatrix} \tag{119}$$

$$\boldsymbol{b}^{[m]} = \begin{bmatrix} -2\Delta s \left(\omega^{[m0]}\omega'^{[m0]} + F_1^{[m0]}\right) \\ \Delta s^2 \left(-F_1'^{[m0]} + \alpha\omega^{[m0]}F_2^{[m0]} - (1+\alpha)\omega^{[m0]}\omega''^{[m0]} - \left(\omega'^{[m0]}\right)^2\right) \\ \vdots \\ \Delta s^2 \left(-F_1'^{[m,N-1]} + \alpha\omega^{[m,N-1]}F_2^{[m,N-1]} - (1+\alpha)\omega^{[m,N-1]}\omega''^{[m,N-1]} - \left(\omega'^{[m,N-1]}\right)^2\right) \\ 0 \end{bmatrix} \tag{120}$$

Evolution

Let $Q = F_2'' + \omega''\Lambda + \omega\Lambda'' + 2\omega'\Lambda' + \frac{1}{\alpha_\parallel}\left(\omega' F_1 + \omega'\Lambda' + 2\omega\omega'^2 + \omega F_1' + \omega\Lambda'' + \omega^2\omega''\right)$

$$\frac{\omega^{[m-1]} - 4\omega^{[m]} + 3\omega^{[m+1]}}{2\Delta t} = -\omega''''^{[m+1]} + Q^{[m+1]} + O(\Delta t^2) \tag{121}$$

$$= -\omega''''^{[m+1]} + 2Q^{[m]} - Q^{[m-1]} + O(\Delta t^2) \tag{122}$$

$$3\omega^{[m+1]} + 2\Delta t\omega''''^{[m+1]} = 4\omega^{[m]} - \omega^{[m-1]} + 4\Delta t Q^{[m]} - 2\Delta t Q^{[m-1]} \tag{123}$$

$$\omega''''^{[m+1,n]} = \frac{\omega^{[m+1,n-2]} - 4\omega^{[m+1,n-1]} + 6\omega^{[m+1,n]} - 4\omega^{[m+1,n+1]} + \omega^{[m+1,n+2]}}{\Delta s^4} + O(\Delta s^2) \tag{124}$$

$$\begin{aligned} 3\Delta s^4 \omega^{[m+1,n]} + 2\Delta t \left(\omega^{[m+1,n-2]} - 4\omega^{[m+1,n-1]} + 6\omega^{[m+1,n]} - 4\omega^{[m+1,n+1]} + \omega^{[m+1,n+2]}\right) \\ = 4\Delta s^4 \omega^{[m,n]} - \Delta s^4 \omega^{[m-1,n]} + 4\Delta s^4 \Delta t Q^{[m,n]} - 2\Delta s^4 \Delta t Q^{[m-1,n]} \end{aligned}$$

$$\boxed{\begin{aligned} 2\Delta t \omega^{[m+1,n-2]} - 8\Delta t \omega^{[m+1,n-1]} + \left(12\Delta t + 3\Delta s^4\right) \omega^{[m+1,n]} - 8\Delta t \omega^{[m+1,n+1]} + 2\Delta t \omega^{[m+1,n+2]} \\ = 4\Delta s^4 \omega^{[m,n]} - \Delta s^4 \omega^{[m-1,n]} + 4\Delta s^4 \Delta t Q^{[m,n]} - 2\Delta s^4 \Delta t Q^{[m-1,n]} \end{aligned}}$$

From $v_2|_{s=0} = 0$:

$$\omega^{[m+1,-1]} - \left(2 + \Delta s^2 \Lambda^{[m+1,0]}\right) \omega^{[m+1,0]} + \omega^{[m+1,1]} = \Delta s^2 F_2^{[m+1,0]} \tag{125}$$

- Need to introduce ghost point $\omega^{[m+1,-1]}$.

Zero angular velocity $\Omega|_{s=0} = v_2'|_{s=0} = 0$:

$$\omega^{[m+1,-2]} - \left(2 + \Delta s^2 \Lambda^{[m+1,0]}\right) \omega^{[m+1,-1]} + \Delta s^2 \left(-\Lambda^{[m+1,-1]} + \Lambda^{[m+1,1]}\right) \omega^{[m+1,0]} + \left(2 + \Delta s^2 \Lambda^{[m+1,0]}\right) \omega^{[m+1,1]} \tag{126}$$

$$= -2\Delta \tag{127}$$

- Need to introduce ghost point $\omega^{[m+1,-2]}$.

What to do with $F_2'^{[m+1,0]}$? Compute using ghost points!

$$-\omega(0)\boldsymbol{d}_1(0) = \boldsymbol{d}_2'(0) = \frac{-\boldsymbol{d}_2(-\Delta s) + \boldsymbol{d}_2(\Delta s)}{2\Delta s} + O(\Delta s^2) \tag{128}$$

$$\boldsymbol{d}_1(0) = \boldsymbol{r}'(0) = \frac{-\boldsymbol{r}(-\Delta s) + \boldsymbol{r}(\Delta s)}{2\Delta s} + O(\Delta s^2) \tag{129}$$

$$\boxed{\boldsymbol{d}_2^{[m+1,-1]} = \boldsymbol{d}_2^{[m+1,1]} + 2\Delta s \omega^{[m+1,0]} \boldsymbol{d}_1^{[m+1,0]}} \tag{130}$$

$$\boxed{\boldsymbol{r}^{[m+1,-1]} = \boldsymbol{r}^{[m+1,1]} - 2\Delta s \boldsymbol{d}_1^{[m+1,0]}} \tag{131}$$

$$\boxed{F_2^{[m+1,-1]} = \boldsymbol{d}_2^{[m+1,-1]} \cdot \int_0^L \boldsymbol{\nabla} G\left(\boldsymbol{r}^{[m+1,-1]} - \boldsymbol{r}(\xi)\right) \mathrm{d}\xi} \tag{132}$$

$$\boxed{F_2'^{[m+1,0]} = \frac{-F_2^{[m+1,-1]} + F_2^{[m+1,1]}}{2\Delta s} + O(\Delta s^2)} \tag{133}$$

Moment free end $\boldsymbol{m}(L) = 0$:

$$\omega^{[m+1,N]} = \sigma \tag{134}$$

$\boldsymbol{d}_2$ component of force free end $f_2(L) = -B\omega'(L) = 0$:

$$\frac{-\omega^{[m+1,N-1]} + \omega^{[m+1,N+1]}}{2\Delta s} = 0 \tag{135}$$

$$\boldsymbol{C}^{[m]} \boldsymbol{\omega}^{[m]} = \boldsymbol{y}^{[m]} \tag{136}$$

$$\omega''''^{[m+1,n]} = \frac{\omega^{[m+1,n-2]} - 4\omega^{[m+1,n-1]} + 6\omega^{[m+1,n]} - 4\omega^{[m+1,n+1]} + \omega^{[m+1,n+2]}}{\Delta s^4} + O(\Delta s^2) \tag{137}$$

$$\omega^{[m]} = \begin{bmatrix} \omega^{[m,-2]} \\ \vdots \\ \omega^{[m,N+1]} \end{bmatrix} \tag{138}$$

$$y^{[m+1]} = \begin{bmatrix} \Delta s^2 F_2^{[m+1,0]} \\ -2\Delta s^3 F_2'^{[m+1,0]} \\ 4\Delta s^4 \omega^{[m,0]} - \Delta s^4 \omega^{[m-1,0]} + 4\Delta s^4 \Delta t Q^{[m,0]} - 2\Delta s^4 \Delta t Q^{[m-1,0]} \\ \vdots \\ 4\Delta s^4 \omega^{[m,N-1]} - \Delta s^4 \omega^{[m-1,N-1]} + 4\Delta s^4 \Delta t Q^{[m,N-1]} - 2\Delta s^4 \Delta t Q^{[m-1,N-1]} \\ \sigma \\ 0 \end{bmatrix} \tag{139}$$

$$\boldsymbol{C}^{[m]} = \tag{140}$$

$$\begin{bmatrix} 0 & 1 & -\left(2+\Delta s^2 \Lambda^{[m+1,0]}\right) & 1 & 0 & \dots & 0 \\ 1 & -\left(2+\Delta s^2 \Lambda^{[m+1,0]}\right) & \Delta s^2\left(-\Lambda^{[m+1,-1]} + \Lambda^{[m+1,1]}\right) & \left(2+\Delta s^2 \Lambda^{[m+1,0]}\right) & -1 & \ddots & \\ 2\Delta t & -8\Delta t & (12\Delta t + 3\Delta s^4) & -8\Delta t & 2\Delta t & 0 & \\ 0 & \ddots & \ddots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 2\Delta t & -8\Delta t & (12\Delta t + 3\Delta s^4) & -8\Delta t & 2\Delta \\ \vdots & \ddots & \ddots & 0 & 0 & 1 & 0 \\ 0 & \dots & \dots & 0 & -1 & 0 & 1 \end{bmatrix} \tag{141}$$

$\boldsymbol{C}^{[m]}$ is $(N+4)$-by-$(N+4)$

- +3 diagonal: $[1, -1, 0, \dots, 0]$
- +2 diagonal: $\left[-\left(2+\Delta s^2 \Lambda^{[m+1,0]}\right), \left(2+\Delta s^2 \Lambda^{[m+1,0]}\right), 2\Delta t, \dots, 2\Delta t\right]$
- +1 diagonal: $\left[1, \Delta s^2\left(-\Lambda^{[m+1,-1]} + \Lambda^{[m+1,1]}\right), -8\Delta t, \dots, -8\Delta t, 0\right]$
- Main diagonal: $\left[0, -\left(2+\Delta s^2 \Lambda^{[m+1,0]}\right), 12\Delta t + 3\Delta s^4, \dots, 12\Delta t + 3\Delta s^4, 1, 1\right]$
- -1 diagonal: $[1, -8\Delta t, \dots, -8\Delta t, 0, 0]$
- -2 diagonal: $[2\Delta t, \dots, 2\Delta t, 0, -1]$

Velocity DEs

The second order equation for Λ can be eliminated in favor of first order equations for v_1, v_2. This has the advantage of simplifying the clamped boundary conditions and the pde for ω. The force balance equation gives

$$\Lambda = \frac{1}{\omega}\left(\alpha_\perp v_2 + B\omega'' - F_{s2}\right) \tag{142}$$

$$\Lambda' = \alpha_\parallel v_1 - \left(\frac{1}{2}B\omega^2 + H\right)' \tag{143}$$

Where we used $B\omega'' + f_2 = 0$ to eliminate f_2 and $H(s) = \int_0^L -c^2 G(\boldsymbol{r}(s) - \boldsymbol{r}(\xi))\,\mathrm{d}\xi$ so that $H'(s) = F_{s1}$. Taking a derivative of the expression for Λ and equating it to the equation for Λ' gives an equation for v_2.

$$v_2' = \frac{\alpha_\parallel}{\alpha_\perp}\omega v_1 + \frac{\omega'}{\omega}v_2 - \frac{1}{\alpha_\perp}\omega\left[\frac{1}{\omega}\left(B\omega'' - F_{s2}\right) + \frac{1}{2}B\omega^2 + H\right]' \tag{144}$$

$$v_2' = \frac{\alpha_\parallel}{\alpha_\perp}\omega v_1 + \frac{\omega'}{\omega}v_2 + Q \tag{145}$$

$$v_1'' = \frac{\alpha_\parallel}{\alpha_\perp}\omega^2 v_1 + 2\frac{\omega'}{\omega}v_1' + \omega Q \tag{146}$$

$$\boxed{\begin{aligned} v_1' &= \omega v_2 \\ v_2' &= \frac{\alpha_\parallel}{\alpha_\perp}\omega v_1 + \frac{\omega'}{\omega}v_2 + Q \end{aligned}} \tag{147}$$

$$Q = -\frac{1}{\alpha_\perp}\omega\left[\frac{1}{\omega}\left(B\omega'' - F_{s2}\right) + \frac{1}{2}B\omega^2 + H\right]' \tag{148}$$

BC's give $v_1 = v_2 = Q = 0$ at $s = 0$.

5 Surface bound filaments

Let the filament be constrained to move along a surface S with $\boldsymbol{d}_3$ equal to the surface unit normal $\boldsymbol{n}$ (this is what I mean when I talk about 'surface confinement' below).

5.1 Ambiguity of constraints

There are about a billion different equivalent sets of constraints, and I'm not convinced that they all lead to identical variational shape equations.

5.1.1 Original constraints

Suppose the surface can be described implicitly by

$$g(\boldsymbol{x}) = 0. \tag{149}$$

Fact 5. *When taken together with inextensibility/unshearbility,*

$$\boldsymbol{r}' - \boldsymbol{d}_1 = 0, \tag{150}$$

the conditions

$$g(\boldsymbol{r}) = 0 \tag{151}$$

$$\boldsymbol{d}_2 \cdot \boldsymbol{n}(\boldsymbol{r}) = 0. \tag{152}$$

are equivalent to surface confinement.

Proof. The centerline is confined to S by (151) so we just need to show that $\boldsymbol{d}_3$ is normal to S. Taking a derivative of (151) and using (150) to replace $\boldsymbol{r}'$ with $\boldsymbol{d}_1$ shows that $\boldsymbol{d}_1$ is tangent to S

$$\boldsymbol{d}_1 \cdot \boldsymbol{\nabla} g(\boldsymbol{r}) = 0. \tag{153}$$

Since (152) means $\boldsymbol{d}_2$ is also tangent to S and we know that the Darboux frame is orthonormal, it must be the case that $\boldsymbol{d}_3$ is normal to S. Technically these constraints allow for either $\boldsymbol{d}_3 = \boldsymbol{n}$ or $\boldsymbol{d}_3 = -\boldsymbol{n}$, but this ambiguity is taken care of by initial conditions and continuity. □

We introduce two additional Lagrange multipliers

$$\lambda, \mu \tag{154}$$

to enforce the surface constraints. The constrained energy is

$$E = \int_0^L \frac{1}{2} \sum_i B_i(\omega_i - \sigma_i)^2 + \boldsymbol{f} \cdot (\boldsymbol{r}' - \boldsymbol{d}_1) + \lambda \frac{g(\boldsymbol{r})}{\|\boldsymbol{\nabla} g\|} + \mu \boldsymbol{d}_2 \cdot \boldsymbol{n}(\boldsymbol{r}) \, \mathrm{d}s + \phi[\boldsymbol{r}] \tag{155}$$

Where $\phi[\boldsymbol{r}]$ is the energy due to some potential which only depends on centerline position. Varying the Darboux frame results in

$$\delta_\psi E = \int_0^L \sum_i B_i(\omega_i - \sigma_i)\delta\omega_i - \boldsymbol{f} \cdot \delta \boldsymbol{d}_1 + \mu \delta \boldsymbol{d}_2 \cdot \boldsymbol{n}(\boldsymbol{r}) \, \mathrm{d}s \tag{156}$$

Independent components of the frame variation can now be expressed in terms of the infinitesimal rotation $\boldsymbol{\psi} = \sum_i \psi_i \boldsymbol{d}_i$,

$$\delta \boldsymbol{d}_i = \boldsymbol{\psi} \times \boldsymbol{d}_i \tag{157}$$

$$\delta \omega_i = \boldsymbol{d}_i \cdot \boldsymbol{\psi}' \tag{158}$$

$$\delta_\psi E = \int_0^L \sum_i B_i(\omega_i - \sigma_i) \boldsymbol{d}_i \cdot \boldsymbol{\psi}' - \boldsymbol{f} \cdot (\boldsymbol{\psi} \times \boldsymbol{d}_1) + \mu \, (\boldsymbol{\psi} \times \boldsymbol{d}_2) \cdot \boldsymbol{n}(\boldsymbol{r}) \, \mathrm{d}s \tag{159}$$

$$= \int_0^L \boldsymbol{m} \cdot \boldsymbol{\psi}' - (\boldsymbol{d}_1 \times \boldsymbol{f}) \cdot \boldsymbol{\psi} + (\mu \boldsymbol{d}_2 \times \boldsymbol{n}(\boldsymbol{r})) \cdot \boldsymbol{\psi} \, \mathrm{d}s \tag{160}$$

$$= \boldsymbol{m} \cdot \boldsymbol{\psi}|_0^L - \int_0^L (\boldsymbol{m} + \boldsymbol{d}_1 \times \boldsymbol{f} - \mu \boldsymbol{d}_2 \times \boldsymbol{n}(\boldsymbol{r})) \cdot \boldsymbol{\psi} \, \mathrm{d}s \tag{161}$$

Enforcing constraints simplifies the last term $\mu \boldsymbol{d}_2 \times \boldsymbol{n}(\boldsymbol{r}) = \mu \boldsymbol{d}_1$,

$$\boxed{\delta_\psi E = \boldsymbol{m} \cdot \boldsymbol{\psi}|_0^L - \int_0^L (\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f} - \mu \boldsymbol{d}_1) \cdot \boldsymbol{\psi} \, \mathrm{d}s} \tag{162}$$

Varying the centerline results in

$$\delta_r E = \int_0^L \boldsymbol{f} \cdot \delta \boldsymbol{r}' + \lambda \left(\boldsymbol{n} - \frac{g \boldsymbol{n} \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|^2} \right) \cdot \delta \boldsymbol{r} + \mu \delta \boldsymbol{r} \cdot \left(\frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|} - \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \boldsymbol{n}\boldsymbol{n}}{\|\boldsymbol{\nabla} g\|} \right) \cdot \boldsymbol{d}_2 + \frac{\delta \phi}{\delta \boldsymbol{r}} \cdot \delta \boldsymbol{r} \, \mathrm{d}s \tag{163}$$

$$= \boldsymbol{f} \cdot \delta \boldsymbol{r}|_0^L - \int_0^L \left[\boldsymbol{f}' - \lambda \boldsymbol{n} \cdot \left(\boldsymbol{\delta} - \frac{g \boldsymbol{\nabla}\boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|^2} \right) - \mu \boldsymbol{d}_2 \cdot (\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) \cdot \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|} - \frac{\delta \phi}{\delta \boldsymbol{r}} \right] \cdot \delta \boldsymbol{r} \, \mathrm{d}s \, . \tag{164}$$

Enforcing constraints $g(\boldsymbol{r}) = 0$ and $\boldsymbol{d}_2 \cdot \boldsymbol{n} = 0$ eliminates a couple terms,

$$\boxed{\delta_r E = \boldsymbol{f} \cdot \delta \boldsymbol{r}|_0^L - \int_0^L \left[\boldsymbol{f}' - \lambda \boldsymbol{n} - \mu \boldsymbol{d}_2 \cdot \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|} - \frac{\delta \phi}{\delta \boldsymbol{r}} \right] \cdot \delta \boldsymbol{r} \, \mathrm{d}s} \tag{165}$$

14 Unknowns:

- 3 independent degrees of freedom the Darboux frame

$$\boldsymbol{E}_i \cdot \boldsymbol{d}_j, \text{ for } i, j = 1, 2, 3$$

- 3 Darboux components of the Darboux vector

$$\boldsymbol{d}_i \cdot \boldsymbol{\omega}, \text{ for } i = 1, 2, 3$$

- 3 Darboux components of the tension

$$\boldsymbol{d}_i \cdot \boldsymbol{f}, \text{ for } i = 1, 2, 3$$

- 3 ambient components of the centerline position

$$\boldsymbol{E}_i \cdot \boldsymbol{r}, \text{ for } i = 1, 2, 3$$

- 2 Lagrange multipliers

$$\lambda, \mu$$

12 first order ODEs and 2 algebraic equations:

$$\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f} - \mu \boldsymbol{d}_1 = 0 \tag{166}$$

$$\boldsymbol{f}' - \lambda \boldsymbol{n} - \mu \boldsymbol{\nabla} \boldsymbol{n} \cdot \boldsymbol{d}_2 - \frac{\delta \phi}{\delta \boldsymbol{r}} = 0 \tag{167}$$

$$\boldsymbol{r}' = \boldsymbol{d}_1 \tag{168}$$

$$\boldsymbol{d}_i' = \boldsymbol{\omega} \times \boldsymbol{d}_i, \text{ for } i = 1, 2, 3 \tag{169}$$

$$g(\boldsymbol{r}) = 0 \tag{170}$$

$$\boldsymbol{d}_2 \cdot \boldsymbol{n}(\boldsymbol{r}) = 0 \tag{171}$$

Where $\boldsymbol{\nabla} \boldsymbol{n} = \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|} \cdot (\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n})$ is almost the (negative) shape operator $\boldsymbol{K} = -(\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) \cdot \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|} \cdot (\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) = (\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) \cdot \boldsymbol{\nabla} \boldsymbol{n}$. The two constraint equations are equivalent saying that the first two components of the Darboux vector must agree with the surface curvatures

$$\omega_1 = \boldsymbol{d}_1 \cdot \boldsymbol{K} \cdot \boldsymbol{d}_2 \tag{172}$$

$$\omega_2 = -\boldsymbol{d}_1 \cdot \boldsymbol{K} \cdot \boldsymbol{d}_1. \tag{173}$$

Then, you can shove these into the ODEs and solve for lagrange multipliers.

5.1.2 Direct enforcement of constraints using the principle of virtual work

A general variation of the fliament configuration is some combination of displacement $\delta \boldsymbol{r}$ and rotation $\delta D = \hat{\boldsymbol{\psi}} D$. The method of Lagrange multipliers allows these variations to be taken independently at the cost of the introduction many new degrees of freedom in the form of Lagrange multipliers. The principle of virtual work states that the equilibrium shape equations can be obtained by restricting $\delta \boldsymbol{r}, \boldsymbol{\psi}$ to the subset of variations which do not violate the constraints (or any s derivative of constraints).

$$\delta \frac{\mathrm{d}^k}{\mathrm{d}s^k} (\boldsymbol{d}_1 - \boldsymbol{r}') = \frac{\mathrm{d}^k}{\mathrm{d}s^k} \delta (\boldsymbol{d}_1 - \boldsymbol{r}') = 0 \tag{174}$$

$$\delta \frac{\mathrm{d}^k}{\mathrm{d}s^k} (g(\boldsymbol{r})) = \frac{\mathrm{d}^k}{\mathrm{d}s^k} \delta (g(\boldsymbol{r})) = 0 \tag{175}$$

$$\delta \frac{\mathrm{d}^k}{\mathrm{d}s^k} (\boldsymbol{d}_3 - \boldsymbol{n}) = \frac{\mathrm{d}^k}{\mathrm{d}s^k} \delta (\boldsymbol{d}_3 - \boldsymbol{n}) = 0 \tag{176}$$

Frame kinematics, adaptedness of the frame to the centerline, and inextensibility are preserved when

$$\delta \boldsymbol{d}_k = \boldsymbol{\psi} \times \boldsymbol{d}_k \tag{177}$$

$$\delta \boldsymbol{\omega} = \boldsymbol{\psi}' + \boldsymbol{\psi} \times \boldsymbol{\omega} \tag{178}$$

$$\delta \boldsymbol{r}' = \boldsymbol{\psi} \times \boldsymbol{d}_1 \tag{179}$$

5.2 Dynamics

If we include viscous forces, moments

$$\boldsymbol{F}_v = -\sum_i \alpha_i v_i \boldsymbol{d}_i, \qquad \boldsymbol{M}_v = -\beta \Omega_1 \boldsymbol{d}_1 \tag{180}$$

force and moment balance say

$$\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f} - \mu \boldsymbol{d}_1 + \boldsymbol{M}_v = 0 \tag{181}$$

$$\boldsymbol{f}' - \lambda \boldsymbol{n} - \mu \boldsymbol{\nabla} \boldsymbol{n} \cdot \boldsymbol{d}_2 + \boldsymbol{F}_s + \boldsymbol{F}_v = 0 \tag{182}$$

Requiring the definitions of $\boldsymbol{v}, \boldsymbol{\omega}, \boldsymbol{\omega}, g, \boldsymbol{d}_i$'s, etc. . . to be consitent gives us some fun new dynamical constraints to play with.

$$\boldsymbol{n} \cdot \boldsymbol{v} = 0 \tag{183}$$

$$\boldsymbol{d}_1 \cdot \boldsymbol{v}' = 0 \tag{184}$$

$$\Omega_1 = \boldsymbol{v} \cdot \boldsymbol{K} \cdot \boldsymbol{d}_2 \tag{185}$$

$$= K_{12} v_1 + K_{22} v_2 \tag{186}$$

$$= \omega_1 v_1 + K_{22} v_2 \tag{187}$$

$$\Omega_2 = -\boldsymbol{v} \cdot \boldsymbol{K} \cdot \boldsymbol{d}_1 \tag{188}$$

$$\dot{\boldsymbol{\omega}} = \boldsymbol{\Omega}' + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{189}$$

As in the case of the free filament, moment balance tells us

$$\boldsymbol{f} = f_1 \boldsymbol{d}_1 + \boldsymbol{d}_1 \times \boldsymbol{m}'. \tag{190}$$

The $\boldsymbol{d}_1$ component of force balance is

$$f_1' + \boldsymbol{d}_1 \cdot (\boldsymbol{\omega} \times \boldsymbol{f}) - \mu \boldsymbol{d}_1 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{d}_2 + F_{s1} + F_{v1} = 0 \tag{191}$$

Dotting moment balance with ω

$$\boldsymbol{\omega} \cdot \boldsymbol{m}' + \boldsymbol{\omega} \cdot (\boldsymbol{d}_1 \times \boldsymbol{f}) - \mu \omega_1 + \boldsymbol{\omega} \cdot \boldsymbol{M}_v = 0 \tag{192}$$

and using the fact that $-\mu \boldsymbol{d}_1 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{d}_2 = \omega_1 \mu = \boldsymbol{\omega} \cdot \boldsymbol{m}' - \boldsymbol{f} \cdot (\boldsymbol{\omega} \times \boldsymbol{d}_1) - \mu \omega_1 + \boldsymbol{\omega} \cdot \boldsymbol{M}_v$ lets us rewrite Eq 191 as

$$f_1' + \boldsymbol{\omega} \cdot \boldsymbol{m}' + F_{s1} + F_{v1} + \boldsymbol{M}_v \cdot \boldsymbol{\omega} = 0 \tag{193}$$

Then, using the kinematic constraints on $\boldsymbol{\Omega}$ and the definitions of $\boldsymbol{F}_v, \boldsymbol{M}_v$, we get

$$\boxed{f_1' + \boldsymbol{\omega} \cdot \boldsymbol{m}' + F_{s1} = \left(\alpha_1 + \beta \omega_1^2\right) v_1 + \beta \omega_1 K_{22} v_2} \tag{194}$$

The μ can be eliminated from the $\boldsymbol{d}_2$ component of force balance

$$\omega_3 f_1 + \boldsymbol{d}_2 \cdot (\boldsymbol{d}_1 \times \boldsymbol{m}')' + \mu K_{22} + F_{s2} + F_{v2} = 0 \tag{195}$$

using the $\boldsymbol{d}_1$ component of moment balance

$$\mu = \boldsymbol{d}_1 \cdot \boldsymbol{m}' - \beta \Omega_1 \tag{196}$$

$$= \boldsymbol{d}_1 \cdot \boldsymbol{m}' - \beta \left(K_{12} v_1 + K_{22} v_2\right). \tag{197}$$

After a wee bit of algebra, you end up with

$$\boxed{\omega_3 f_1 + \boldsymbol{d}_2 \cdot (\boldsymbol{d}_1 \times \boldsymbol{m}')' + K_{22} \boldsymbol{d}_1 \cdot \boldsymbol{m}' + F_{s2} = \beta K_{12} K_{22} v_1 + \left[\alpha_2 + \beta \left(K_{22}\right)^2\right] v_2} \tag{198}$$

$$\begin{bmatrix} f_1' + \boldsymbol{\omega} \cdot \boldsymbol{m}' + F_{s1} \\ \omega_3 f_1 + \boldsymbol{d}_2 \cdot (\boldsymbol{d}_1 \times \boldsymbol{m}')' + K_{22} \boldsymbol{d}_1 \cdot \boldsymbol{m}' + F_{s2} \end{bmatrix} = \begin{bmatrix} \alpha_1 + \beta \omega_1^2 & \beta \omega_1 K_{22} \\ \beta K_{12} K_{22} & \alpha_2 + \beta \left(K_{22}\right)^2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \tag{199}$$

Lets define some new things:

$$X = \begin{bmatrix} f_1' + \boldsymbol{\omega} \cdot \boldsymbol{m}' + F_{s1} \\ \omega_3 f_1 + \boldsymbol{d}_2 \cdot (\boldsymbol{d}_1 \times \boldsymbol{m}')' + K_{22} \boldsymbol{d}_1 \cdot \boldsymbol{m}' + F_{s2} \end{bmatrix} \tag{200}$$

$$Z = \begin{bmatrix} f_1' \\ \omega_3 f_1 \end{bmatrix}, \quad Y = \begin{bmatrix} \boldsymbol{\omega} \cdot \boldsymbol{m}' + F_{s1} \\ \boldsymbol{d}_2 \cdot (\boldsymbol{d}_1 \times \boldsymbol{m}')' + K_{22} \boldsymbol{d}_1 \cdot \boldsymbol{m}' + F_{s2} \end{bmatrix}, \quad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \tag{201}$$

$$A = \begin{bmatrix} \alpha_1 + \beta \omega_1^2 & \beta \omega_1 K_{22} \\ \beta K_{12} K_{22} & \alpha_2 + \beta \left(K_{22}\right)^2 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \beta \left(K_{12}\right)^2 & \beta K_{12} K_{22} \\ \beta K_{12} K_{22} & \alpha_2 + \beta \left(K_{22}\right)^2 \end{bmatrix} \tag{202}$$

$$A^{-1} = \frac{1}{\alpha_1 \alpha_2 + \beta \alpha_1 \left(K_{22}\right)^2 + \beta \alpha_2 \left(K_{12}\right)^2} \begin{bmatrix} \alpha_2 + \beta \left(K_{22}\right)^2 & -\beta K_{12} K_{22} \\ -\beta K_{12} K_{22} & \alpha_1 + \beta \left(K_{12}\right)^2 \end{bmatrix} \tag{203}$$

Note that Y, A are entirely determined by the geometry of the filament and A^{-1} always exists since $\det A = \alpha_1 \alpha_2 + \beta \alpha_1 \left(K_{22}\right)^2 + \beta \alpha_2 \left(K_{12}\right)^2 > 0$.

From constraints we know that $v_1' = \omega_3 v_2$ and from all that stuff above we know

$$v = A^{-1} Z + A^{-1} Y \tag{204}$$

$$v' = A^{-1} Z' + A^{-1\prime} (Z + Y) + A^{-1} Y'. \tag{205}$$

Taking the appropriate components of the matrix equations:

$$v_1' = A_{1i}^{-1} Z_i' + A_{1i}^{-1\prime} (Z_i + Y_i) + A_{1i}^{-1} Y_i' \tag{206}$$

$$= A_{11}^{-1} Z_1' + A_{12}^{-1} Z_2' + A_{1i}^{-1\prime} (Z_i + Y_i) + A_{1i}^{-1} Y_i' \tag{207}$$

$$\omega_3 v_2 = \omega_3 A_{2i}^{-1} (Z_i + Y_i) \tag{208}$$

equating the two expressions:

$$A_{11}^{-1} Z_1' + A_{12}^{-1} Z_2' + A_{1i}^{-1\prime} (Z_i + Y_i) + A_{1i}^{-1} Y_i' = \omega_3 A_{2i}^{-1} (Z_i + Y_i) \tag{209}$$

$$\Rightarrow Z_1' + \frac{1}{A_{11}^{-1}} \left[A_{12}^{-1} Z_2' + A_{1i}^{-1\prime} Z_i - \omega_3 A_{2i}^{-1} Z_i \right] = \frac{1}{A_{11}^{-1}} \left[\omega_3 A_{2i}^{-1} Y_i - A_{1i}^{-1\prime} Y_i - A_{1i}^{-1} Y_i' \right] \tag{210}$$

$$\Rightarrow f_1'' + \frac{1}{A_{11}^{-1}} \left[A_{12}^{-1} Z_2' + A_{1i}^{-1\prime} Z_i - \omega_3 A_{2i}^{-1} Z_i \right] = \frac{1}{A_{11}^{-1}} \left[\omega_3 A_{2i}^{-1} Y_i - A_{1i}^{-1\prime} Y_i - A_{1i}^{-1} Y_i' \right] \tag{211}$$

In terms of $f = f_1, \omega = \omega_3$

$$\boxed{f'' + \frac{A_{11}^{-1\prime}}{A_{11}^{-1}} f' + \frac{\left(A_{12}^{-1}\omega\right)' - \omega^2 A_{22}^{-1}}{A_{11}^{-1}} f = \frac{\omega A_{2i}^{-1} Y_i - \left(A_{1i}^{-1} Y_i\right)'}{A_{11}^{-1}}} \tag{212}$$

$$A_{11}^{-1} f'' + A_{11}^{-1\prime} f' + \left[\left(A_{12}^{-1}\omega\right)' - \omega^2 A_{22}^{-1}\right] f = \omega A_{2i}^{-1} Y_i - \left(A_{1i}^{-1} Y_i\right)' \tag{213}$$

$$\left[A_{11}^{-1} f'\right]' + \left[\left(A_{12}^{-1}\omega\right)' - \omega^2 A_{22}^{-1}\right] f = \omega A_{2i}^{-1} Y_i - \left(A_{1i}^{-1} Y_i\right)' \tag{214}$$

$$\frac{\mathrm{d}}{\mathrm{d}s}\left[p(s)\frac{\mathrm{d}f}{\mathrm{d}s}\right] + q(s) f = h(s) \tag{215}$$

At the clamped end the velocity/angular velocity is zero, so we have

$$f'(0) = -\boldsymbol{\omega}(0) \cdot \boldsymbol{m}'(0) \tag{216}$$

$$= -B_1\omega_1(0)\omega_1'(0) - B_2\omega_2(0)\omega_2'(0) - B_3\omega_3(0)\omega_3'(0) \tag{217}$$

$$= \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}\sum_k B_k \omega_k^2\right]\Bigg|_{s=0} \tag{218}$$

At the free end we have

$$f(L) = -B_1(\omega_1(L) - \sigma_1)\omega_1(L) - B2(\omega_2(L) - \sigma_2)\omega_2(L) \tag{219}$$

When $\omega_3(L) = \sigma_3$ (the free end BC for ω_3) this is equivalent to

$$f(L) = -\boldsymbol{\omega}(L) \cdot \boldsymbol{m}(L), \tag{220}$$

but if the initial shape donesn't satisfy the free end BC this won't work!

$$p(s) = A_{11}^{-1} \tag{221}$$

$$q(s) = \left(A_{12}^{-1}\omega\right)' - \omega^2 A_{22}^{-1} \tag{222}$$

$$h(s) = \omega A_{2i}^{-1} Y_i - \left(A_{1i}^{-1} Y_i\right)' \tag{223}$$

$$\tag{224}$$

$$Y = \begin{bmatrix} \boldsymbol{\omega} \cdot \boldsymbol{m}' \\ (K_{11} + K_{22})\boldsymbol{d}_1 \cdot \boldsymbol{m}' - \boldsymbol{d}_3 \cdot \boldsymbol{m}'' \end{bmatrix} \tag{225}$$

Let $\psi(s) = f(s) - f'(0)(s - L)$.

$$\mathcal{L} \cdot \psi = H \tag{226}$$

$$\psi(= H \tag{227}$$

Boundary conditions

Suppose the filament is clamped at $s = 0$ and free at $s = L$. The clamped end needs to satisfy

$$\boldsymbol{v}_S|_{s=0} = 0, \Omega_3|_{s=0} = 0 \qquad \text{(OR } \boldsymbol{\Omega}|_{s=0} = 0) \tag{228}$$

$$X = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{229}$$

numerics

- compute some initial shape from specified boundary data or just specify omega

$$??? \mapsto (\omega^{[0]}, \boldsymbol{r}^{[0]})$$

- compute tension

—Still need to update equations below here—

5.3 1st integrals

$$\frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}\sum_i B_i\omega_i^2 + f_1\right] = 0 \tag{230}$$

Proof. Since

$$\boldsymbol{d}_1 \cdot \boldsymbol{\nabla} g(\boldsymbol{r}) = 0 \qquad \boldsymbol{d}_1 \cdot \left[\frac{\partial^2 g(\boldsymbol{r})}{\partial x_i \partial x_j}\boldsymbol{E}_i\boldsymbol{E}_j^T\right] \cdot \boldsymbol{d}_2 = -\|\boldsymbol{\nabla} g(\boldsymbol{r})\|\omega_1,$$

$\boldsymbol{d}_1\cdot$(force balance equation) gives

$$\boldsymbol{d}_1 \cdot \boldsymbol{f}' = -\mu\|\boldsymbol{\nabla} g(\boldsymbol{r})\|\omega_1.$$

Then, taking the inner product of $\boldsymbol{\omega}$ with the torque balance equation,

$$\begin{aligned}
0 &= \boldsymbol{\omega} \cdot [\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f} - \mu\|\boldsymbol{\nabla} g(\boldsymbol{r})\|\boldsymbol{d}_1] \\
&= \boldsymbol{\omega} \cdot \left[\sum_i B_i\omega_i'\boldsymbol{d}_i + \boldsymbol{\omega} \times \boldsymbol{m}\right] + \boldsymbol{\omega} \cdot [\boldsymbol{d}_1 \times \boldsymbol{f}] - \mu\|\boldsymbol{\nabla} g(\boldsymbol{r})\|\omega_1 \\
&= \sum_i B_i\omega_i\omega_i' + [\boldsymbol{\omega} \times \boldsymbol{d}_1] \cdot \boldsymbol{f} + \boldsymbol{d}_1 \cdot \boldsymbol{f}' \\
&= \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}\sum_i B_i\omega_i^2 + \boldsymbol{d}_1 \cdot \boldsymbol{f}\right].
\end{aligned}$$

□

5.4 Boundary conditions

Suppose $\Gamma_0 = (\boldsymbol{r}, D)$ and $\Gamma_\epsilon = \Gamma_0 + \delta\Gamma + O(\epsilon^2) = (\boldsymbol{r} + \delta\boldsymbol{r}, D + \delta D) + O(\epsilon^2)$ are two stationary solutions of the energy functional functional satisfying different boundary conditions. Then The 1st variation of the energy functional evaluated at solution of the shape equations satisfies

$$\delta E \tag{231}$$

The total work due during an infinitesimal translation/rotatation of an endpoint is

$$\delta W = \delta W_{\text{external}} + \boldsymbol{m} \cdot \boldsymbol{\psi} + \boldsymbol{f} \cdot \delta\boldsymbol{r}. \tag{232}$$

In equilibrium, this must vanish for all variations satisfying the constraints.

$$\delta \boldsymbol{r}' = \boldsymbol{\psi} \times \boldsymbol{d}_1 \quad \delta \boldsymbol{r} \cdot \boldsymbol{\nabla} g = 0 \quad [\boldsymbol{\psi} \times \boldsymbol{d}_2] \cdot \boldsymbol{\nabla} g + \boldsymbol{d}_2 \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \delta \boldsymbol{r} = 0. \tag{233}$$

These three conditions ensure that the variations are consitent with all kinematic constraints which the filament is subject to. The third condition can be written as

$$\psi_1 = -\frac{1}{\|\boldsymbol{\nabla} g\|} \boldsymbol{d}_2 \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \delta \boldsymbol{r}. \tag{234}$$

The second constraint just lets us write $\delta \boldsymbol{r} = a_1 \boldsymbol{d}_1 + a_2 \boldsymbol{d}_2$, which we can differentiate and then set the $\boldsymbol{d}_3$ component equal to that of $\boldsymbol{\psi} \times \boldsymbol{d}_1$.

$$\psi_2 = \frac{1}{\|\boldsymbol{\nabla} g\|} \boldsymbol{d}_1 \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \delta \boldsymbol{r} \tag{235}$$

$$\boldsymbol{\psi} = \boldsymbol{n} \times \left(\frac{1}{|\boldsymbol{\nabla} g|} \boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \delta \boldsymbol{r} \right) + \psi_3 \boldsymbol{n} \tag{236}$$

If $\boldsymbol{F}, \boldsymbol{M}$ are the net force and couple on the cross-section, then the work is

$$\delta W = \boldsymbol{M} \cdot \boldsymbol{\psi} + \boldsymbol{F} \cdot \delta \boldsymbol{r} \tag{237}$$

$$= [(\boldsymbol{n} \times \boldsymbol{M}) \cdot \boldsymbol{K} + \boldsymbol{F}] \cdot \delta \boldsymbol{r} + \boldsymbol{M} \cdot \boldsymbol{n} \psi_3 \tag{238}$$

Free end:

$$m_3 = 0 \tag{239}$$

$$\omega_1 m_1 + \omega_2 m_2 + f_1 = 0 \tag{240}$$

$$\frac{1}{\|\boldsymbol{\nabla} g\|} \boldsymbol{d}_2 \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \boldsymbol{d}_2 m_1 + \omega_1 m_2 - f_2 = 0 \tag{241}$$

'Pinned' end: (fixed position, but free to rotate about)

5.5 Boundary conditions

The total work due during an infinitesimal translation/rotatation of an endpoint is

$$\delta W = \delta W_{\text{external}} + \boldsymbol{m} \cdot \boldsymbol{\psi} + \boldsymbol{f} \cdot \delta \boldsymbol{r}. \tag{242}$$

In equilibrium, this must vanish for all variations satisfying the constraints.

$$\delta \boldsymbol{r}' = \boldsymbol{\psi} \times \boldsymbol{d}_1 \quad \delta \boldsymbol{r} \cdot \boldsymbol{\nabla} g = 0 \quad [\boldsymbol{\psi} \times \boldsymbol{d}_2] \cdot \boldsymbol{\nabla} g + \boldsymbol{d}_2 \cdot \left[\frac{\partial^2 g(\boldsymbol{r})}{\partial x_i \partial x_j} \boldsymbol{E}_i \boldsymbol{E}_j^T \right] \cdot \delta \boldsymbol{r} = 0. \tag{243}$$

These three conditions ensure that the variations are consitent with all kinematic constraints which the filament is subject to. The third condition can be written as

$$\psi_1 = -\frac{1}{\|\boldsymbol{\nabla} g\|} \boldsymbol{d}_2 \cdot \left[\frac{\partial^2 g(\boldsymbol{r})}{\partial x_i \partial x_j} \boldsymbol{E}_i \boldsymbol{E}_j^T \right] \cdot \delta \boldsymbol{r}. \tag{244}$$

The second constraint just lets us write $\delta \boldsymbol{r} = a_1 \boldsymbol{d}_1 + a_2 \boldsymbol{d}_2$, which we can differentiate and then set the $\boldsymbol{d}_3$ component equal to that of $\boldsymbol{\psi} \times \boldsymbol{d}_1$.

$$\psi_2 = \frac{1}{\|\boldsymbol{\nabla} g\|} \boldsymbol{d}_1 \cdot \left[\frac{\partial^2 g(\boldsymbol{r})}{\partial x_i \partial x_j} \boldsymbol{E}_i \boldsymbol{E}_j^T \right] \cdot \delta \boldsymbol{r} \tag{245}$$

Free end:

$$m_3 = 0 \tag{246}$$

$$\omega_1 m_1 + \omega_2 m_2 + f_1 = 0 \tag{247}$$

$$\frac{1}{\|\boldsymbol{\nabla} g\|} \boldsymbol{d}_2 \cdot \left[\frac{\partial^2 g(\boldsymbol{r})}{\partial x_i \partial x_j} \boldsymbol{E}_i \boldsymbol{E}_j^T \right] \cdot \boldsymbol{d}_2 m_1 + \omega_1 m_2 - f_2 = 0 \tag{248}$$

'Pinned' end: (fixed position, but free to rotate about)

5.6 Quadratic Surfaces

Pay attention to where Darboux vs ambient components are used. I'm using the constraint with $\boldsymbol{n}$ rather than $\boldsymbol{\nabla} g$.

wiki page for quadratic surfaces

$$g(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x} \cdot \boldsymbol{A} \cdot \boldsymbol{x} + \boldsymbol{b} \cdot \boldsymbol{x} + c \tag{249}$$

$$\boldsymbol{\nabla} g(\boldsymbol{x}) = \boldsymbol{A} \cdot \boldsymbol{x} + \boldsymbol{b} \tag{250}$$

$$\boldsymbol{\nabla}\boldsymbol{\nabla} g(\boldsymbol{x}) = \boldsymbol{A} \tag{251}$$

Where $A^T = A$ is constant.

Component matrices in Darboux basis!

$$K_{ij} := \boldsymbol{d}_i \cdot \frac{1}{\|\boldsymbol{\nabla} g\|}\boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \boldsymbol{d}_j \tag{252}$$

$$\hat{\boldsymbol{\omega}} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_1 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \tag{253}$$

$$\omega_1 = -K_{12} \qquad \omega_2 = K_{11} \tag{254}$$

$$K'_{ij} = (\boldsymbol{\omega} \times \boldsymbol{d}_i) \cdot \frac{1}{\|\boldsymbol{\nabla} g\|}\boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \boldsymbol{d}_j + \boldsymbol{d}_i \cdot \frac{1}{\|\boldsymbol{\nabla} g\|}\boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot (\boldsymbol{\omega} \times \boldsymbol{d}_j) + \boldsymbol{d}_i \cdot \left(\frac{-\boldsymbol{d}_1 \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \boldsymbol{\nabla} g}{\|\boldsymbol{\nabla} g\|^3}\right)\boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot \boldsymbol{d}_j \tag{255}$$

$$= (\hat{\boldsymbol{\omega}} \cdot \boldsymbol{d}_i)^T \cdot \boldsymbol{K} \cdot \boldsymbol{d}_j + \boldsymbol{d}_i \cdot \boldsymbol{K} \cdot (\hat{\boldsymbol{\omega}} \cdot \boldsymbol{d}_j) - K_{13}K_{ij} \tag{256}$$

$$= -\boldsymbol{d}_i \cdot \hat{\boldsymbol{\omega}} \cdot \boldsymbol{K} \cdot \boldsymbol{d}_j + \boldsymbol{d}_i \cdot \boldsymbol{K} \cdot \hat{\boldsymbol{\omega}} \cdot \boldsymbol{d}_j - K_{13}K_{ij} \tag{257}$$

$$\tag{258}$$

$$\boxed{\boldsymbol{K}' = -K_{13}\boldsymbol{K} + [\boldsymbol{K}, \hat{\boldsymbol{\omega}}]} \tag{259}$$

$[\boldsymbol{K}, \hat{\boldsymbol{\omega}}]$ is symmetric and traceless...

$$\omega_1' = -K_{12}' = K_{13}K_{12} - [\boldsymbol{K}, \hat{\boldsymbol{\omega}}]_{12} = -2K_{13}\omega_1 - K_{22}\omega_3 + K_{23}\omega_2 + \omega_2\omega_3 \tag{260}$$

$$\omega_2' = K_{11}' = -K_{13}K_{11} + [\boldsymbol{K}, \hat{\boldsymbol{\omega}}]_{11} = -3K_{13}\omega_2 - \omega_1\omega_3 \tag{261}$$

$$\boxed{\omega_1' = -2K_{13}\omega_1 - K_{22}\omega_3 + K_{23}\omega_2 + \omega_2\omega_3} \tag{262}$$

$$\boxed{\omega_2' = -3K_{13}\omega_2 - 2\omega_1\omega_3} \tag{263}$$

Want expressions for μ, λ.

$$\boldsymbol{m}' + \boldsymbol{d}_1 \times \boldsymbol{f} - \mu \boldsymbol{d}_1 = 0 \tag{264}$$

$$\boldsymbol{f}' - \lambda \boldsymbol{\nabla} g - \mu \boldsymbol{K} \cdot \boldsymbol{d}_2 = 0 \tag{265}$$

From moment balance:

$$\mu = B_1\omega_1' + \boldsymbol{d}_1 \cdot (\boldsymbol{\omega} \times \boldsymbol{m}) = B_1\omega_1' + (B_3 - B_2)\,\omega_2\omega_3 + B_2\sigma_2\omega_3 - B_3\sigma_3\omega_2 \tag{266}$$

Using the expression for ω_1' above:

$$\mu = B_1\left(-2K_{13}\omega_1 - K_{22}\omega_3 + K_{23}\omega_2 + \omega_2\omega_3\right) + (B_3 - B_2)\,\omega_2\omega_3 + B_2\sigma_2\omega_3 - B_3\sigma_3\omega_2 \tag{267}$$

$$= -2B_1K_{13}\omega_1 + (B_1K_{23} - B_3\sigma_3)\,\omega_2 + (B_2\sigma_2 - B_1K_{22})\,\omega_3 + (B_1 - B_2 + B_3)\,\omega_2\omega_3 \tag{268}$$

$$\boxed{\mu = -2B_1K_{13}\omega_1 + (B_1K_{23} - B_3\sigma_3)\,\omega_2 + (B_2\sigma_2 - B_1K_{22})\,\omega_3 + (B_1 - B_2 + B_3)\,\omega_2\omega_3} \tag{269}$$

From the normal component of force balance:

$$\|\boldsymbol{\nabla} g\|\lambda = \boldsymbol{d}_3 \cdot (\boldsymbol{f}' - \mu \boldsymbol{K} \cdot \boldsymbol{d}_2) \tag{270}$$

$$= f_3' + \omega_1 f_2 - \omega_2 f_1 - \mu K_{23} \tag{271}$$

Need f_3'. From $\boldsymbol{d}_2$ component of moment balance:

$$\begin{aligned} f_3 &= B_2\omega_2' + (B_1 - B_3)\,\omega_1\omega_3 + B_3\sigma_3\omega_1 - B_1\sigma_1\omega_3 && (272)\\ &= (B_1 - 2B_2 - B_3)\,\omega_1\omega_3 + B_3\sigma_3\omega_1 - 3B_2K_{13}\omega_2 - B_1\sigma_1\omega_3 && (273)\\ &= [(B_1 - 2B_2 - B_3)\,\omega_1 - B_1\sigma_1]\,\omega_3 + B_3\sigma_3\omega_1 - 3B_2K_{13}\omega_2 && (274) \end{aligned}$$

Taking a derivative...

$$f_3' = (B_1 - 2B_2 - B_3)\,(\omega_1'\omega_3 + \omega_1\omega_3') + B_3\sigma_3\omega_1' - 3B_2\,(K_{13}'\omega_2 + K_{13}\omega_2') - B_1\sigma_1\omega_3' \tag{275}$$

$$\begin{aligned} \omega_1' &= -2K_{13}\omega_1 - K_{22}\omega_3 + K_{23}\omega_2 + \omega_2\omega_3 && (276)\\ \omega_2' &= -3K_{13}\omega_2 - 2\omega_1\omega_3 && (277)\\ \omega_3' &= \frac{B_1 - B_2}{B_3}\omega_1\omega_2 + \frac{B_2\sigma_2}{B_3}\omega_1 - \frac{B_1\sigma_1}{B_3}\omega_2 - f_2/B_3 && (278)\\ K_{13}' &= -\left(K_{13}\right)^2 + \omega_1^2 + \omega_2^2 - K_{33}\omega_2 + K_{23}\omega_3 && (279) \end{aligned}$$

$$\boxed{\mu = -2B_1K_{13}\omega_1 + (B_1K_{23} - B_3\sigma_3)\,\omega_2 + (B_2\sigma_2 - B_1K_{22})\,\omega_3 + (B_1 - B_2 + B_3)\,\omega_2\omega_3} \tag{280}$$

$$\boxed{\|\boldsymbol{\nabla} g\|\lambda = f_3' + \omega_1 f_2 - \omega_2 f_1 - \mu K_{23}} \tag{281}$$

5.7 Plane

$$g(\boldsymbol{x}) = z \tag{282}$$

$$\boldsymbol{\nabla} g = \boldsymbol{E}_z \tag{283}$$

$$\boldsymbol{\nabla}\boldsymbol{\nabla} g = 0 \tag{284}$$

5.8 Paraboloid

5.9 Sphere

$$\frac{1}{2}B\left(\frac{\mathrm{d}\omega}{\mathrm{d}s}\right)^2 + V(\omega) = E \tag{285}$$

$$s(\omega) = \int_{\omega(0)}^{\omega} \frac{\mathrm{d}\omega}{\sqrt{\frac{2}{B}\left(E - V(\omega)\right)}} \tag{286}$$

If the filament is constrained to the surface of a sphere with radius R and outward unit normal $\boldsymbol{n} = \boldsymbol{d}_3$, then the normal component of the Darboux vector satisfies

$$\frac{1}{2}\omega'^2 + \frac{1}{8}\omega^4 - a\omega^2 - b\omega = E \tag{287}$$

where a, b, and E are constants given by

$$a = \frac{1}{2B_3}\left[\frac{1}{2}B_3\omega_3(0)^2 + f_1(0) - \frac{1}{R^2}\left(B_3 - B_2\right) - \frac{B_2\sigma_2}{R}\right] \tag{288}$$

$$b = \frac{\sigma_3}{R^2} \tag{289}$$

$$E = \frac{1}{2}\left[\frac{f_2(0)}{B_3} + \frac{B_1\sigma_1}{B_3 R}\right]^2 + \frac{1}{8}\omega(0)^4 - a\omega(0)^2 - b\omega(0). \tag{290}$$

Solutions are given in terms of Jacobi elliptic functions in Ref. [**?**].

5.9.1 Messy details...

$$g(\boldsymbol{r}) = \boldsymbol{r}\cdot\boldsymbol{r} - R^2 \tag{291}$$

$$\boldsymbol{\nabla} g(\boldsymbol{r}) = 2\boldsymbol{r} \qquad \frac{\partial^2 g(\boldsymbol{r})}{\partial x_i \partial x_j}\boldsymbol{E}_i\boldsymbol{E}_j^T = 2\boldsymbol{E}_i\boldsymbol{E}_i^T \tag{292}$$

Constraints imply $\boldsymbol{d}_1 = \boldsymbol{r}', \boldsymbol{d}_2 = \frac{1}{R}\boldsymbol{r}\times\boldsymbol{r}', \boldsymbol{d}_3 = \frac{1}{R}\boldsymbol{r}, \omega_1 = 0, \omega_2 = 1/R$, the force and torque balance equations are

$$\boldsymbol{f}' - 2R\lambda\boldsymbol{d}_3 - 2\mu\boldsymbol{d}_2 = 0 \qquad \boldsymbol{m}' + \boldsymbol{d}_1\times\boldsymbol{f} - 2R\mu\boldsymbol{d}_1 = 0 \tag{293}$$

and the system satisfies

$$\frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}B_3\omega_3^2 + f_1\right] = 0 \tag{294}$$

$$\frac{\mathrm{d}}{\mathrm{d}s}\left[\boldsymbol{m} + \boldsymbol{r}\times\boldsymbol{f}\right] = 0. \tag{295}$$

Proof. The first one is proved for an arbitrary surface in a previous section.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}s}\left[\boldsymbol{m}+\boldsymbol{r}\times\boldsymbol{f}\right] &= \boldsymbol{m}'+\boldsymbol{d}_1\times\boldsymbol{f}+\boldsymbol{r}\times\boldsymbol{f}' \\
&= \boldsymbol{m}'+\boldsymbol{d}_1\times\boldsymbol{f}+R\boldsymbol{d}_3\times\boldsymbol{f}' \\
&= \boldsymbol{m}'+\boldsymbol{d}_1\times\boldsymbol{f}-2R\mu\boldsymbol{d}_1 && \text{(force balance)} \\
&= 0 && \text{(torque balance)}
\end{aligned}$$

□

Taking components of the force and torque equations in the Darboux frame gives

$$f_1'+f_3/R-\omega_3 f_2=0 \tag{296}$$
$$f_2'+\omega_3 f_1-2\mu=0 \tag{297}$$
$$f_3'-f_1/R-2R\lambda=0 \tag{298}$$

$$\left[(B_3-B_2)/R+B_2\sigma_2\right]\omega_3-B_3\sigma_3/R-2R\mu=0 \tag{299}$$
$$-B_1\sigma_1\omega_3-f_3=0 \tag{300}$$
$$B_3\omega_3'+B_1\sigma_1/R+f_2=0. \tag{301}$$

This lets us solve for f_3,μ,λ in terms of the other variables.

$$f_3=-B_1\sigma_1\omega_3 \tag{302}$$
$$\lambda=-\frac{1}{2R^2}f_1+\frac{B_1\sigma_1}{2RB_3}f_2+\frac{B_1^2\sigma_1^2}{2B_3R^2} \tag{303}$$
$$\mu=\frac{1}{2}\left[\frac{1}{R^2}\left(B_3-B_2\right)+\frac{B_2\sigma_2}{R}\right]\omega_3-\frac{B_3\sigma_3}{2R^2} \tag{304}$$

and the remaining variables satisfy

$$f_1'=\left[f_2+\frac{B_1\sigma_1}{R}\right]\omega_3 \tag{305}$$
$$f_2'=\left[-f_1+\frac{1}{R^2}\left(B_3-B_2\right)+\frac{B_2\sigma_2}{R}\right]\omega_3-\frac{B_3\sigma_3}{R^2} \tag{306}$$
$$\omega_3'=-\frac{1}{B_3}f_2-\frac{B_1\sigma_1}{B_3R}. \tag{307}$$

Next, use the fact that

$$-f_1 = \frac{1}{2}B_3\omega_3^2 - \frac{1}{2}B_3\omega_3(0)^2 - f_1(0) \tag{308}$$

$$\tag{309}$$

to eliminate f_1, which leaves

$$f_2' = \frac{1}{2}B_3\omega_3^3 + \left[-\frac{1}{2}B_3\omega_3(0)^2 - f_1(0) + \frac{1}{R^2}\left(B_3 - B_2\right) + \frac{B_2\sigma_2}{R}\right]\omega_3 - \frac{B_3\sigma_3}{R^2} \tag{310}$$

$$\omega_3' = -\frac{1}{B_3}f_2 - \frac{B_1\sigma_1}{B_3R}. \tag{311}$$

Taking a derivative of the second equation and substituting the expression for f_2' into the result gives

$$\omega_3'' + \frac{1}{2}\omega_3^3 - \frac{1}{B_3}\left[\frac{1}{2}B_3\omega_3(0)^2 + f_1(0) - \frac{1}{R^2}\left(B_3 - B_2\right) - \frac{B_2\sigma_2}{R}\right]\omega_3 - \frac{\sigma_3}{R^2} = 0 \tag{312}$$

Multiplying by ω_3' lets us write this as

$$0 = \omega_3'\omega_3'' + \frac{1}{2}\omega_3^3\omega_3' - \frac{1}{B_3}\left[\frac{1}{2}B_3\omega_3(0)^2 + f_1(0) - \frac{1}{R^2}\left(B_3 - B_2\right) - \frac{B_2\sigma_2}{R}\right]\omega_3\omega_3' - \frac{\sigma_3}{R^2}\omega_3' \tag{313}$$

$$= \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}\left(\omega_3'\right)^2 + \frac{1}{8}\left(\omega_3\right)^4 - \frac{1}{2B_3}\left[\frac{1}{2}B_3\omega_3(0)^2 + f_1(0) - \frac{1}{R^2}\left(B_3 - B_2\right) - \frac{B_2\sigma_2}{R}\right]\left(\omega_3\right)^2 - \frac{\sigma_3}{R^2}\omega_3\right]. \tag{314}$$

After dropping the subscript on ω_3 and defining constants

$$a = \frac{1}{2B_3}\left[\frac{1}{2}B_3\omega_3(0)^2 + f_1(0) - \frac{1}{R^2}\left(B_3 - B_2\right) - \frac{B_2\sigma_2}{R}\right] \tag{315}$$

$$b = \frac{\sigma_3}{R^2} \tag{316}$$

this can be written

$$\frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{2}\omega'^2 + \frac{1}{8}\omega^4 - a\omega^2 - b\omega\right] = 0. \tag{317}$$

After integrating we get

$$\frac{1}{2}\omega'^2 + \frac{1}{8}\omega^4 - a\omega^2 - b\omega = E \tag{318}$$

where the constant E can be determined using the equation relating ω' and f_2.

$$E = \frac{1}{2}\left[\frac{f_2(0)}{B_3} + \frac{B_1\sigma_1}{B_3 R}\right]^2 + \frac{1}{8}\omega(0)^4 - a\omega(0)^2 - b\omega(0). \tag{319}$$

5.10 Minimal surfaces

5.11 Nondimensionalization

Denote dimensional stuff with an asterisk and non-dimensional stuff without. Use the $\boldsymbol{d}_3$ (normal) components of the bending stiffness B_3^* (Energy-Length) and spontaneous curvature σ_3^* (1/Length) to define natural units.

- Length: $1/\sigma_3^*$
- Energy: $B_3^*\sigma_3^*$
- Force: $B_3^*\left[\sigma_3^*\right]^2$

Dimensional surface quantities, like sphere radius, can be written in natural units and constraint equations can be replaced with nondimensional versions. For example, if

$$g^*(\boldsymbol{r}^*) = \boldsymbol{r}^* \cdot \boldsymbol{r}^* - (R^*)^2,$$

then

$$g^*(\boldsymbol{r}^*) = g^*(\boldsymbol{r}/\sigma_3^*) = \left[g^*_{\text{dim}}\right]\left[g(\boldsymbol{r})\right] = \left[1/\sigma_3^*\right]^2\left[\boldsymbol{r}\cdot\boldsymbol{r} - R^2\right]$$

$$g(\boldsymbol{r}) = \frac{1}{\left[g^*_{\text{dim}}\right]} g^*(\boldsymbol{r}^*) \qquad \boldsymbol{\nabla} g(\boldsymbol{r}) = \frac{1}{\sigma_3^*\left[g^*_{\text{dim}}\right]}\boldsymbol{\nabla}^* g^*(\boldsymbol{r}^*) \tag{320}$$

$$\partial_i\partial_j g(\boldsymbol{r}) = \frac{1}{\left[\sigma_3^*\right]^2\left[g^*_{\text{dim}}\right]}\partial_i^*\partial_j^* g^*(\boldsymbol{r}^*) \tag{321}$$

$$\sigma_1 = \sigma_1^*/\sigma_3^* \qquad \sigma_2 = \sigma_2^*/\sigma_3^* \qquad \sigma_3 = 1 \tag{322}$$

$$B_1 = B_1^*/B_3^* \qquad B_2 = B_2^*/B_3^* \qquad B_3 = 1 \tag{323}$$

$$\tag{324}$$

$$\boldsymbol{\omega} = \boldsymbol{\omega}^*/\sigma_3^* \qquad \boldsymbol{r} = \sigma_3^* \boldsymbol{r}^* \qquad s = \sigma_3^* s^* \tag{325}$$

$$\boldsymbol{m} = \frac{1}{B_3^* \sigma_3^*} \boldsymbol{m}^* \qquad \lambda_{\mathrm{dim}}^* = \frac{B_3^* \left[\sigma_3^*\right]^2}{g_{\mathrm{dim}}^*} \qquad \mu_{\mathrm{dim}}^* = \frac{B_3^* \sigma_3^*}{g_{\mathrm{dim}}^*} \tag{326}$$

$$\tag{327}$$

5.11.1 Reduces variables in shooting method for free end

Conservred: $f_1(L) = f_1(0) + \sum_i \frac{1}{2} B_i \left(\omega_i(0)^2 - \omega_i(L)^2\right)$

For free end: $\omega_3(L) = \sigma_3$, $f_1(L) = -B_2(1/R - \sigma_2)/R$, $f_2(L) = -B_1 \sigma_1/R$.

If you choose

$$f_1(0) = -B_2(1/R - \sigma_2)/R - \frac{1}{2} B_3 \left(\omega_3(0)^2 - \sigma_3^2\right) \tag{328}$$

then you can reduce the dimension of the shooting problem by eliminating $f_1(L)$.

6 Membranes

A Groups and representations

Let H be some finite dimensional Hilbert space, let G be a closed subgroup of $GL(H)$, and let $\mathfrak{g} = T_eG$ be the Lie algebra of G with Lie bracket $[\cdot,\cdot]$. Let $Aut(G)$ be the group of homomorphisms on G and define $C : G \to Aut(G)$ by $C_g(h) = ghg^{-1}$. The adjoint representation of G is

$$Ad : G \to GL(\mathfrak{g}) \qquad Ad_g(\hat{\boldsymbol{u}}) = g\hat{\boldsymbol{u}}g^{-1} \tag{329}$$

The derivative of Ad at the identity gives you the associated representation of $\mathfrak{g}$.

$$ad_{\hat{\boldsymbol{u}}} : \mathfrak{g} \to \mathfrak{g} \qquad ad_{\hat{\boldsymbol{u}}}(\hat{\boldsymbol{v}}) = [\hat{\boldsymbol{u}}, \hat{\boldsymbol{v}}] \tag{330}$$

adjoint/exponential stuff... $e^{ad_{\hat{\boldsymbol{u}}}} = Ad_{e^{\hat{\boldsymbol{u}}}}$

Rotations

The rotation group, denoted $SO(3)$, is the group of length, orientation, and origin-preserving transformations of three-dimensional Euclidean space. The Lie algebra $\mathfrak{so}(3)$ is the tangent space of $SO(3)$ at the identity. They have matrix representations

$$SO(\mathbb{R}^3) = \{U \in GL(\mathbb{R}^3) | UU^T = I, \det(U) = 1\} \quad \mathfrak{so}(\mathbb{R}^3) = \{\hat{\boldsymbol{u}} \in \mathfrak{gl}(\mathbb{R}^3) | \hat{\boldsymbol{u}}^T = -\hat{\boldsymbol{u}}\}. \tag{331}$$

Let $\{\boldsymbol{e}_i\}$ be an orthonormal frame related to the ambient frame $\{\boldsymbol{E}_i\}$ by the continuous rotation $R : I \to SO(3)$. Denote the corresponding coframes by $\{\boldsymbol{E}^i\}$ and $\{\boldsymbol{e}^i\}$.

$$\boldsymbol{e}_i = R \cdot \boldsymbol{E}_i \qquad \boldsymbol{E}_i = R^T \cdot \boldsymbol{e}_i \tag{332}$$

$$\boldsymbol{e}^i = \boldsymbol{E}^i \cdot R^T \qquad \boldsymbol{E}^i = \boldsymbol{e}^i \cdot R \tag{333}$$

$$R = \boldsymbol{e}_i \boldsymbol{E}^i \qquad R^T = \boldsymbol{E}_i \boldsymbol{e}^i \tag{334}$$

The matrix elements of R (in either basis) are just the ambient components of the $\boldsymbol{e}_i$'s.

$$R^{i\cdot}_{\cdot j} = \boldsymbol{e}^i \cdot R \cdot \boldsymbol{e}_j = \boldsymbol{E}^i \cdot R \cdot \boldsymbol{E}_j = \boldsymbol{E}^i \cdot \boldsymbol{e}_j \quad \left(R^T\right)^{i\cdot}_{\cdot j} = \boldsymbol{e}^i \cdot R^T \cdot \boldsymbol{e}_j = \boldsymbol{E}^i \cdot R^T \cdot \boldsymbol{E}_j = \boldsymbol{e}^i \cdot \boldsymbol{E}_j \tag{335}$$

$$\boldsymbol{E}^i \cdot R \cdot \boldsymbol{E}_j = \boldsymbol{E}^i \cdot \boldsymbol{e}_j \tag{336}$$

$$\begin{pmatrix} \boldsymbol{E}^1 \cdot \boldsymbol{e}_1 & \boldsymbol{E}^1 \cdot \boldsymbol{e}_2 & \boldsymbol{E}^1 \cdot \boldsymbol{e}_3 \\ \boldsymbol{E}^2 \cdot \boldsymbol{e}_1 & \boldsymbol{E}^2 \cdot \boldsymbol{e}_2 & \boldsymbol{E}^2 \cdot \boldsymbol{e}_3 \\ \boldsymbol{E}^3 \cdot \boldsymbol{e}_1 & \boldsymbol{E}^3 \cdot \boldsymbol{e}_2 & \boldsymbol{E}^3 \cdot \boldsymbol{e}_3 \end{pmatrix} \tag{337}$$

$$\boldsymbol{e}_j = \boldsymbol{E}_i \boldsymbol{E}^i \cdot \boldsymbol{e}_j = \boldsymbol{E}_i R^i_j \qquad \boldsymbol{e}^i = \boldsymbol{e}^i \cdot \boldsymbol{E}_j \boldsymbol{E}^j = \left(R^T\right)^i_j \boldsymbol{E}^j \tag{338}$$

$$\boldsymbol{E}_j = \boldsymbol{e}_i \boldsymbol{e}^i \cdot \boldsymbol{E}_j = \boldsymbol{e}_i \left(R^T\right)^i_j \qquad \boldsymbol{E}^i = \boldsymbol{E}^i \cdot \boldsymbol{e}_j \boldsymbol{e}^j = R^i_j \boldsymbol{E}^j \tag{339}$$

The hat map

The hat map $\hat{\cdot}$ is a Lie algebra isomorphism between 3-dimensional Euclidean space $\mathbb{E}$ with cross product $\times$ and the Lie algebra $\mathfrak{so}(3)$ with commutator $[\cdot, \cdot]$, and it even plays nice with rotations. Let $\{\boldsymbol{E}_i\}$ be an orthonormal basis for $\mathbb{E}$, define $\hat{\boldsymbol{E}}_i$ by

$$\boldsymbol{E}_n \mapsto \hat{\boldsymbol{E}}_n = \varepsilon^i_{nj} \boldsymbol{E}_i \boldsymbol{E}^j, \tag{340}$$

and extend linearly to the rest of $\mathbb{E}$. Note that $\{\hat{\boldsymbol{E}}_i\}$ is a basis for $\mathfrak{so}(3)$ so

$$\boldsymbol{w} = \boldsymbol{u} \times \boldsymbol{v} \iff \hat{\boldsymbol{w}} = [\hat{\boldsymbol{u}}, \hat{\boldsymbol{v}}]. \tag{341}$$

For any basis related to $\boldsymbol{E}_i$ by rotation (i.e. any other right-handed orthonormal basis) we can apply the hat map without thinking too hard.

$$\boldsymbol{e}_n \mapsto \hat{\boldsymbol{e}}_n = \varepsilon^i_{nj} \boldsymbol{e}_i \boldsymbol{e}^j \tag{342}$$

Proof. Suppose $\boldsymbol{e}_i = R^j_i \boldsymbol{E}_j$ for rotation R.

$$\begin{aligned} R^j_i \hat{\boldsymbol{E}}_j &= \varepsilon^m_{jn} R^j_i \boldsymbol{E}_m \boldsymbol{E}^n \\ &= \varepsilon^m_{jn} \left(R^T\right)^a_m R^j_i R^n_b \boldsymbol{e}_a \boldsymbol{e}^b \\ &= \det(R) \varepsilon^a_{ib} \boldsymbol{e}_a \boldsymbol{e}^b \\ &= \varepsilon^a_{ib} \boldsymbol{e}_a \boldsymbol{e}^b \end{aligned}$$

□

If we let $SO(3)$ act on $\mathfrak{so}(3)$ by conjugation (adjoint representation), then we end up with something that looks just like its natural action on $\mathbb{E}$.

$$\boldsymbol{u} = R \cdot \boldsymbol{v} \iff \hat{\boldsymbol{u}} = R \cdot \hat{\boldsymbol{v}} \cdot R^T \tag{343}$$

$$\tag{344}$$

The standard dot product in $\mathbb{E}$ is equivalent to

$$\langle \hat{\boldsymbol{u}}, \hat{\boldsymbol{v}} \rangle_{\mathfrak{so}(3)} = -\frac{1}{2}\operatorname{tr}(\hat{\boldsymbol{u}} \cdot \hat{\boldsymbol{v}}) = \frac{1}{2}\operatorname{tr}\big(\hat{\boldsymbol{u}}^T \cdot \hat{\boldsymbol{v}}\big) \tag{345}$$

in $\mathfrak{so}(3)$. Various other hat related things to remember...

- Things in $\mathfrak{so}(3)$ are antisymmetric $\hat{\boldsymbol{u}}^T = -\hat{\boldsymbol{u}}$
- cross products $\hat{\boldsymbol{u}} \cdot \boldsymbol{v} = \boldsymbol{u} \times \boldsymbol{v} = -\hat{\boldsymbol{v}} \cdot \boldsymbol{u}$
- If $A^i = \boldsymbol{E}^i \cdot \boldsymbol{A}$ and $A^i_j = \boldsymbol{E}^i \cdot \hat{\boldsymbol{A}} \cdot \boldsymbol{E}_j$, then $A^i = \frac{1}{2}\varepsilon^{ij\cdot}_{\cdot\cdot k} A^k_j$ and $A^i_j = \varepsilon^i_{nj} A^n$

A.1 Evolution operators

Let H be a finite dimensional Hilbert space, let G be a closed subgroup of $GL(H)$, and let $\mathfrak{g}$ be the Lie algebra of G.[4] Suppose $\{U(\alpha,\beta) \in G | \alpha, \beta \in I\}$, where I is some real interval, is a smooth-ish family of operators satisfying the composition law

$$U(\alpha,\beta)U(\beta,\xi) = U(\alpha,\xi) \tag{346}$$

and define the generator of the family to be

$$\hat{\boldsymbol{\omega}}(\xi) := \frac{\partial}{\partial\alpha} U(\xi+\alpha,\xi)\Big|_{\alpha=0}. \tag{347}$$

Fact 6.

- *The composition law implies $U(\xi,\xi) = 1$ and $U(\alpha,\beta)^{-1} = U(\beta,\alpha)$*
- *$\hat{\boldsymbol{\omega}} \in \mathfrak{g}$ so if the subgroup is unitary then $\hat{\boldsymbol{\omega}}^\dagger = -\hat{\boldsymbol{\omega}}$ is anti-Hermitian.*
- *The family of operators satisfies*

$$\frac{\partial}{\partial\alpha} U(\alpha,\xi) = \hat{\omega}(\alpha) U(\alpha,\xi) \tag{348}$$

- *Families with $U(\alpha,\beta) = U(\alpha-\beta)$ are called one-parameter subgroups, have a constant generator $\hat{\boldsymbol{\omega}}$, and can be written as an exponential $U(\alpha) = e^{\alpha\hat{\boldsymbol{\omega}}}$.*

[4] This probably still works if H is separable and G is a group of (bounded?) linear operators. compact subgroup of bounded linear operators $\Rightarrow$ Lie group??

Proof.

$$\frac{\partial}{\partial\beta}U(\beta,\xi) = \frac{\partial}{\partial\alpha}\left[U(\alpha+\beta,\xi)\right]\Big|_{\alpha=0} = \frac{\partial}{\partial\alpha}\left[U(\alpha+\beta,\beta)U(\beta,\xi)\right]\Big|_{\alpha=0} = \frac{\partial}{\partial\alpha}\left[U(\alpha+\beta,\beta)\right]\Big|_{\alpha=0}U(\beta$$

□

B Framed curves

Let $\boldsymbol{x} = \boldsymbol{x}(\xi)$ be a parametrized curve and let $s(\xi) = \int_0^\xi \|\boldsymbol{x}'(\xi^*)\|\,\mathrm{d}\xi^*$ be its arclength. Primes denote ξ derivatives. Note: $s' = \|\boldsymbol{x}'\|$.

The Frenet-Serret frame

The prototypical example of an adapted frame is the Frenet-Serret frame.

Tangent	$\boldsymbol{T} = \boldsymbol{x}'/s'$	(349)
(Principal) Normal	$\boldsymbol{N} = \boldsymbol{T}'/\|\boldsymbol{T}'\|$	(350)
Binormal	$\boldsymbol{B} = \boldsymbol{T}\times\boldsymbol{N} = \dfrac{\boldsymbol{x}'\times\boldsymbol{x}''}{\|\boldsymbol{x}'\times\boldsymbol{x}''\|}$	(351)

Proof. of Binormal formula
Since $\boldsymbol{T}\cdot\boldsymbol{T}' = 0$ we know $\|\boldsymbol{T}'\| = \|\boldsymbol{T}\times\boldsymbol{T}'\|$ and since $\boldsymbol{T}\times\boldsymbol{T}' = \boldsymbol{T}\times(\boldsymbol{x}''/s' - s''\boldsymbol{x}'/s'^2) = \boldsymbol{x}'\times\boldsymbol{x}''/s'^2$ it follows that $\boldsymbol{T}\times\boldsymbol{T}'/\|\boldsymbol{T}'\| = \frac{\boldsymbol{x}'\times\boldsymbol{x}''}{\|\boldsymbol{x}'\times\boldsymbol{x}''\|}$. □

The Frenet-Serret formulas are

$$\begin{bmatrix}\boldsymbol{T}\\ \boldsymbol{N}\\ \boldsymbol{B}\end{bmatrix}' = s'\begin{bmatrix}0 & \kappa & 0\\ -\kappa & 0 & \tau\\ 0 & -\tau & 0\end{bmatrix}\begin{bmatrix}\boldsymbol{T}\\ \boldsymbol{N}\\ \boldsymbol{B}\end{bmatrix} = s'\begin{bmatrix} & \kappa\boldsymbol{N} & \\ -\kappa\boldsymbol{T} & & +\tau\boldsymbol{B}\\ & -\tau\boldsymbol{N} & \end{bmatrix} \tag{352}$$

The curvature, torsion, and parametrization determine a curve up to a rigid motion. The rigid motion can be determined by initial conditions for the curve. If we let $\boldsymbol{\kappa} = s'\tau\boldsymbol{T} + s'\kappa\boldsymbol{B}$ the Frenet-Serret formulas can be written as

$$\boldsymbol{T}' = \boldsymbol{\kappa}\times\boldsymbol{T} \qquad \boldsymbol{N}' = \boldsymbol{\kappa}\times\boldsymbol{N} \qquad \boldsymbol{B}' = \boldsymbol{\kappa}\times\boldsymbol{B}. \tag{353}$$

More general adapted frames

An orthonormal frame $\{\boldsymbol{d}_1, \boldsymbol{d}_2, \boldsymbol{d}_3\}$ defined along a curve $\boldsymbol{x}$ is said to be

adapted[5] to the curve if one of the frame vectors is tangent to $\boldsymbol{x}$.

$$\boldsymbol{d}_1 = \boldsymbol{x}'/s' \tag{354}$$

The frame can be expressed in terms of a evolution operator $U : I \to SO(3)$

$$\boldsymbol{d}_i(\xi) = U(\xi)\boldsymbol{d}_i(0), \tag{355}$$

an as a result, it must satisfy

$$\boldsymbol{d}_i' = \hat{\boldsymbol{\omega}}\boldsymbol{d}_i = \boldsymbol{\omega} \times \boldsymbol{d}_i \tag{356}$$

where $\hat{\boldsymbol{\omega}} : I \to \mathfrak{so}(3)$ is defined by $\hat{\boldsymbol{\omega}} = U'U^T$ and $\boldsymbol{\omega}$ is called the Darboux vector.

Invariant Darboux vector and vector curvature tensor

As we have defined it, the Darboux vector $\boldsymbol{\omega}$ is parametrization dependent. Under reparametrization $\xi \mapsto \xi = \xi(\xi^*)$ it transforms as

$$\boldsymbol{\omega}(\xi) \mapsto \boldsymbol{\omega}^*(\xi^*) = \frac{\partial \xi}{\partial \xi^*}\boldsymbol{\omega}(\xi(\xi^*)) \tag{357}$$

making it a relative tensor of weight one on the manifold $\boldsymbol{x}(I)$ with metric $r_{\Theta\Phi} = \frac{\partial \boldsymbol{r}}{\partial \xi^\Theta} \cdot \frac{\partial \boldsymbol{r}}{\partial \xi^\Phi}$ inherited from its embedding $\boldsymbol{x}$ in the ambient space $\mathbb{E}$. Since $s' = \sqrt{\det(r_{\Phi\Theta})}$ is the invariant 1-dimensional volume element, dividing the Darboux vector by s' gives an invariant quantity.

$$\boldsymbol{\Omega} = \frac{1}{s'}\boldsymbol{\omega} \tag{358}$$

Alternatively, we can think of the Darboux vector as having single covariant filament index

$$\boldsymbol{\omega}_\Phi \tag{359}$$

and contract it with components of the tetrad $\boldsymbol{d}_{(1)} = d^\Phi_{(1)}\boldsymbol{r}_\Phi = \frac{1}{s'}\boldsymbol{r}'$ (i.e. the unit tangent Darboux frame vector).

$$\boldsymbol{\omega}_{(\Theta)} = d^\Phi_{(\Theta)}\boldsymbol{\omega}_\Phi \tag{360}$$

[5]These are sometimes referred to as Darboux frames, but be careful assuming this is what someone means when they say 'Darboux frame'. Some authors exclusively use 'Darboux frame' to refer to a frame which is adapted to a curve through an embedded surface.

Either approach results in the same invariant quantity since the space is one-dimensional.

$$\boldsymbol{f}_i' = s'\boldsymbol{\Omega} \times \boldsymbol{f}_i \tag{361}$$

Changing frames

Let $\{\boldsymbol{f}_i = Q\boldsymbol{f}_i(0)\}$ and $\{\boldsymbol{d}_i = U\boldsymbol{d}_i(0)\}$ be two adapted frames with respective Darboux vectors $\boldsymbol{\eta}$ and $\boldsymbol{\omega}$ and which satisfy $\boldsymbol{f}_i(0) = \boldsymbol{d}_i(0)$. Since both frames must include the unit tangent, they can only differ by a rotation about the tangent. Let ϕ be the angle about $\boldsymbol{d}_1 = \boldsymbol{f}_1$ which the $\boldsymbol{d}_i$ frame is rotated with respect to the $\boldsymbol{f}_i$ frame.

$$\boldsymbol{d}_1 = \boldsymbol{f}_1 \quad \boldsymbol{d}_2 = \cos\phi\boldsymbol{f}_2 + \sin\phi\boldsymbol{f}_3 \quad \boldsymbol{d}_3 = -\sin\phi\boldsymbol{f}_2 + \cos\phi\boldsymbol{f}_3 \quad U = e^{\phi\hat{\boldsymbol{f}}_1}Q \tag{362}$$

$$\boldsymbol{f}_2 = \cos\phi\boldsymbol{d}_2 - \sin\phi\boldsymbol{d}_3 \qquad \boldsymbol{f}_3 = \sin\phi\boldsymbol{d}_2 + \cos\phi\boldsymbol{d}_3 \tag{363}$$

$$\boldsymbol{\omega} = \phi'\boldsymbol{d}_1 + \boldsymbol{\eta} \tag{364}$$

In particular, if $\boldsymbol{f}_i$ is the Frenet-Serret frame ($\boldsymbol{\eta} = \boldsymbol{\kappa}$) this means

$$\omega^1 = \phi' + s'\tau \qquad \omega^2 = s'\kappa\sin\phi \qquad \omega^3 = s'\kappa\cos\phi \tag{365}$$

$$s'\kappa = \sqrt{\left[\omega^2\right]^2 + \left[\omega^3\right]^2} \qquad s'\tau = \omega^1 - \frac{\partial}{\partial\xi}\left[\arctan\left(\omega^2/\omega^3\right)\right] \tag{366}$$

Proof. Let $\Phi = e^{\phi\hat{\boldsymbol{f}}_1} = Qe^{\phi\hat{\boldsymbol{f}}_1(0)}Q^T$. Then $\hat{\boldsymbol{\omega}} = U'U^T = \Phi'\Phi^T + \Phi\hat{\boldsymbol{\eta}}\Phi^T = Q'Q^T + \Phi QQ'^T\Phi^T + \phi'\frac{\partial\Phi}{\partial\phi}\Phi^T + \Phi\hat{\boldsymbol{\eta}}\Phi^T$ and since $QQ'^T = -\hat{\boldsymbol{\eta}}$ we get $\hat{\boldsymbol{\omega}} = \phi'\hat{\boldsymbol{f}}_1 + \hat{\boldsymbol{\eta}}$. □

Bishop frame

frame with zero twist...

Derivatives and variations

Let $\{\boldsymbol{E}_i\}$ be the ambient orthonormal frame, let $\{\boldsymbol{d}_i = R\cdot\boldsymbol{E}_i\}$ adapted to $\boldsymbol{r}$, and let $\hat{\boldsymbol{\omega}}$ be the generator of R.

Derivatives with respect to parameter ξ, denoted by primes, are related to the Darboux vector $\boldsymbol{\omega}$.[6]

$$\boldsymbol{d}_i' = \boldsymbol{\omega} \times \boldsymbol{d}_i = \hat{\boldsymbol{\omega}} \cdot \boldsymbol{d}_i \qquad \boldsymbol{d}'^i = -\boldsymbol{d}^i \cdot \hat{\boldsymbol{\omega}} \tag{367}$$

$$R' = \hat{\boldsymbol{\omega}} \cdot R \qquad R'^T = -R^T \cdot \hat{\boldsymbol{\omega}} \tag{368}$$

Proof. $0 = \frac{\mathrm{d}}{\mathrm{d}s}\left(\boldsymbol{d}^i \cdot \boldsymbol{d}_k\right) = \boldsymbol{d}'^i \cdot \boldsymbol{d}_k + \boldsymbol{d}^i \cdot \boldsymbol{d}_k' = \boldsymbol{d}'^i \cdot \boldsymbol{d}_k + \boldsymbol{d}^i \cdot \hat{\boldsymbol{\omega}} \cdot \boldsymbol{d}_k$ □

B.1 Frames depending on multiple parameters

$$\partial_\xi \boldsymbol{d}_i = \boldsymbol{\omega}_\xi \times \boldsymbol{d}_i, \qquad \partial_\tau \boldsymbol{d}_i = \boldsymbol{\omega}_\tau \times \boldsymbol{d}_i \tag{369}$$

and $\partial_\xi \partial_\tau \boldsymbol{d}_i = \partial_\tau \partial_\xi \boldsymbol{d}_i$ then

$$\partial_\xi \boldsymbol{\omega}_\tau = \partial_\tau \boldsymbol{\omega}_\xi + \boldsymbol{\omega}_\xi \times \boldsymbol{\omega}_\tau \tag{370}$$

or in terms of components

$$\partial_\xi \omega_\tau^i - \partial_\tau \omega_\xi^i = \varepsilon_{jk}^i \omega_\xi^j \omega_\tau^k \tag{371}$$

- A frame is called rotation minimizing wrt $\boldsymbol{c}$ if $\boldsymbol{\omega} \cdot \boldsymbol{c} = 0$

B.2 Adapted surface curves

$\boldsymbol{r}(s)$ adapted to surface with normal $\boldsymbol{n} = \boldsymbol{d}_3$.

- geodesic $\omega_3 = 0$
- line of curvature $\omega_1 = 0$
- asymptotic line $\omega_2 = 0$

Can you show that (constrained to surface) stationary points of

$$E = \frac{1}{2} \int \sum_i \omega_i^2 \, \mathrm{d}s \tag{372}$$

[6] If you imagine the frame moving along the filament at speed $s' = \|\boldsymbol{r}'\|$, then $\boldsymbol{\omega}$ would just be its angular velocity.

or maybe

$$E = \frac{1}{2}\int \omega_3^2 \,\mathrm{d}s \tag{373}$$

are geodesics? Does this generalize to higher dimensions? Geodesics in$M \subset \mathbb{R}^n$ are (adapted) curves whose Darboux frames are rotation minimizing wrt anything in the normal bundle of M? Minimize

$$E = \frac{1}{2}\int \sum_i (\boldsymbol{n}_i \cdot \boldsymbol{\omega})^2 \,\mathrm{d}s \tag{374}$$

where $\boldsymbol{n}_i$ is a basis for normal bundle.

C The screended Poisson equation and Yukawa potential

$$\boxed{\left[\nabla^2 - 1/a^2\right]\phi(\boldsymbol{x}) = -\rho(\boldsymbol{x})} \tag{375}$$

The potential due to a point charge is called the Yukawa potential

$$\boxed{\left[\nabla^2 - 1/a^2\right]G(\boldsymbol{x}) = -\delta(\boldsymbol{x})} \tag{376}$$

$$\boxed{G(\boldsymbol{x}) = \frac{1}{4\pi|\boldsymbol{x}|}e^{-|\boldsymbol{x}|/a}} \tag{377}$$

$$G(\boldsymbol{x}) = \frac{1}{4\pi|\boldsymbol{x}|}e^{-|\boldsymbol{x}|/a} \tag{378}$$

$$= \frac{1}{(2\pi)^3}\int \frac{1}{|\boldsymbol{k}|^2 + 1/a^2}e^{i\boldsymbol{k}\cdot\boldsymbol{x}}\,\mathrm{d}V(\boldsymbol{k}) \tag{379}$$

$$= \frac{1}{2\pi^2 r}\int_0^\infty \frac{k\sin(kr)}{k^2 + 1/a^2}\,\mathrm{d}k \tag{380}$$

and is the free space Green's function for 375. The solution to 375 with $\phi(\infty) = 0$ is

$$\boxed{\phi(\boldsymbol{x}) = \int G(\boldsymbol{x} - \boldsymbol{x}')\rho(\boldsymbol{x}')\,\mathrm{d}V(\boldsymbol{x}')} \tag{381}$$

Electric Field:

$$-\boldsymbol{\nabla} G(\boldsymbol{x}) = \frac{(r+a)e^{-r/a}}{4\pi a r^3}\boldsymbol{x} \tag{382}$$

$$\boldsymbol{E}(\boldsymbol{x}) = -\boldsymbol{\nabla}\phi \tag{383}$$

The self energy of a distribution ρ with potential ϕ is

$$E[\rho] = \frac{1}{2}\int \rho(\boldsymbol{x})\phi(\boldsymbol{x})\,\mathrm{d}V(\boldsymbol{x}) \tag{384}$$

$$= \frac{1}{2}\int\int \rho(\boldsymbol{x})\rho(\boldsymbol{x}')G(\boldsymbol{x}-\boldsymbol{x}')\,\mathrm{d}V(\boldsymbol{x})\,\mathrm{d}V(\boldsymbol{x}') \tag{385}$$

D Mechanics

D.1 Nonholonomic constraints

Holonomic constraints restrict the system to the tangent bundle of a submanifold of the configuration space. Nonholonomic constraints also carve out a subset of the tangent bundle, but the piece that is carved out is not necessarily just the tangent bundle of a submanifold.

D.2 Principal of virtual work

Consider a system with internal energy E_{int} in mechanical equilibrium with the surrounding environement. Variation in the system's configuration leads to a change in the internal energy δE_{int} and work by external forces δW. The principle of virtual work states that in equilibrium these two quatities are equal for all virtual displacements.

$$\delta E_{\text{int}} = \delta W. \tag{386}$$

The change in energy of the environment is $\delta E_{\text{env}} = -\delta W$ so we can think of this as saying that in equilibrium variations of the total energy $E_{\text{tot}} = E_{\text{int}} + E_{\text{env}}$ vanish for all virtual displacements.

$$\boxed{\delta E_{\text{tot}} = \delta E_{\text{int}} - \delta W = 0} \tag{387}$$

This principle can be used to find expressions for constraint forces.

D.3 Parameter-invariant actions

We call the action

$$A[\gamma] = \int \mathscr{L}(x^i, \frac{\partial x^i}{\partial \xi}, ..., \frac{\partial^N x^i}{\partial \xi^N}, \xi)\, \mathrm{d}\xi \tag{388}$$

parameter-invariant if it is independent of the choice of parametrization in the following sense: For each curve γ between each pair of points in the configuration space,

$$\int \mathscr{L}(x^i, \frac{\partial x^i}{\partial \xi}, ..., \frac{\partial^N x^i}{\partial \xi^N}, \xi)\, \mathrm{d}\xi = \int \mathscr{L}(x^i, \frac{\partial x^i}{\partial \xi^*}, ..., \frac{\partial^N x^i}{\partial {\xi^*}^N}, \xi^*)\, \mathrm{d}\xi^* \tag{389}$$

where $\xi^* = \xi^*(\xi)$ with $\frac{\partial \xi^*}{\partial \xi} > 0$. For this to hold, $\mathcal{L}$ can have no explicit parameter dependence

$$\mathscr{L} = \mathscr{L}(x^i, \frac{\partial x^i}{\partial \xi}, ..., \frac{\partial^N x^i}{\partial \xi^N}) \tag{390}$$

and it must satisfy

$$\mathscr{L}(x^i, \frac{\partial x^i}{\partial \xi}, ..., \frac{\partial^N x^i}{\partial \xi^N})\lambda = \mathscr{L}(x^i, \lambda\frac{\partial x^i}{\partial \xi}, ..., \lambda^N\frac{\partial^N x^i}{\partial \xi^N}). \tag{391}$$

where everything is evaluated at either ξ or $\xi^*(\xi)$. These conditions are sufficient when the Lagrangian is only a function of the generalized coordinates and their first derivatives (i.e. when $N = 1$). For $N = 1$ the second condition is just the statement that $\mathscr{L}$ is homogeneous of degree 1 in the coordinate velocities.

Proof. Since Eq (389) holds for arbitrary curves between arbitrary points it's equivalent to

$$\mathscr{L}(x^i, \frac{\partial x^i}{\partial \xi}, ..., \frac{\partial^N x^i}{\partial \xi^N}, \xi)\, \mathrm{d}\xi = \mathscr{L}(x^i, \frac{\partial x^i}{\partial \xi^*}, ..., \frac{\partial^N x^i}{\partial {\xi^*}^N}, \xi^*)\, \mathrm{d}\xi^* \tag{392}$$

which 'clearly' prohibits elxplicit parameter dependence. For the $N = 1$ case just apply the chain rule, and for $N > 1$ consider reparametrizations of the form $\xi^*(\xi) = \xi/\lambda$. □

E Tensors and embedded surfaces/curves

Weingarten's formula:

$$\boldsymbol{\nabla}_S \boldsymbol{n} = -\boldsymbol{K} \tag{393}$$

$$\boldsymbol{\nabla}_\alpha N^i = -Z^i_\beta K^\beta_\alpha \tag{394}$$

You can compute the action of a 2nd order symmetric tensor a vector by just forgeting about the fact that the 'coefficients' are really vector-valued and taking a dot product without thinking too hard.

$$\boldsymbol{K} = K_{11}\boldsymbol{d}_1\boldsymbol{d}_1 + K_{12}\left(\boldsymbol{d}_1\boldsymbol{d}_2 + \boldsymbol{d}_2\boldsymbol{d}_1\right) + K_{22}\boldsymbol{d}_2\boldsymbol{d}_2 \tag{395}$$

$$= \left(K_{11}\boldsymbol{d}_1 + K_{12}\boldsymbol{d}_2\right)\boldsymbol{d}_1 + \left(K_{12}\boldsymbol{d}_1 + K_{22}\boldsymbol{d}_2\right)\boldsymbol{d}_2 \tag{396}$$

$$= \boldsymbol{d}_1\left(K_{11}\boldsymbol{d}_1 + K_{12}\boldsymbol{d}_2\right) + \boldsymbol{d}_2\left(K_{12}\boldsymbol{d}_1 + K_{22}\boldsymbol{d}_2\right) \tag{397}$$

The vector-valued coefficients of $\boldsymbol{d}_1, \boldsymbol{d}_2$, when grouped together on either the left or right are the same. This means dotting $\boldsymbol{K}$ on either side by a vector $\boldsymbol{u} = u_1\boldsymbol{d}_1 + u_2\boldsymbol{d}_2$ gives you

$$\boldsymbol{u}\cdot\boldsymbol{K} = \boldsymbol{K}\cdot\boldsymbol{u} = u_1\left(K_{11}\boldsymbol{d}_1 + K_{12}\boldsymbol{d}_2\right) + u_2\left(K_{12}\boldsymbol{d}_1 + K_{22}\boldsymbol{d}_2\right). \tag{398}$$

What about cross-products? Let $\boldsymbol{k}_1 = K_{11}\boldsymbol{d}_1 + K_{12}\boldsymbol{d}_2$, $\boldsymbol{k}_2 = K_{12}\boldsymbol{d}_1 + K_{22}\boldsymbol{d}_2$ so that.

$$\boldsymbol{K} = \boldsymbol{k}_1\boldsymbol{d}_1 + \boldsymbol{k}_2\boldsymbol{d}_2 = \boldsymbol{d}_1\boldsymbol{k}_1 + \boldsymbol{d}_2\boldsymbol{k}_2. \tag{399}$$

$$\hat{\boldsymbol{u}}\cdot\boldsymbol{K} = (\boldsymbol{u}\times\boldsymbol{k}_i)\boldsymbol{d}_i \tag{400}$$

$$\boldsymbol{K}\cdot\hat{\boldsymbol{u}} = -(\hat{\boldsymbol{u}}\cdot\boldsymbol{K})^T = -\boldsymbol{d}_i(\boldsymbol{u}\times\boldsymbol{k}_i) = \boldsymbol{d}_i(\boldsymbol{k}_i\times\boldsymbol{u}) \tag{401}$$

$$\boldsymbol{K} = (\boldsymbol{n}\times\boldsymbol{\omega})\boldsymbol{d}_1 + (\boldsymbol{\omega}_S + 2\overline{\kappa}\boldsymbol{d}_2)\boldsymbol{d}_2 \tag{402}$$

- minimizing the energy $E = \frac{1}{2}\int \partial_\xi\boldsymbol{\gamma}\cdot\partial_\xi\boldsymbol{\gamma}\,\mathrm{d}\xi$ of a curve instead of its' length gets you affine parameterization (parameter proportional to arclength $\xi = cs$) apparently? Something about an einbein?

Let S be a surface in $\mathbb{R}^3$ parameterized by some nice manifold M. Suppose S can be described implicitly as a level set of

$$g : \mathbb{R}^3 \to \mathbb{R} \tag{403}$$

and explicitly by the (isometric) embedding

$$\boldsymbol{h} : M \to S \subset \mathbb{R}^3 \tag{404}$$
$$u \mapsto \boldsymbol{h}(u) \tag{405}$$

$$S = \{\boldsymbol{x} \in \mathbb{R}^3 | g(\boldsymbol{x}) = 0\} = \{\boldsymbol{h}(u) | u \in M\} \tag{406}$$

$$\boldsymbol{n} = \frac{\boldsymbol{\nabla} g}{|\boldsymbol{\nabla} g|}, \quad \boldsymbol{K} = -(\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) \cdot \boldsymbol{\nabla}\boldsymbol{n} = -(\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) \cdot \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{|\boldsymbol{\nabla} g|} \cdot (\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) \tag{407}$$

- The pushforward of the coordinate basis in M by $\boldsymbol{h}$ is just the covariant derivative on the surface..

$$d\boldsymbol{h}_u : T_u M \to T_{\boldsymbol{h}(u)} S \tag{408}$$
$$d\boldsymbol{h}_u(\partial_\alpha) = \boldsymbol{h}_\alpha \cdot \boldsymbol{\nabla} = \boldsymbol{\nabla}_\alpha \tag{409}$$

- The pushforward of the metric tensor in M at u by $\boldsymbol{h}$ is just the 1st fundamental form on S at $\boldsymbol{h}(u)$. Since S is a surface in $\mathbb{R}^3$, this is just the projection onto the tangent space

$$\boldsymbol{P}_S = \boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n} \tag{410}$$

$$\boldsymbol{n} = \frac{\boldsymbol{\nabla} g}{|\boldsymbol{\nabla} g|} \tag{411}$$

$$\boldsymbol{n} = \frac{\boldsymbol{h}_1 \times \boldsymbol{h}_2}{|\boldsymbol{h}_1 \times \boldsymbol{h}_2|} \tag{412}$$

- Mean curvature: $2\overline{\kappa} = \mathrm{tr}(\boldsymbol{K}) = K_i^i = -\boldsymbol{\nabla} \cdot \boldsymbol{n}$
- Gaussian curvature: $\kappa_G = \det(\boldsymbol{K}) = \frac{1}{2}\epsilon^{ij}\epsilon^{kl}K_{ik}K_{jl}$

$$\boldsymbol{K} \cdot \hat{\boldsymbol{n}} \cdot \boldsymbol{K} = \kappa_G \hat{\boldsymbol{n}} \tag{413}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{n} = -\operatorname{tr}(\boldsymbol{K}) = -2\overline{\kappa} \tag{414}$$

Fact 7. *Let $\boldsymbol{r}, \{\boldsymbol{d}_i\}$ be a framed curve with Darboux vector $\boldsymbol{\omega} = \omega_1 \boldsymbol{d}_1 + \omega_2 \boldsymbol{d}_2 + \omega_3 \boldsymbol{d}_3$ which is adapted to S so that $\boldsymbol{d}_3(s) = \boldsymbol{n}(\boldsymbol{r}(s))$.*

(i) $\boldsymbol{r}$ is a geodesic on S iff the geodesic curvature ω_3 is zero. Geodesics really are the straightest possible curves since the are the ones which don't rotate about the surface normal.

(i) *Proof.* We want to minimize the functional

$$L[\boldsymbol{\gamma}] = \int_0^1 \sqrt{\partial_\xi \boldsymbol{\gamma} \cdot \partial_\xi \boldsymbol{\gamma}} \, \mathrm{d}\xi$$

over curves between fixed points $\boldsymbol{\gamma}(0) = \boldsymbol{r}(0), \boldsymbol{\gamma}(1) = \boldsymbol{r}(L) \in \mathbb{R}^3$ subject to the constraint $g(\boldsymbol{\gamma}(\xi)) = 0$ so introduce a Lagrange multiplier!

$$L_S[\boldsymbol{\gamma}, \lambda] = \int_0^1 \sqrt{\partial_\xi \boldsymbol{\gamma} \cdot \partial_\xi \boldsymbol{\gamma}} - \lambda g(\boldsymbol{\gamma}) \, \mathrm{d}\xi$$

$$\begin{aligned} \delta L_S &= \int \frac{\partial_\xi \boldsymbol{\gamma} \cdot \partial_\xi \delta \boldsymbol{\gamma}}{\sqrt{\partial_\xi \boldsymbol{\gamma} \cdot \partial_\xi \boldsymbol{\gamma}}} - \lambda \delta \boldsymbol{\gamma} \cdot \boldsymbol{\nabla} g(\boldsymbol{\gamma}) \, \mathrm{d}\xi \\ &= -\int \left[\partial_\xi \left(\frac{\partial_\xi \boldsymbol{\gamma}}{\sqrt{\partial_\xi \boldsymbol{\gamma} \cdot \partial_\xi \boldsymbol{\gamma}}} \right) + \lambda \boldsymbol{\nabla} g(\boldsymbol{\gamma}) \right] \cdot \delta \boldsymbol{\gamma} \, \mathrm{d}\xi \end{aligned}$$

$$\Rightarrow \partial_\xi \left(\frac{\partial_\xi \boldsymbol{\gamma}}{\sqrt{\partial_\xi \boldsymbol{\gamma} \cdot \partial_\xi \boldsymbol{\gamma}}} \right) + \lambda \boldsymbol{\nabla} g(\boldsymbol{\gamma}) = 0$$

The stuff in parenthesis is just the derivative of the unit tangent so reparametrize by arclength to get

$$\frac{\mathrm{d}s}{\mathrm{d}\xi} \boldsymbol{\gamma}'' + \lambda \boldsymbol{\nabla} g(\boldsymbol{\gamma}) = 0.$$

Since we know the curve is in the surface we can introduce an adapted Darboux frame $\boldsymbol{d}_1 = \gamma', \boldsymbol{d_2} = \boldsymbol{n} \times \boldsymbol{d}_1, \boldsymbol{d}_3 = \boldsymbol{n}$ and the geodesic equation can be written as

$$\boldsymbol{\omega} \times \boldsymbol{d}_1 + \tilde{\lambda} \boldsymbol{d}_3 = 0$$

where $\tilde{\lambda} = \frac{|\boldsymbol{\nabla} g|}{\frac{\mathrm{d}s}{\mathrm{d}\xi}} \lambda$ and we used $\boldsymbol{\gamma}'' = \boldsymbol{d}_1' = \boldsymbol{\omega} \times \boldsymbol{d}_1$. We conclude that the rescaled Lagrange multiplier is determined by the surface curvature and that the geodesic curvature is zero.

$$\tilde{\lambda} = \omega_2 \qquad\qquad \omega_3 = 0$$

□

E.1 Closest point to an implicit surface

Let S_ξ be the level set of a function $g : \mathbb{R}^3 \to \mathbb{R}$ with $g(\boldsymbol{x}) = \xi$

$$S_\xi = \left\{ \boldsymbol{x} \in \mathbb{R}^3 \middle| \, g(\boldsymbol{x}) = \xi \right\} = g^{-1}(\xi), \tag{415}$$

and let $\boldsymbol{n}$ denote the unit normal to the level sets of g. Call the point $\boldsymbol{x}$ outside S_ξ when $g(\boldsymbol{x}) > \xi$.

$$\boldsymbol{n}(\boldsymbol{x}) = \frac{\boldsymbol{\nabla} g(\boldsymbol{x})}{|\boldsymbol{\nabla} g(\boldsymbol{x})|} \tag{416}$$

Let $d_\xi(\boldsymbol{x})$ be the distance between $\boldsymbol{x}$ and the closest point on S_ξ to $\boldsymbol{x}$.

$$d_\xi(\boldsymbol{x}) = \inf_{\boldsymbol{y} \in S_\xi} |\boldsymbol{x} - \boldsymbol{y}| \tag{417}$$

Define $cp_\xi(\boldsymbol{x})$ to be the (maybe not unique) closest point on S_ξ to $\boldsymbol{x}$.

$$d_\xi(\boldsymbol{x}) = |\boldsymbol{x} - cp_\xi(\boldsymbol{x})| \tag{418}$$

Theorem 1. *Let* $\boldsymbol{x}^* = cp_\xi(\boldsymbol{x})$.

$$\begin{aligned} \boldsymbol{x} - \boldsymbol{x}^* &= sgn(g(\boldsymbol{x}) - \xi)|\boldsymbol{x} - \boldsymbol{x}^*|\boldsymbol{n}(\boldsymbol{x}^*) && (419) \\ &= (\boldsymbol{x} - \boldsymbol{x}^*) \cdot \boldsymbol{n}(\boldsymbol{x}^*)\boldsymbol{n}(\boldsymbol{x}^*) && (420) \\ &= \pm|\boldsymbol{x} - \boldsymbol{x}^*|\boldsymbol{n}(\boldsymbol{x}^*) && (421) \end{aligned}$$

Proof. Fix $\boldsymbol{x}$ and minimize $|\boldsymbol{x}-\boldsymbol{y}|$ subject to the constraint $g(\boldsymbol{y})=\xi$.

$$f(\boldsymbol{y},\lambda)=|\boldsymbol{x}-\boldsymbol{y}|+\lambda(g(\boldsymbol{y})-\xi)$$

$$\begin{aligned}\delta f &= -\frac{(\boldsymbol{x}-\boldsymbol{y})}{|\boldsymbol{x}-\boldsymbol{y}|}\cdot\delta\boldsymbol{y}+\lambda\boldsymbol{\nabla}g(\boldsymbol{y})\cdot\delta\boldsymbol{y}\\ \Rightarrow\frac{(\boldsymbol{x}-\boldsymbol{y})}{|\boldsymbol{x}-\boldsymbol{y}|} &= \lambda\boldsymbol{\nabla}g(\boldsymbol{y})\\ \Rightarrow\frac{(\boldsymbol{x}-\boldsymbol{y})}{|\boldsymbol{x}-\boldsymbol{y}|} &= \lambda|\boldsymbol{\nabla}g|\boldsymbol{n}\end{aligned}$$

Since the left hand side is a unit vector, $|\lambda|=\frac{1}{|\boldsymbol{\nabla}g(\boldsymbol{y})|}$. If the point $\boldsymbol{x}$ is outside the surface $g(\boldsymbol{x})-\xi>0$ and $\boldsymbol{n}\cdot(\boldsymbol{x}-\boldsymbol{y})>0$. If the point $\boldsymbol{x}$ is inside the surface $g(\boldsymbol{x})-\xi<0$ and $\boldsymbol{n}\cdot(\boldsymbol{x}-\boldsymbol{y})<0$. □

Theorem 2. *Let $\boldsymbol{x}_1\in S_{\xi_1}$ and $\boldsymbol{x}_2\in S_{\xi_2}$. Then, the following are equivalent (locally):*

(i) $cp_{\xi_1}(\boldsymbol{x}_2)=\boldsymbol{x}_1$ *and* $cp_{\xi_2}(\boldsymbol{x}_1)=\boldsymbol{x}_2$

(ii) $d_{\xi_1}(\boldsymbol{x}_2)=d_{\xi_2}(\boldsymbol{x}_1)$

(iii) $\boldsymbol{n}(\boldsymbol{x}_2)=\boldsymbol{n}(\boldsymbol{x}_1)$

Theorems are great and all, but let's say you have the implicit function g, some fixed point $\boldsymbol{r}\in\mathbb{R}^3$, and you want to actually compute $\boldsymbol{z}^*=cp_{\xi^*}(\boldsymbol{r})$, the closest point to $\boldsymbol{r}$ on the surface $S_{\xi^*}=g^{-1}(\xi^*)$. You can find $\boldsymbol{z}^*$ by computing zeros of the function

$$\boldsymbol{F}(\boldsymbol{x})=\boldsymbol{P}(\boldsymbol{x})\cdot(\boldsymbol{r}-\boldsymbol{x})-\frac{g(\boldsymbol{x})-\xi^*}{|\boldsymbol{\nabla}g(\boldsymbol{x})|}\boldsymbol{n}(\boldsymbol{x}) \tag{422}$$

where $P(\boldsymbol{x})=\boldsymbol{\delta}-\boldsymbol{n}(\boldsymbol{x})\boldsymbol{n}(\boldsymbol{x})$ is the projection onto $T_{\boldsymbol{x}}S_{g(\boldsymbol{x})}$, the tangent space of $S_{g(\boldsymbol{x})}=g^{-1}(\boldsymbol{x})$ at the point $\boldsymbol{x}$. Since the two terms in $\boldsymbol{F}(\boldsymbol{x})$ are orthogonal, $\boldsymbol{x}$ is zero iff

$$\boldsymbol{P}(\boldsymbol{x})\cdot(\boldsymbol{r}-\boldsymbol{x})=0 \qquad \text{and} \qquad g(\boldsymbol{x})=\xi^* \tag{423}$$

which are the two conditions required for $\boldsymbol{x}=\boldsymbol{z}^*=cp_{\xi^*}(\boldsymbol{r})$. You can find the closest point by iterating $\boldsymbol{F}$ or using Newton's method:

$$\boldsymbol{F}(\boldsymbol{x}_m)+(\boldsymbol{x}_{m+1}-\boldsymbol{x}_m)\cdot\boldsymbol{\nabla}\boldsymbol{F}(\boldsymbol{x}_m)=0. \tag{424}$$

E.2 Closest point flow

Fix a point $\boldsymbol{r}_0 \in \mathbb{R}^3$ and let $\boldsymbol{r}(\xi)$ be the closest point to $\boldsymbol{r}_0$ on the surface $g^{-1}(\xi)$. Then $\boldsymbol{r}$ satisfies

$$g(\boldsymbol{r}) = \xi \tag{425}$$

$$\boldsymbol{r} - \boldsymbol{r}_0 = \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0)\, \boldsymbol{n}. \tag{426}$$

Differentiating (425) and (426) with respect to ξ gives

$$\partial_\xi \boldsymbol{r} \cdot \boldsymbol{\nabla} g = 1 \tag{427}$$

$$\partial_\xi \boldsymbol{r} = (\partial_\xi \boldsymbol{r} \cdot \boldsymbol{\nabla} \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0) + \boldsymbol{n} \cdot \partial_\xi \boldsymbol{r})\, \boldsymbol{n} + \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0)\, \partial_\xi \boldsymbol{r} \cdot \boldsymbol{\nabla} \boldsymbol{n} \tag{428}$$

$$= (\boldsymbol{n} \cdot \partial_\xi \boldsymbol{r})\, \boldsymbol{n} + \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0)\, \partial_\xi \boldsymbol{r} \cdot \boldsymbol{\nabla} \boldsymbol{n} \qquad \text{using: } \boldsymbol{\nabla} \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0) = 0 \tag{429}$$

$$= \frac{1}{|\boldsymbol{\nabla} g|} \boldsymbol{n} + \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0)\, \partial_\xi \boldsymbol{r} \cdot \boldsymbol{\nabla} \boldsymbol{n} \qquad \text{using: } \partial_\xi \boldsymbol{r} \cdot \boldsymbol{\nabla} g = 1 \tag{430}$$

Let $\delta r := \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0)$ and $P = \boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}$.

$$\partial_\xi \boldsymbol{r} = \frac{1}{|\boldsymbol{\nabla} g|} \boldsymbol{n} + \frac{\delta r}{|\boldsymbol{\nabla} g|} \boldsymbol{P} \cdot \boldsymbol{\nabla} \boldsymbol{\nabla} g \cdot \partial_\xi \boldsymbol{r} \tag{431}$$

$$\Rightarrow \left(\boldsymbol{\delta} - \frac{\delta r}{|\boldsymbol{\nabla} g|} \boldsymbol{P} \cdot \boldsymbol{\nabla} \boldsymbol{\nabla} g \right) \cdot \partial_\xi \boldsymbol{r} = \frac{1}{|\boldsymbol{\nabla} g|} \boldsymbol{n} \tag{432}$$

$$\Rightarrow \partial_\xi \boldsymbol{r} = \frac{1}{|\boldsymbol{\nabla} g|} \left(\boldsymbol{\delta} - \frac{\delta r}{|\boldsymbol{\nabla} g|} \boldsymbol{P} \cdot \boldsymbol{\nabla} \boldsymbol{\nabla} g \right)^{-1} \cdot \boldsymbol{n} \tag{433}$$

Turns out, to compute the action of the action of the nasty inverse on $\boldsymbol{n}$ you can get away with only inverting something that is rank 2.

$$\left(\boldsymbol{\delta} - \frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\right)^{-1} = \sum_{k=0}^{\infty}\left(\frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\right)^{k} \tag{434}$$

$$= \boldsymbol{\delta} + \sum_{k=1}^{\infty}\left(\frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\right)^{k-1}\frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g \tag{435}$$

$$= \boldsymbol{\delta} + \sum_{k=1}^{\infty}\left(\frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\cdot\boldsymbol{P}\right)^{k-1}\frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g \tag{436}$$

The projection operators let you replace most of the $\boldsymbol{\nabla}\boldsymbol{\nabla} g$'s with the second fundamental form

$$\boldsymbol{K} = -\frac{1}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\cdot\boldsymbol{P}. \tag{437}$$

$$\left(\boldsymbol{\delta} - \frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\right)^{-1} = \boldsymbol{\delta} + \frac{\delta r}{|\boldsymbol{\nabla} g|}\sum_{k=1}^{\infty}\left(-\delta r\boldsymbol{K}\right)^{k-1}\cdot\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g \tag{438}$$

$$= \boldsymbol{\delta} + \frac{\delta r}{|\boldsymbol{\nabla} g|}\sum_{k=1}^{\infty}\left(-\delta r\boldsymbol{K}\right)^{k-1}\cdot\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g \tag{439}$$

$$= \boldsymbol{\delta} + \frac{\delta r}{|\boldsymbol{\nabla} g|}\sum_{k=0}^{\infty}\left(-\delta r\boldsymbol{K}\right)^{k}\cdot\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g \tag{440}$$

$$= \boldsymbol{\delta} + \frac{\delta r}{|\boldsymbol{\nabla} g|}\left(\boldsymbol{P} + \delta r\boldsymbol{K}\right)^{-1}\cdot\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g \tag{441}$$

Where the inverse of the operator $\boldsymbol{P} + \delta r\boldsymbol{K}$ is computed in $T_{\boldsymbol{r}}S_\xi$, the (2-dimensional) tangent space of $S_\xi = g^{-1}(\xi)$ at $\boldsymbol{r}$. Since $(\boldsymbol{P} + \delta r\boldsymbol{K})^{-1}$ is already a projection, you can even get rid of that last $\boldsymbol{P}$ multiplying $\boldsymbol{\nabla}\boldsymbol{\nabla} g$.

$$\left(\boldsymbol{\delta} - \frac{\delta r}{|\boldsymbol{\nabla} g|}\boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\right)^{-1} = \boldsymbol{\delta} + \frac{\delta r}{|\boldsymbol{\nabla} g|}\left(\boldsymbol{P} + \delta r\boldsymbol{K}\right)^{-1}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g \tag{442}$$

All that means that $\boldsymbol{r}$ satisfies

$$\boxed{\partial_\xi\boldsymbol{r} = \frac{1}{|\boldsymbol{\nabla} g|}\left(\boldsymbol{\delta} + \frac{\delta r}{|\boldsymbol{\nabla} g|}\left(\boldsymbol{P} + \delta r\boldsymbol{K}\right)^{-1}\cdot\boldsymbol{\nabla}\boldsymbol{\nabla} g\right)\cdot\boldsymbol{n}} \tag{443}$$

$$\boldsymbol{r}(\xi_0) = \boldsymbol{r}_0, \qquad \delta r = \boldsymbol{n} \cdot (\boldsymbol{r} - \boldsymbol{r}_0) \tag{444}$$

and the closest point to $\boldsymbol{r}_0$ on any level set $S_\xi = g^{-1}(\xi)$ is

$$cp_\xi(\boldsymbol{r}_0) = \boldsymbol{r}(\xi). \tag{445}$$

If you don't care about inverting the rank 3 operator its' easier to just use

$$\partial_\xi \boldsymbol{r} = \frac{1}{|\boldsymbol{\nabla} g|} \left(\boldsymbol{\delta} - \frac{\delta r}{|\boldsymbol{\nabla} g|} \boldsymbol{P} \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g \right)^{-1} \cdot \boldsymbol{n} \tag{446}$$

E.3 Curves adapted to surfaces

Let's just imagine that $(\boldsymbol{r}, \{\boldsymbol{d}_i\})$ is a framed curve with Darboux vector $\boldsymbol{\omega}$ which is adapted to a surface $S = \{\boldsymbol{x} \in \mathbb{R}^3 | g(\boldsymbol{x}) = 0\} = \{\boldsymbol{h}(u) | u \in P\}$ with curvature $\boldsymbol{K}$ and normal $\boldsymbol{n} = \boldsymbol{d}_3$. Let $\boldsymbol{\omega}_S = \omega_1 \boldsymbol{d}_1 + \omega_2 \boldsymbol{d}_2$ be the projection of $\boldsymbol{\omega}$ onto the surface.

$$\begin{aligned} \boldsymbol{K} &= K_{11} \boldsymbol{d}_1 \boldsymbol{d}_1 + K_{12} \left(\boldsymbol{d}_1 \boldsymbol{d}_2 + \boldsymbol{d}_2 \boldsymbol{d}_1 \right) + K_{22} \boldsymbol{d}_2 \boldsymbol{d}_2 && (447) \\ &= \left(\boldsymbol{n} \times \boldsymbol{\omega} \right) \boldsymbol{d}_1 + \boldsymbol{\omega}_S \boldsymbol{d}_2 + 2\overline{\kappa} \boldsymbol{d}_2 \boldsymbol{d}_2 && (448) \\ &= \boldsymbol{d}_1 \left(\boldsymbol{n} \times \boldsymbol{\omega} \right) + \boldsymbol{d}_2 \boldsymbol{\omega}_S + 2\overline{\kappa} \boldsymbol{d}_2 \boldsymbol{d}_2 && (449) \end{aligned}$$

$$\begin{aligned} \boldsymbol{K}^{-1} &= \frac{1}{\kappa_G} \left[2\overline{\kappa} \boldsymbol{d}_1 \boldsymbol{d}_1 - \left(\boldsymbol{n} \times \boldsymbol{\omega} \right) \boldsymbol{d}_1 - \boldsymbol{\omega}_S \boldsymbol{d}_2 \right] && (450) \\ &= \frac{1}{\kappa_G} \left[2\overline{\kappa} \boldsymbol{d}_1 \boldsymbol{d}_1 - \boldsymbol{d}_1 \left(\boldsymbol{n} \times \boldsymbol{\omega} \right) - \boldsymbol{d}_2 \boldsymbol{\omega}_S \right] && (451) \end{aligned}$$

- Mean curvature: $2\overline{\kappa} = -\omega_2 + K_{22}$
- Gaussian curvature: $\kappa_G = -(\omega_1^2 + \omega_2 K_{22}) = -\boldsymbol{\omega} \cdot \boldsymbol{K} \cdot \boldsymbol{d}_2$

Implicit/explicit surface variations

Let $\Gamma : S \to S'$ be a diffeomorphism mapping a surface S defined implicitly by

$$g(\boldsymbol{x}) = 0 \tag{452}$$

to a new surface $\tilde{S}$ defined by

$$\tilde{g}(\boldsymbol{x}) = 0. \tag{453}$$

The implicit functions must be related by

$$\tilde{g}(\boldsymbol{x}) = g(\Gamma^{-1}(\boldsymbol{x})). \tag{454}$$

Now suppose you have a family of deformed surfaces $\tilde{S}_\epsilon$ and diffeomorphisms Γ_ϵ indexed by a smooth paramter ϵ. Denote the inverse transformation by

$$\boldsymbol{q}(\boldsymbol{x}) = \Gamma_\epsilon^{-1}(\boldsymbol{x}). \tag{455}$$

Suppose that the transformed surface coincides with the original when $\epsilon = 0$.

$$\boldsymbol{q}(\boldsymbol{x}) = \boldsymbol{x} - \epsilon \boldsymbol{J}(\boldsymbol{x}) + O(\epsilon^2) \tag{456}$$

where $\boldsymbol{J}(\boldsymbol{x})$ is called the generator of the point transformation $\Gamma_\epsilon(\boldsymbol{x})$.

$$\tilde{g}(\boldsymbol{x}) = g(\boldsymbol{q}(\boldsymbol{x})) = g(\boldsymbol{x}) - \epsilon \boldsymbol{J}(\boldsymbol{x}) \cdot \boldsymbol{\nabla} g(\boldsymbol{x}) + O(\epsilon^2) \tag{457}$$

$$\boldsymbol{\nabla}\tilde{g}(\boldsymbol{x}) = \boldsymbol{\nabla}\boldsymbol{q}(\boldsymbol{x}) \cdot \boldsymbol{\nabla}_q g(\boldsymbol{q}(\boldsymbol{x})) = \boldsymbol{\nabla} g(\boldsymbol{x}) - \epsilon \left(\boldsymbol{\nabla}\boldsymbol{J}(\boldsymbol{x}) \cdot \boldsymbol{\nabla} g(\boldsymbol{x}) + \boldsymbol{J}(\boldsymbol{x}) \cdot \boldsymbol{\nabla}\boldsymbol{\nabla} g(\boldsymbol{x})\right) + O(\epsilon^2) \tag{458}$$

This shows that λ only gives you a force, but μ gives both force and moment.

If you describe the same surface explicitly by

$$\boldsymbol{h}(u^1, u^2) \tag{459}$$

then the action of Γ in this representation is given by

$$\Gamma \cdot \boldsymbol{h}(u) = \boldsymbol{h}(u) + \epsilon \boldsymbol{J}(\boldsymbol{h}(u)) + O(\epsilon^2), \tag{460}$$

so, the variation of th eexplicit surface is just

$$\delta \boldsymbol{h}(u) = \boldsymbol{J}(\boldsymbol{h}(u)) \tag{461}$$

Note that we are assuming the surface coordinates are 'material' labels not affected by the transformation.

E.4 Tensor calculus

- ambient position vector $\boldsymbol{g} = \boldsymbol{g}(x^i)$, coordinates x^i, coordinate basis $\boldsymbol{g}_i = \frac{\partial \boldsymbol{g}}{\partial x^i}$, and metric $g_{ij} = \boldsymbol{g}_i \cdot \boldsymbol{g}_j$
  - may write $\partial_i = \frac{\partial}{\partial x^i}$
  - Latin letters a, b, i, j, etc... used for ambient indices and range from 1 to 3.
- surface embedding $\boldsymbol{g}(x^i(u^\alpha)) = \boldsymbol{h}(u^1, u^2)$, coordinate basis $\boldsymbol{h}_\alpha = \frac{\partial \boldsymbol{h}}{\partial u^\alpha}$, and metric $h_{\mu\nu} = \boldsymbol{h}_i \cdot \boldsymbol{h}_j$
  - shift tensor $x^i_\mu = \frac{\partial x^i}{\partial u^\mu}$ used to juggle ambient/surface indices and whatnot
  - may also write: $\partial_\mu = \frac{\partial}{\partial u^\mu}$, $\boldsymbol{h}_\alpha = \mathrm{d}\boldsymbol{y}\,(\partial_\alpha)$
  - Greek letters α, β, μ, ν, etc... used for surface indices and range from 1 to 2.
- surface normal $\boldsymbol{n} = \boldsymbol{h}_3 = \frac{\boldsymbol{h}_1 \times \boldsymbol{h}_2}{\|\boldsymbol{h}_1 \times \boldsymbol{h}_2\|}$ and surface adapted orthonormal frame $\boldsymbol{e}_1 = \boldsymbol{h}_1/\|\boldsymbol{h}_1\|$, $\boldsymbol{e}_2 = \boldsymbol{n} \times \boldsymbol{e}_1$, $\boldsymbol{e}_3 = \boldsymbol{n}$ surface adapted orthonormal frame
- curve embedding $\boldsymbol{r}(s) = \boldsymbol{h}(u^\alpha(s)) = \boldsymbol{g}(x^i(s))$ and orthonormal Darboux frame $\boldsymbol{d}_{(i)}$ with $\boldsymbol{d}_{(3)} = \boldsymbol{n}$ and $\boldsymbol{d}_1 = \boldsymbol{r}'(s)$
  - let $d^j_{(i)} = \boldsymbol{g}^j \cdot \boldsymbol{d}_{(i)}$, $d^{(i)}_j = \boldsymbol{d}^{(i)} \cdot \boldsymbol{g}_j$ and note: $d^j_{(i)} d^{(i)}_k = \delta^j_k$, $d^k_{(i)} d^{(n)}_k = \delta^{(n)}_{(i)}$
  - similar relationship holds for surface components of $\boldsymbol{d}_{(\alpha)}$
  - can juggle $(i), (j)$,... with $\delta_{(ij)}$

Orthogonal transformations and orthonormal bases

Let $\{\boldsymbol{e}_i\}$ and $\{\boldsymbol{E}_i\}$ be orthonormal bases for $\mathbb{E}^3$ related by the rotation R.

$$\boldsymbol{e}_i = R \cdot \boldsymbol{E}_i = R^j_i \boldsymbol{E}_j \qquad \boldsymbol{e}^i = \boldsymbol{E}^i \cdot R^T = \left(R^T\right)^i_j \boldsymbol{E}^j \tag{462}$$

$$\boldsymbol{E}_i = R^T \cdot \boldsymbol{e}_i = \left(R^T\right)^j_i \boldsymbol{e}_j \qquad \boldsymbol{E}^i = \boldsymbol{e}^i \cdot R = R^i_j \boldsymbol{e}^j \tag{463}$$

The induced coordinate change $x^i = x^i(X)$ has a Jacobian matrix whose components equal the components of R^T.

$$\boldsymbol{E}_j = \frac{\partial \boldsymbol{r}}{\partial X^j} = \frac{\partial x^i}{\partial X^j}\frac{\partial \boldsymbol{r}}{\partial x^i} = \frac{\partial x^i}{\partial X^j}\boldsymbol{e}_i \tag{464}$$

$$\frac{\partial x^i}{\partial X^j} = \boldsymbol{e}^i \cdot \boldsymbol{E}_j = \boldsymbol{e}^i \cdot R^T \cdot \boldsymbol{e}_j \tag{465}$$

When transforming $\boldsymbol{E}$ components of tensors into $\boldsymbol{e}$ components, covariant (lower) indices are contracted with R^i_j and contravariant (upper) components are contracted with $\left(R^T\right)^i_{\ j}$. The opposite is true for the inverse tranformation.

$$\boldsymbol{T} = t^{ij\cdot\cdot}_{\cdot\cdot kl}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}^k \otimes \boldsymbol{e}^l = T^{ab\cdot\cdot}_{\cdot\cdot cd}\boldsymbol{E}_a \otimes \boldsymbol{E}_b \otimes \boldsymbol{E}^c \otimes \boldsymbol{E}^d \tag{466}$$

$$\tag{467}$$

$$t^{ij\cdot\cdot}_{\cdot\cdot kl} = T^{ab\cdot\cdot}_{\cdot\cdot cd}\left(R^T\right)^i_{\ a}\left(R^T\right)^j_{\ b}R^c_k R^d_l \qquad T^{ab\cdot\cdot}_{\cdot\cdot cd} = t^{ij\cdot\cdot}_{\cdot\cdot kl}R^a_i R^b_j\left(R^T\right)^k_{\ c}\left(R^T\right)^l_{\ d} \tag{468}$$

Covariant cross products

Use e to denote the permutation symbol, ε to denote the Levi-Civita tensor. In two and three dimensions we have

$$e_{\mu\nu} = e^{\mu\nu} = \begin{cases} 1 & \text{If } \mu\nu = 12 \\ -1 & \text{If } \mu\nu = 21 \\ 0 & \text{otherwise} \end{cases} \qquad \varepsilon_{\mu\nu} = \sqrt{h}e_{\mu\nu} \qquad \varepsilon^{\mu\nu} = \frac{1}{\sqrt{h}}e^{\mu\nu} \tag{469}$$

and

$$e_{ijk} = e^{ijk} = \begin{cases} 1 & \text{If } ijk \text{ is an even permutation of } 123 \\ -1 & \text{If } ijk \text{ is an odd permutation of } 123 \\ 0 & \text{otherwise} \end{cases} \qquad \varepsilon_{ijk} = \sqrt{g}e_{ijk} \quad \varepsilon^{ijk} = \frac{1}{\sqrt{g}}e^{ijk} \tag{470}$$

where $h = \det\left(\begin{smallmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{smallmatrix}\right)$ and $g = \det\left(\begin{smallmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{smallmatrix}\right)$ are the determinants of the ambient and surface metric component matrices. Cross products of the coordinate basis can be written

$$\boldsymbol{g}_i \times \boldsymbol{g}_j = \varepsilon_{ijk}\boldsymbol{g}^k \qquad\qquad \boldsymbol{g}^i \times \boldsymbol{g}^j = \varepsilon^{ijk}\boldsymbol{g}_k \tag{471}$$

and components of the surface normal are given by

$$\boldsymbol{n} = \frac{1}{\sqrt{h}}\boldsymbol{h}_1 \times \boldsymbol{h}_2 = \frac{1}{2}\varepsilon_{\alpha\beta}\varepsilon^{ijk}x_i^\alpha x_j^\beta \boldsymbol{g}_k \tag{472}$$

Covariant differentiation and surface curvature
In a curvelinear coordinate system we have

$$\partial_i \boldsymbol{g}_i = \Gamma_{ij}^k \boldsymbol{g}_k \qquad \partial_i \boldsymbol{g}^j = -\Gamma_{ik}^j \boldsymbol{g}^k \qquad \Gamma_{ij}^k = \Gamma_{ji}^k \tag{473}$$

$$\Gamma_{ab}^n = \frac{1}{2}g^{nc}\left(\partial_a g_{bc} + \partial_b g_{ca} - \partial_c g_{ab}\right) \tag{474}$$

The (ambient) covariant derivative $\boldsymbol{\nabla}_k$ acting on components gives components of the partial derivative of the corresponding invariant. For example,

$$\boldsymbol{\nabla}_j v^i = \boldsymbol{g}^i \cdot \partial_j \boldsymbol{v} = \partial_j v^i + \Gamma_{jk}^i v^k \quad \boldsymbol{\nabla}_j v_i = \boldsymbol{g}_i \cdot \partial_j \boldsymbol{v} = \partial_j v_i - \Gamma_{ji}^k v_k \tag{475}$$

The corresponding surface quantities are slightly more complicated since derivatives of basis vectors may have normal components. The rates of change of the basis in the normal direction is captured by the second fundamental form K.

$$\partial_\beta \boldsymbol{h}_\gamma = \Gamma_{\beta\gamma}^\alpha \boldsymbol{h}_\alpha + K_{\beta\gamma}\boldsymbol{n} \quad \boldsymbol{\nabla}_\beta \boldsymbol{h}_\gamma = K_{\beta\gamma}\boldsymbol{n} \quad \partial_\beta \boldsymbol{n} = \boldsymbol{\nabla}_\beta \boldsymbol{n} = -K_\beta^\alpha \boldsymbol{h}_\alpha \tag{476}$$

Eigenvalues of the matrix K_α^β are principal curvatures of the surface.

Curves as embedded submanifolds

- curve embedding $\boldsymbol{r}(t^\Theta) = \boldsymbol{g}(x^i(t))$, coordinate t^Θ $(\Theta = 1)$
- coordinate basis $\boldsymbol{r}_\Theta = \partial_\Theta \boldsymbol{r}$, and metric $r_{\Theta\Phi} = \boldsymbol{r}_\Theta \cdot \boldsymbol{r}_\Phi$
  - $\boldsymbol{r}_\Theta = s'\boldsymbol{d}_{(1)}$ and $\boldsymbol{r}^\Theta = \frac{1}{s'}\boldsymbol{d}_{(1)}$, $\boldsymbol{d}_{(1)} = \varepsilon^\Theta \boldsymbol{r}_\Theta$
- ambient/curve shift tensor $x_\Theta^i = \frac{\partial x^i}{\partial t^\Theta}$
  - $r_{\Theta\Phi} = x_\Theta^i x_\Phi^j g_{ij}$
- $s' = \sqrt{\det(r_{\Theta\Phi})}$ is the length element

- Levi-Civita symbols $\varepsilon_\Theta = d^{(1)}_\Theta = s'\ \varepsilon^\Theta = d^\Theta_{(1)} = 1/s'$ used to build invariants
- $d^i_{(1)} = \varepsilon^\Theta x^i_\Theta$

- Christoffel $\Gamma^\Theta_{\Phi\Psi} = \boldsymbol{r}^\Theta \cdot \partial_\Phi \boldsymbol{r}_\Psi = -\partial_\Phi \boldsymbol{r}^\Theta \cdot \boldsymbol{r}_\Psi = s''/s'$

$\boldsymbol{\nabla}_\Theta \boldsymbol{r}_\Phi = \partial_\Theta \boldsymbol{r}_\Phi - \Gamma^\Psi_{\Theta\Phi} \boldsymbol{r}_\Psi = \boldsymbol{\omega}_\Theta \times \boldsymbol{r}_\Phi$

Define $\boldsymbol{\nabla}_\Psi x^\alpha_\Phi =: k_{\Phi\Psi} d^\alpha_{(2)}$ so $\boldsymbol{h}_\alpha \boldsymbol{\nabla}_\Psi x^\alpha_\Phi =: k_{\Phi\Psi} \boldsymbol{d}_{(2)}$. The invariant $k^\Phi_\Phi = k_{(11)}$ is called the geodesic curvature.

$$\boldsymbol{\nabla}_s \boldsymbol{d}_{(1)} = \Omega^{(3)} \boldsymbol{d}_{(2)} - \Omega^{(2)} \boldsymbol{d}_{(3)} \quad \boldsymbol{\nabla}_s \boldsymbol{d}_{(2)} = \Omega^{(1)} \boldsymbol{d}_{(3)} - \Omega^{(3)} \boldsymbol{d}_{(1)} \quad \boldsymbol{\nabla}_s \boldsymbol{d}_{(3)} = \Omega^{(2)} \boldsymbol{d}_{(1)} - \Omega^{(1)} \boldsymbol{d}_{(2)} \tag{477}$$

The first two components of the Darboux vector $\Omega^{(1)}$ and $\Omega^{(2)}$ depend on how the surface is embedded in ambient space, but the third component $\Omega^{(3)}$ is independent of extrinsic surface curvatures.

$$\Omega^{(1)} = K_{\alpha\beta} d^\alpha_{(1)} d^\beta_{(2)} \quad \Omega^{(2)} = -K_{\alpha\beta} d^\alpha_{(1)} d^\beta_{(1)} \quad \Omega^{(3)} = d^{(2)}_\alpha \boldsymbol{\nabla}_s d^\alpha_{(1)} = k^\Theta_\Theta = k_{(11)} \tag{478}$$

Proof. $\Omega^{(1)} = -\boldsymbol{d}_{(2)} \cdot (\boldsymbol{\nabla}_s \boldsymbol{d}_{(3)}) = -\boldsymbol{d}_{(2)} \cdot (d^\alpha_{(1)} \boldsymbol{\nabla}_\alpha \boldsymbol{n}) = K^\beta_\alpha d^\alpha_{(1)} d^{(2)}_\beta$

$\boldsymbol{\nabla}_\Psi \boldsymbol{r}_\Phi = \boldsymbol{h}_\alpha \boldsymbol{\nabla}_\Psi x^\alpha_\Phi + x^\alpha_\Phi x^\beta_\Psi \boldsymbol{\nabla}_\beta \boldsymbol{h}_\alpha = k_{\Phi\Psi} \boldsymbol{d}_{(2)} + x^\alpha_\Phi x^\beta_\Psi K_{\alpha\beta} \boldsymbol{d}_{(3)}$, so, in the Darboux basis $\boldsymbol{\nabla}_s \boldsymbol{d}_{(1)} = \boldsymbol{\Omega} \times \boldsymbol{d}_{(1)} = k_{(11)} \boldsymbol{d}_{(2)} + K_{(11)} \boldsymbol{d}_{(3)}$ □

$$\boldsymbol{d}_{(2)} = d^\alpha_{(2)} \boldsymbol{h}_\alpha = \varepsilon^{\alpha\beta} \varepsilon_\Phi x^\Phi_\beta \boldsymbol{h}_\alpha = \varepsilon^{\alpha\beta} d^{(1)}_\beta \boldsymbol{h}_\alpha \tag{479}$$

E.5 Geometry of $SO(3)$

Let θ^α be coordinates for $SO(3)$ and $\partial_\alpha D = \frac{\partial D}{\partial \theta^\alpha} = D_\alpha$ be the associated covariant basis. The covariant basis can be expressed in terms of the $\mathfrak{so}(3)$-valued fields

$$\hat{\boldsymbol{\eta}}_\alpha = D^T D_\alpha \tag{480}$$

$$D_\alpha = D \hat{\boldsymbol{\eta}}_\alpha. \tag{481}$$

The covariant $\hat{\boldsymbol{\eta}}_\alpha$'s can be associated with contravariant counterparts $\hat{\boldsymbol{\eta}}^\alpha$ using the hat map/hodge dual to map everything to $\mathbb{R}^3$ and back. The contravariant vectors in $SO(3)$

$$D^\alpha = D\hat{\boldsymbol{\eta}}^\alpha \tag{482}$$

satisfy

$$\begin{aligned}
\langle D^\alpha, D_\beta \rangle &= \frac{1}{2}\,\mathrm{tr}\Big((D^\alpha)^T\, D_\beta \Big) && (483)\\
&= \frac{1}{2}\,\mathrm{tr}\Big((\boldsymbol{\eta}^\alpha)^T\, D^T D\boldsymbol{\eta}_\beta \Big) && (484)\\
&= \frac{1}{2}\,\mathrm{tr}\Big((\boldsymbol{\eta}^\alpha)^T\, \boldsymbol{\eta}_\beta \Big) && (485)\\
&= \langle \boldsymbol{\eta}^\alpha, \boldsymbol{\eta}_\beta \rangle && (486)\\
&= \delta^\alpha_\beta && (487)
\end{aligned}$$

The metric is given by

$$\eta_{\alpha\beta} = \langle \boldsymbol{\eta}_\alpha, \boldsymbol{\eta}_\beta \rangle \tag{488}$$

In a coordinate basis we need to have

$$\partial_\alpha \partial_\beta D = \partial_\beta \partial_\alpha D \tag{489}$$

which implies that

$$\partial_\alpha \boldsymbol{\eta}_\beta - \partial_\beta \boldsymbol{\eta}_\alpha + [\boldsymbol{\eta}_\alpha, \boldsymbol{\eta}_\beta] = 0 \tag{490}$$

This is just the structural equation for the Maurer-Cartan form in disguise! The Maurer-Cartan form is the $\mathfrak{so}(3)$-valued one form given by

$$\hat{\boldsymbol{\psi}} = \hat{\boldsymbol{\eta}}_\alpha \delta\theta^\alpha \tag{491}$$

$$\delta D = D\hat{\boldsymbol{\eta}}_\alpha \delta\theta^\alpha = D\hat{\boldsymbol{\psi}}. \tag{492}$$

It satisfies the structural equation

$$\delta\hat{\boldsymbol{\psi}} = \partial_\beta \hat{\boldsymbol{\eta}}_\alpha \delta\theta^\beta \wedge \delta\theta^\alpha = -\partial_\alpha \hat{\boldsymbol{\eta}}_\beta \delta\theta^\beta \wedge \delta\theta^\alpha \tag{493}$$

$$\delta\hat{\boldsymbol{\psi}} = \frac{1}{2}\left(\partial_\beta \hat{\boldsymbol{\eta}}_\alpha - \partial_\alpha \hat{\boldsymbol{\eta}}_\beta\right)\delta\theta^\beta \wedge \delta\theta^\alpha \tag{494}$$

$$\tag{495}$$

by the structure equation in disguise, this is

$$\delta\hat{\boldsymbol{\psi}} = \frac{1}{2}[\hat{\boldsymbol{\eta}}_\alpha, \hat{\boldsymbol{\eta}}_\beta]\delta\theta^\beta \wedge \delta\theta^\alpha = \frac{1}{2}\left[\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\psi}}\right] \tag{496}$$

where the first $[,]$ is the normal matrix commutator and the second is the bracket of Lie algebra-valued forms.

Suppose you have a surface U in θ^α space parameterized by real coordinates $(s^1, s^2) = (s, t)$. By restricting the Maurer-Cartan form to S we obtain a field of $\mathfrak{so}(3)$-valued differential forms

$$\hat{\boldsymbol{\psi}} = \hat{\boldsymbol{\eta}}_\alpha \theta^\alpha_{,a} \delta s^a = \hat{\boldsymbol{\eta}}_\alpha \frac{\partial \theta^\alpha}{\partial s^a}\delta s^a, \tag{497}$$

whose components in the (s, t)-coordinate system are the Darboux vector and the angular velocity.

$$\hat{\boldsymbol{\psi}}_\alpha = \hat{\boldsymbol{\eta}}_\alpha \theta^\alpha_{,a} = \begin{cases} \hat{\boldsymbol{\omega}} & \text{, if } a = 1 \\ \hat{\boldsymbol{\Omega}} & \text{, if } a = 2 \end{cases} \tag{498}$$

This means that the Darboux and angular velocity tensors in a body fixed frame are given by

$$\hat{\boldsymbol{\omega}} = \hat{\boldsymbol{\psi}}(\theta'^\alpha \partial_\alpha) = \theta'^\alpha \hat{\boldsymbol{\eta}}^\alpha \tag{499}$$

$$\hat{\boldsymbol{\Omega}} = \hat{\boldsymbol{\psi}}(\dot{\theta}^\alpha \partial_\alpha) = \dot{\theta}^\alpha \hat{\boldsymbol{\eta}}^\alpha \tag{500}$$

You can do something similar for the position vector to get the

F Various useful things

$$\omega_1 = \boldsymbol{r}' \cdot \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{|\boldsymbol{\nabla} g|^2} \cdot (\boldsymbol{r}' \times \boldsymbol{\nabla} g) \tag{501}$$

$$\omega_2 = \boldsymbol{r}' \cdot \frac{\boldsymbol{\nabla}\boldsymbol{\nabla} g}{|\boldsymbol{\nabla} g|} \cdot \boldsymbol{r}' \tag{502}$$

$$\omega_3 = (\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \frac{\boldsymbol{\nabla} g}{|\boldsymbol{\nabla} g|} \tag{503}$$

If $\boldsymbol{n} = \frac{\boldsymbol{\nabla} g}{|\boldsymbol{\nabla} g|}$, then

$$\boldsymbol{\nabla}\boldsymbol{n} = \frac{1}{|\boldsymbol{\nabla} g|}\boldsymbol{\nabla}\boldsymbol{\nabla} g \cdot (\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) \tag{504}$$

$$\boldsymbol{\omega} = \omega_1 \boldsymbol{r}' + \boldsymbol{r}' \times \boldsymbol{r}'' \tag{505}$$

$$\boldsymbol{\omega}_\perp = \boldsymbol{d}_1 \times (\boldsymbol{\omega} \times \boldsymbol{d}_1) = \boldsymbol{r}' \times \boldsymbol{r}'' \tag{506}$$

$$\omega_2 = -\boldsymbol{d}_3 \cdot \boldsymbol{r}'' \tag{507}$$

$$\omega_3 = \boldsymbol{d}_2 \cdot \boldsymbol{r}'' \tag{508}$$

$$\boldsymbol{\omega} = \frac{1}{2}\boldsymbol{d}^{(a)} \times \boldsymbol{d}'_{(a)} \tag{509}$$

$$\boldsymbol{v}_\parallel = \boldsymbol{v} - (\boldsymbol{n} \cdot \boldsymbol{v})\boldsymbol{n} = -\boldsymbol{n} \times (\boldsymbol{n} \times \boldsymbol{v}) \tag{510}$$

$$\varepsilon_{\mu\nu} = \sqrt{h} e_{\mu\nu} \qquad \varepsilon^{\mu\nu} = \frac{1}{\sqrt{h}} e^{\mu\nu} \tag{511}$$

$$\boldsymbol{n} = \frac{1}{2}\varepsilon^{\mu\nu}\boldsymbol{h}_\mu \times \boldsymbol{h}_\nu \qquad \boldsymbol{n} \times \boldsymbol{h}_\alpha = \varepsilon_{\alpha\beta}\boldsymbol{h}^\beta \qquad \boldsymbol{n} \times \boldsymbol{h}^\alpha = \varepsilon^{\alpha\beta}\boldsymbol{h}_\beta \tag{512}$$

$$\partial_\alpha \boldsymbol{n} = -K_\alpha^\beta \boldsymbol{h}_\beta \qquad \partial_\alpha \boldsymbol{h}_\beta = \Gamma^\mu_{\alpha\beta}\boldsymbol{h}_\mu + K_{\alpha\beta}\boldsymbol{n} \tag{513}$$

Proof. Second equation is just a definition. The first follows from $\boldsymbol{h} \cdot \boldsymbol{n} = 0 \Rightarrow \boldsymbol{n} \cdot \partial_\beta \boldsymbol{h}_\alpha = -\boldsymbol{h}_\alpha \cdot \partial_\beta \boldsymbol{n}$. □

Fact 8. *In terms of the Darboux vector, the torsion and curvature of a space curve (parametrized by arclength) with* $\boldsymbol{r}' = \boldsymbol{d}_{(1)}$ *is given by*

$$\tau = \frac{(\boldsymbol{r}' \times \boldsymbol{r}'') \cdot \boldsymbol{r}'''}{\|\boldsymbol{r}' \times \boldsymbol{r}''\|^2} = \omega^{(1)} + \frac{\omega^{(2)}\omega'^{(3)} - \omega^{(3)}\omega'^{(2)}}{(\omega^{(2)})^2 + (\omega^{(3)})^2} \tag{514}$$

$$\kappa = \frac{\|\boldsymbol{r}' \times \boldsymbol{r}''\|}{\|\boldsymbol{r}'\|^3} = (\omega^{(2)})^2 + (\omega^{(3)})^2 \tag{515}$$

G Quaternions

$q = q_0 + q_1 i + q_2 j + q_3 k$, $\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3 = 1, i, j, k$

$$q = q_0 + \boldsymbol{q}, \qquad p = p_0 + \boldsymbol{p} \tag{516}$$

3D vector stuff

$$qp = q_0 p_0 - \boldsymbol{q} \cdot \boldsymbol{p} + q_0 \boldsymbol{p} + p_0 \boldsymbol{q} + \boldsymbol{q} \times \boldsymbol{p} \tag{517}$$

In particular, for pure quaternions (vectors)

$$\boldsymbol{qp} = -\boldsymbol{q} \cdot \boldsymbol{p} + \boldsymbol{q} \times \boldsymbol{p} \tag{518}$$

$$\boldsymbol{q} \cdot \boldsymbol{p} = -\frac{1}{2}\left(\boldsymbol{qp} + \boldsymbol{pq}\right), \qquad \boldsymbol{q} \times \boldsymbol{p} = \frac{1}{2}\left(\boldsymbol{qp} - \boldsymbol{pq}\right) \tag{519}$$

$$\begin{aligned}
\boldsymbol{pq} &= \sum_{i,j=1}^{3} p_i q_j \boldsymbol{e}_i \boldsymbol{e}_j \\
&= -\sum_{k=1}^{3} p_k q_k + \sum_{i \neq j}^{3} p_i q_j \boldsymbol{e}_i \boldsymbol{e}_j \\
&= -\boldsymbol{p} \cdot \boldsymbol{q} + \boldsymbol{p} \times \boldsymbol{q}
\end{aligned}$$

Quaternions can also be thought of as a REAL 4D inner product space

$$\langle q, p \rangle = q_0 p_0 + \boldsymbol{q} \cdot \boldsymbol{p} = \frac{1}{2}\left(q^\dagger p + p^\dagger q\right) \tag{520}$$

$$= Re(q^\dagger p) \tag{521}$$

$$q^\dagger = q_0 - \boldsymbol{q} \tag{522}$$

$$(pq)^\dagger = q^\dagger p^\dagger \tag{523}$$

Quaternions can be thought of a linear operators on this space and $\dagger$ gives adjoints.

$$\langle q, p \rangle = \langle 1, q^\dagger p \rangle \tag{524}$$

G.1 Rotations

Rotation by ϕ radians about unit vector $\boldsymbol{n}$.

$$\boldsymbol{r} \mapsto \boldsymbol{r}' \tag{525}$$

$$\begin{aligned}
\boldsymbol{r}' &= [\cos\phi\,(\boldsymbol{\delta} - \boldsymbol{n}\boldsymbol{n}) + \sin\phi\hat{\boldsymbol{n}} + \boldsymbol{n}\boldsymbol{n}] \cdot \boldsymbol{r} && (526)\\
&= (\boldsymbol{r} - (\boldsymbol{n}\cdot\boldsymbol{r})\boldsymbol{n})\cos\phi + (\boldsymbol{n}\times\boldsymbol{r})\sin\phi + (\boldsymbol{n}\cdot\boldsymbol{r})\boldsymbol{n} && (527)\\
&= \left(\cos^2\frac{\phi}{2} - \sin^2\frac{\phi}{2}\right)\boldsymbol{r} + 2\sin\frac{\phi}{2}\cos\frac{\phi}{2}\,(\boldsymbol{n}\times\boldsymbol{r}) + 2\sin^2\frac{\phi}{2}\,(\boldsymbol{n}\cdot\boldsymbol{r})\,\boldsymbol{n} && (528)
\end{aligned}$$

Where we used $\cos\phi = \cos^2\frac{\phi}{2} - \sin^2\frac{\phi}{2}$ and $\sin\phi = 2\sin\frac{\phi}{2}\cos\frac{\phi}{2}$.

$$q\boldsymbol{r}q^\dagger = \left(q_0^2 - \boldsymbol{q}\cdot\boldsymbol{q}\right)\boldsymbol{r} + 2q_0\,(\boldsymbol{q}\times\boldsymbol{r}) + 2\,(\boldsymbol{q}\cdot\boldsymbol{r})\,\boldsymbol{q} \tag{529}$$

So if $q_0 = \cos\frac{\phi}{2}$ and $\boldsymbol{q} = \sin\frac{\phi}{2}\boldsymbol{n}$ we get the rotation formula above. For non-unit quaternions $|q|^2 = q^\dagger q \neq 1$, you need to use $q^{-1} = \frac{1}{|q|^2}q^\dagger$

$$q\boldsymbol{r}q^{-1} = \frac{1}{|q|^2}\left(q_0^2 - \boldsymbol{q}\cdot\boldsymbol{q}\right)\boldsymbol{r} + 2\frac{1}{|q|^2}q_0\,(\boldsymbol{q}\times\boldsymbol{r}) + 2\frac{1}{|q|^2}\,(\boldsymbol{q}\cdot\boldsymbol{r})\,\boldsymbol{q} \tag{530}$$

$$Q\boldsymbol{e}_i = \boldsymbol{d}_i, \qquad\qquad q\boldsymbol{e}_i q^\dagger = \boldsymbol{d}_i \tag{531}$$

$$Q = (q_0^2 - \boldsymbol{q}\cdot\boldsymbol{q})\boldsymbol{\delta} + 2q_0\hat{\boldsymbol{q}} + 2\boldsymbol{q}\otimes\boldsymbol{q}^T \tag{532}$$

Note that $Q\boldsymbol{q} = \boldsymbol{q}$.

For computations it's super convenient to use (4-by-4) matrices $\boldsymbol{F}_{ij}$

$$Q_{ij} = \boldsymbol{e}_i\cdot\boldsymbol{d}_j = [q, \boldsymbol{q}]\,\boldsymbol{F}_{ij}\begin{bmatrix} q_0 \\ \boldsymbol{q}^T\end{bmatrix} \tag{533}$$

$$\dot{Q} = \hat{\boldsymbol{\omega}}Q, \qquad\qquad \dot{q} = \frac{1}{2}\boldsymbol{\omega}q \tag{534}$$

Proof. Note that $q'q^\dagger$ has no scalar part since $qq^\dagger = 1$ implies that $q'q^\dagger = -qq'^\dagger = -(q^\dagger q')^\dagger$ □

If a quaternion $q = q_0 + \sum_n q_n \boldsymbol{e}_n$ relates two bases by rotation, $\boldsymbol{d}_n = q\boldsymbol{e}_n q^\dagger$ then the components of it's vector part are the same in both bases.

$$q = q_0 + \sum_n q_n \boldsymbol{e}_n = q_0 + \sum_n q_n \boldsymbol{d}_n \tag{535}$$

Proof. $q_0 + \sum_n q_n \boldsymbol{e}_n = q = qqq^\dagger = q_0 + \sum_n q_n \boldsymbol{d}_n$ □

Rotation about unit vector $\boldsymbol{u}$ by angle ϕ.

- Rodrigues rotation formula

$$Q = \boldsymbol{u}\boldsymbol{u} + \cos\phi\,(\boldsymbol{\delta} - \boldsymbol{u}\boldsymbol{u}) + \sin\phi\hat{\boldsymbol{u}} \tag{536}$$

$$= \boldsymbol{u}\boldsymbol{u} - \cos\phi\hat{\boldsymbol{u}}^2 + \sin\phi\hat{\boldsymbol{u}} \tag{537}$$

G.2 Filaments

$\boldsymbol{m} = \sum_i B_i(\omega_i - \sigma_i)\boldsymbol{e}_i$

$$\mathscr{E} = \frac{1}{2}\sum_i B_i(\omega_i - \sigma_i)^2 + \langle g, q' - \frac{1}{2}q\boldsymbol{\Omega}\rangle + \langle \boldsymbol{f}, \boldsymbol{r}' - \boldsymbol{d}_1\rangle \tag{538}$$

$$\delta_{\omega_i}\mathscr{E} = \left(B_i(\omega_i - \sigma_i) - \langle g, \frac{1}{2}q\boldsymbol{e}_i\rangle\right)\delta\omega_i \tag{539}$$

$$= \left(\langle \boldsymbol{m}, \boldsymbol{e}_i\rangle - \langle \frac{1}{2}q^\dagger g, \boldsymbol{e}_i\rangle\right)\delta\omega_i \tag{540}$$

$$\boldsymbol{m} = Vec(\frac{1}{2}q^\dagger g) = \frac{1}{4}(q^\dagger g - g^\dagger q) \tag{541}$$

$$\delta_q\mathscr{E} = \langle g, \delta q' - \frac{1}{2}\delta q\boldsymbol{\Omega}\rangle - \langle \boldsymbol{f}, \delta q i q^\dagger + q i \delta q^\dagger\rangle \tag{542}$$

$$= \langle g, \delta q'\rangle - \langle g, \frac{1}{2}\delta q\boldsymbol{\Omega}\rangle - \langle \boldsymbol{f}, \delta q i q^\dagger\rangle - \langle \boldsymbol{f}, q i \delta q^\dagger\rangle \tag{543}$$

$$= \langle g, \delta q'\rangle + \langle \frac{1}{2}g\boldsymbol{\Omega}, \delta q\rangle - \langle \boldsymbol{f}, \delta q i q^\dagger\rangle - \langle \boldsymbol{f}, \delta q i q^\dagger\rangle \tag{544}$$

$$= \langle g, \delta q'\rangle + \langle \frac{1}{2}g\boldsymbol{\Omega}, \delta q\rangle + \langle 2\boldsymbol{f} q i, \delta q\rangle \tag{545}$$

After integrating by parts we get...

$$= \langle -g' + \frac{1}{2} g\boldsymbol{\Omega} + 2\boldsymbol{f} qi, \delta q \rangle \tag{546}$$

$$g' = \frac{1}{2} g\boldsymbol{\Omega} + 2\boldsymbol{f} qi \tag{547}$$

Using the expression for g' and q' we get

$$\boldsymbol{m}' + \boldsymbol{\Omega} \times \boldsymbol{m} + q^{\dagger}(\boldsymbol{d}_1 \times \boldsymbol{f})q = 0 \tag{548}$$

$$\delta_{\boldsymbol{r}} \mathscr{E} = \langle \boldsymbol{f}, \delta \boldsymbol{r}' \rangle \tag{549}$$

integrating by parts...

$$\boldsymbol{f}' = 0 \tag{550}$$

$$\boldsymbol{m} = \frac{1}{4}(q^{\dagger} g - g^{\dagger} q) \qquad g' = \frac{1}{2} g\boldsymbol{\Omega} + 2\boldsymbol{f} qi \tag{551}$$

now we just need g0

G.3 Filament with drag

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{r} & \boldsymbol{q} & \boldsymbol{\omega} & \boldsymbol{f} & \boldsymbol{\Omega} \end{bmatrix}^T \tag{552}$$

- Ambient cartesian components of centerline position:

$$\boldsymbol{r} = \begin{bmatrix} x & y & z \end{bmatrix}^T \tag{553}$$

- Ambient/Darboux[7] components of Darboux quaternion:

$$q = \begin{bmatrix} q_0 & q_1 & q_2 & q_3 \end{bmatrix}^T \tag{554}$$

[7] They are the same!

- Darboux components of Darboux vector:

$$\underline{\boldsymbol{\omega}} = \begin{bmatrix}\omega_1 & \omega_2 & \omega_3\end{bmatrix}^T \tag{555}$$

- Darboux components of internal force on front cross-sections:

$$\underline{\mathbf{f}} = \begin{bmatrix}f_1 & f_2 & f_3\end{bmatrix}^T \tag{556}$$

- Darboux (?) components of cross-section angular velocity:

$$\underline{\boldsymbol{\Omega}} = \begin{bmatrix}\Omega_1 & \Omega_2 & \Omega_3\end{bmatrix}^T \tag{557}$$

Viscous torque per unit length

$$\boldsymbol{M}_v = -\beta\Omega_1\boldsymbol{d}_1 \tag{558}$$

Viscous force per unit length

$$\begin{aligned}\boldsymbol{F}_v &= -\alpha_\parallel v_1\boldsymbol{d}_1 - \alpha_\perp\left(v_2\boldsymbol{d}_2 + v_3\boldsymbol{d}_3\right) && (559)\\ &= -\left[\alpha_\parallel\boldsymbol{d}_1\boldsymbol{d}_1 + \left(\boldsymbol{\delta} - \boldsymbol{d}_1\boldsymbol{d}_1\right)\right]\cdot\boldsymbol{v} && (560)\\ &= -\boldsymbol{A}\cdot\boldsymbol{v} && (561)\end{aligned}$$

Steric force per unit length

$$\begin{aligned}\boldsymbol{F}_{st}[\boldsymbol{r}] &= -\int_0^L c^2\boldsymbol{\nabla}G(\boldsymbol{r}(s) - \boldsymbol{r}(\xi))\,\mathrm{d}\xi && (562)\\ &= \int_0^L \frac{c^2\left(|\boldsymbol{r}(s) - \boldsymbol{r}(\xi)| + a\right)e^{-|\boldsymbol{r}(s)-\boldsymbol{r}(\xi)/a|}}{4\pi a|\boldsymbol{r}(s) - \boldsymbol{r}(\xi)|^3}\left(\boldsymbol{r}(s) - \boldsymbol{r}(\xi)\right)\mathrm{d}\xi && (563)\end{aligned}$$

Centerline-Darboux frame

$$\frac{\partial\boldsymbol{r}}{\partial s} = \boldsymbol{d}_1(q) \tag{564}$$

Rotation-Darboux vector

$$\frac{\partial q}{\partial s} = \frac{1}{2}q\underline{\boldsymbol{\omega}} \tag{565}$$

Moment balance:

$$\frac{\partial \underline{\mathbf{m}}}{\partial s} + \underline{\boldsymbol{\omega}} \times \underline{\mathbf{m}} + \underline{\mathbf{d}}_1 \times \underline{\mathbf{f}} + \underline{\mathbf{M}}_v = 0 \tag{566}$$

Force balance

$$\frac{\partial \underline{\mathbf{f}}}{\partial s} + \underline{\boldsymbol{\omega}} \times \underline{\mathbf{f}} + \underline{\mathbf{F}}_v + q^\dagger \boldsymbol{F}_{st} q = 0 \tag{567}$$

Rotation-angular velocity

$$\frac{\partial q}{\partial t} = \frac{1}{2} q \underline{\boldsymbol{\Omega}} \tag{568}$$

Darboux vector-angular velocity

$$\frac{\partial \underline{\boldsymbol{\omega}}}{\partial t} = \frac{\partial \underline{\boldsymbol{\Omega}}}{\partial s} + \underline{\boldsymbol{\omega}} \times \underline{\boldsymbol{\Omega}} \tag{569}$$

H Generalized functions

Curve delta:
E.g. a 1D charge distribution.

$$\delta_{\boldsymbol{r}(s)}(\boldsymbol{x}) = \int_0^L \delta(\boldsymbol{x} - \boldsymbol{r}(s))\,\mathrm{d}s \tag{570}$$

$$\int_{\mathbb{R}^3} f(\boldsymbol{x}) \delta_{\boldsymbol{r}(s)}(\boldsymbol{x})\,\mathrm{d}V(\boldsymbol{x}) = \int_0^L f(\boldsymbol{r}(s))\,\mathrm{d}s \tag{571}$$

I Numerics